General Organic Chemistry

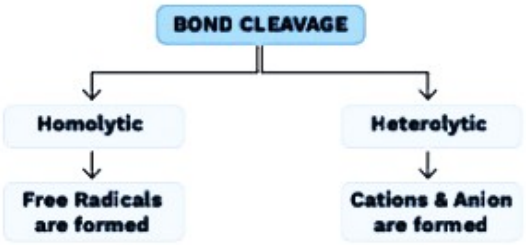
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Bond Breaking and Intermediate

Bond Cleavage

A process in which breaking and formation of covalent bonds take place. A bond can be cracked by two ways :



Homolytic Fission or Homolysis

Homolysis generates free radical in the reaction.



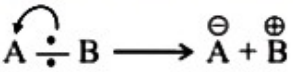
The factor which supports homolysis is zero or a small modification in electronegativity between A and B.

$\mu \approx 0$ or $\Delta EN = 0$

Homolytic bond breaking takes place in gaseous phase or in the attendance of non polar solvents (CCl_4, CS_2).

Heterolytic Bond Fission or Heterolysis

In heterolysis, the covalent bond is cracked in such away that one species (i.e., less electronegative) loss its own electron, while the other species (i.e., more electronegative) gains equal electrons.

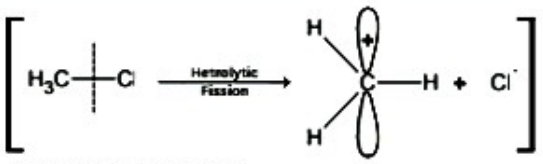


Reaction Intermediates

(1) Carbocation (2) Carbanion (3) Free radical

1. Carbocations:

Carbon atom bearing six electrons in its furthest shell and has a positive charge is called a carbocation. key intermediates in several reactions and mainly in nucleophilic substitution reactions.



* Hetrolytic fission
* Octet incomplete ($6e^-$)
* Electron deficient
* Hybridization – sp^2
* Shape – Trigonal planar

Characteristics of Carbocation :

a) Carbocation have a positive charge on carbon.
b) Positively charged carbocation has only six electrons in the outermost shell. It has an incomplete octet. So it behaves like a Lewis acid.
c) Since all electrons are in paired state so carbocations are diamagnetic.
d) Carbocations are sp^2 hybridised.
e) Carbocations are planar.
f) Carbocations are formed in polar solvent.
g) It stabilize by presence of electron donating group (EDG)

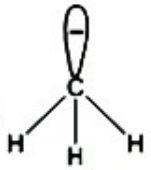
2. Carbanions :

Carbanions are anions of carbon, produced by the removal of one of the groups attached to a carbon without removing the bonding electrons.

Formation of carbanion

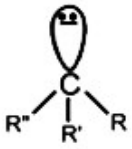


* Hetrolysis fission
* Octet complete
* Hybridization – sp^3
* Shape – Trigobnal pyramidal
* Geometry – Tetrahedral



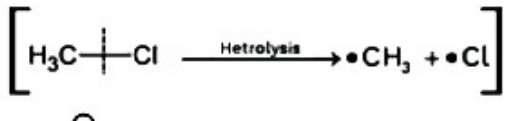
Characteristic of Carbanions:

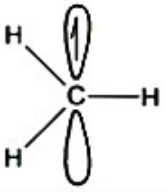
a) It is formed by heterolytic bond fission.
b) It behaves as charged nucleophile.
c) It is diamagnetic in character because all eight electrons are paired.
d) Hybridisation and geometry : Alkyl carbanion has three bond pairs and one lone pair. Thus hybridisation is sp^3 and geometry is pyramidal



e) It reacts with electrophiles.
f) It stabilize by presence of electron withdrawing group (EWG)

3. Carbon Free Radical :





* Hetrolysis fission
*Octet incomplete($7e^-$)
*Hybridization – sp^2
*Shape – Trigonal planar

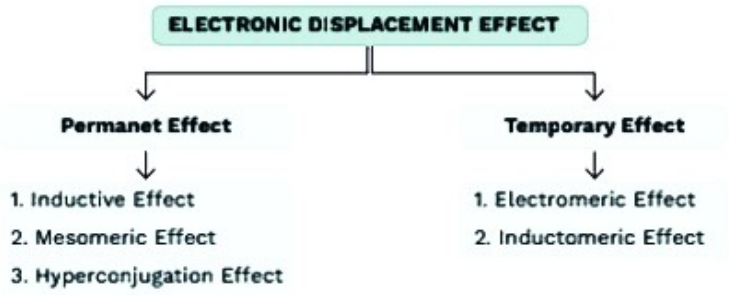
a) Carbon free radicals are odd electron species in which carbon atom bears the odd electron.
b) Homolytic bond fission of a covalent single bond gives rise to free radicals.
c) There are seven electrons in the outer most shell of carbon free radicals.
d) Owing to the presence of an odd electron; a carbon radical is paramagnetic in nature. Due to this reason free radicals are highly reactive.
e) Carbon free radicals are normally sp^2 hybridized
f) Free radicals are neutral electrophiles.
g) It stabilize by presence of electron donating group (EDG)

Inductive Effect

Electron Displacement in covalent bonded molecules :

Dissimilar electronic effects which operate in covalent bonds are

1. Inductive effect
2. Mesomeric and Resonance effect
3. Electrometric effect
4. Hyper conjugation



Inductive Effect (I-Effect)

The permanent, partial displacement of sigma electrons due to electronegativity difference between groups towards more electronegative atom is known as inductive effect.

$$\overset{\delta\,\delta\,\delta\,\delta+}{C}\text{---}\overset{\delta\,\delta\delta+}{C}\text{---}\overset{\delta\delta+}{C}\text{---}\overset{\delta+}{C}\text{---}\overset{\delta-}{Cl}$$

* (Cl : functional group)

This effect is spread through the chain of σ bonds and diminishes, with increasing chain length. Inductive effect is decreases.

Characteristic of Inductive Effect

a) It is a permanent effect and distance dependent.
b) After three carbon effect is decreases.
c) Operates through σ bonds
d) Polarization of electrons is always in single direction.
e) It is generally observed in saturated compounds.
f) For hydrogen inductive is zero.
g) Non polar molecules do not have polarity so inductive is not working.
h) In general alkyl group show (+I) effect.
i) Electron do not changes it's atomic orbital.
j) Inductive effect is additive effect.
k) Its magnitude (i.e., electron withdrawing or donating power) decreases with increase in distance.
On the basis of electron releasing group (ERG) or electron withdrawing group (EWG) inductive effect. There are two types of inductive effect



1. **–I groups :** The group which withdraws electrons is known as –I group and its effect is known as – I effect.

–I = SERIES:

$-\overset{\oplus}{N}F_3 > -\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2 > -CN > -COOH > -F > -Cl- > -Br > -I >$

$-COOR > -OR > -OH > -NH_2 > -C \equiv CH > -Ph > -CH = CH_2$

2. **+I group :** The group which contributes or gives electron is recognized as + I group and effect is acknowledged as + I effect.

e.g., Alkyl groups, $-\bar{O}, -\overset{\overset{O}{\|}}{C}-\overset{\ominus}{O}$ are + I groups. (Generally negative charged atoms or groups)

(a) +I power of different type groups and anions :

* $-\bar{N}H > -O^- > -COO^-$ > tertiary alkyl > secondary alkyl > primary-alkyl $> -CH_3 > -H$
* + I power in decreasing order with reference to H–atom

(b) + I power of same type of alkyl groups :

+ I power $\propto$ number of C's in same type of alkyl group

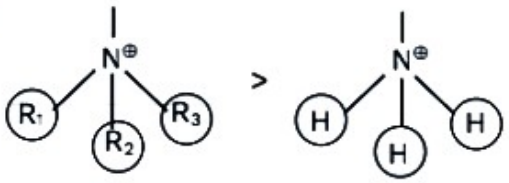
For example:

CH_3-CH_2-
$CH_3-CH_2-CH_2-$
$CH_3-CH_2-CH_2-CH_2-$

↓ + I power is increasing order

Special Note :

1.



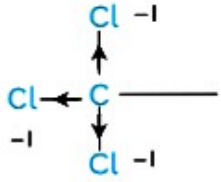
Size of R_1, R_2 and R_3 increases that's why molecule tend to planar, planarity increases, S%↑, sp^2 character increases, EN value increases.

2. +I order of isotopes increases as atomic weight increases

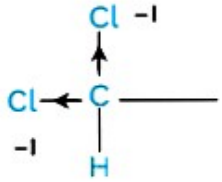
Example : C–H < C–D < C–T

* Less vibrational energy
* Effective collision
* Strong bond formation
* Less bond length
* Inductive effect $\propto \left(\frac{1}{\text{distance}}\right)$

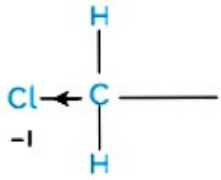
3. Inductive effect is additive effect



3 times (–I)



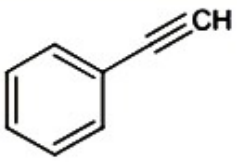
2 times (–I)

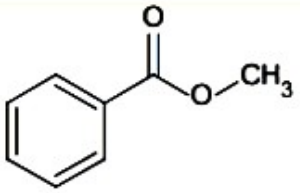


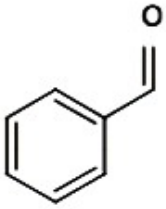
(–I) of single 'Cl' atom only.

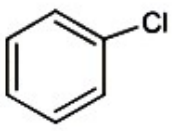
Identify the type of **inductive effect** of **atom or group** which is **directly attached to the cyclic system?**

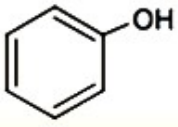
Q1.

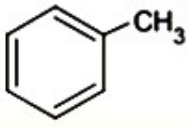
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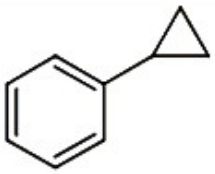
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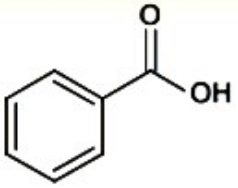
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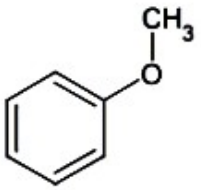
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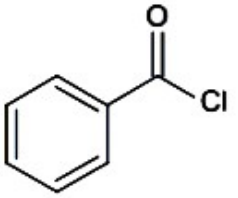
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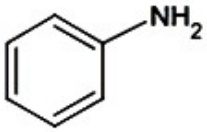
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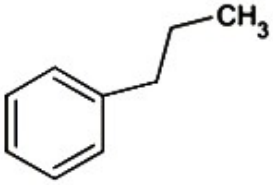
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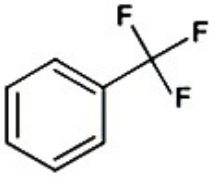
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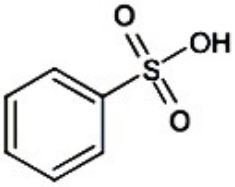
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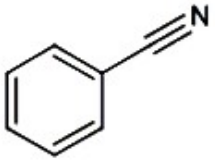
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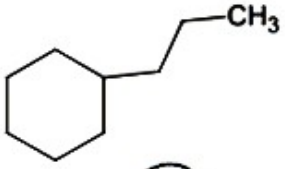
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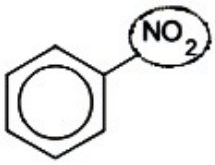
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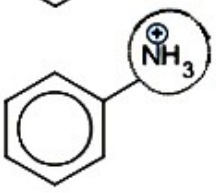
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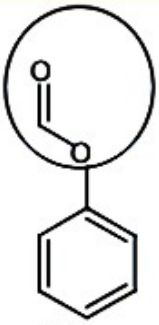
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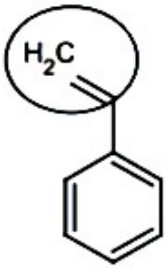
A1.

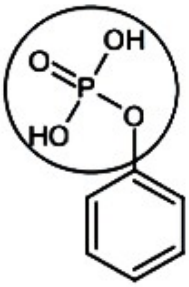
1.	–I	2.	–I	3.	–I	4.	–I	5.	–I	6.	+I
7.	+I	8.	–I	9.	–I	10.	–I	11.	–I	12.	+I
13.	–I	14.	–I	15.	–I	16.	No effect			17.	–I
18.	–I										

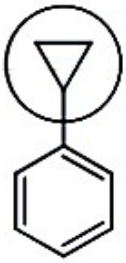
Identify the type of inductive effect of atom or group which is circled?

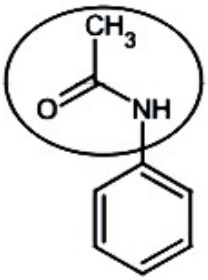
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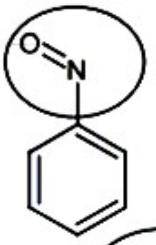
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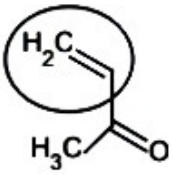
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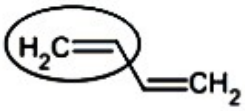
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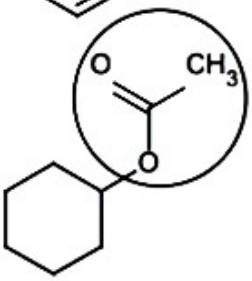
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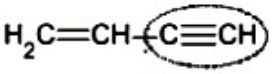
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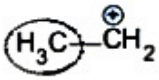
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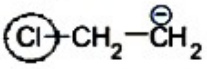
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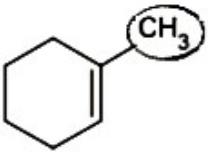
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10. 

11. 

12. 

13. 

A2.

1.	−I	2.	+I	3.	−I	4.	+I	5.	−I	6.	−I
7.	+I	8.	Zero	9.	−I	10.	−I	11.	+I	12.	−I
13.	+I										

Application of Inductive Effect

Applications of Inductive Effect

1. Stability of carbocation
2. Stability of free Radical
3. Stability of carbanion

1. Stability of carbocation

Stability of carbocation $\propto$ Presence of electron releasing group

$\propto$ + I group $\propto \frac{\text{I}}{-\text{I group}}$.

Concept Building Exercise

Q3. $\underset{3^\circ}{(CH_3)_3C^{\oplus}}, \underset{2^\circ}{(CH_3)_2CH^{\oplus}}, \underset{1^\circ}{CH_3CH_2^{\oplus}}, CH_3^{\oplus}$

A3. 3° > 2° > 1° , reason – Higher number of alkyl groups, higher will be stability of carbocation.

Q4. $\overset{\oplus}{C}H_2 - CH_3, Br - CH_2 - \overset{\oplus}{C}H_2, Cl - CH_2 - \overset{\oplus}{C}H_2, F - CH_2 - \overset{\oplus}{C}H_2$

A4. (i) > (ii) > (iii) > (iv), reason - –I power of elements

Q5. $\overset{\oplus}{C}H_2 - \underset{\underset{F}{|}}{C}H - CH_2CH_3, \overset{\oplus}{C}H_2 - CH_2 - \underset{\underset{F}{|}}{C}H - CH_3, \overset{\oplus}{C}H_2 - CH_2 - CH_2 - CH_2 - F$

Maximum-I
Minimum distance

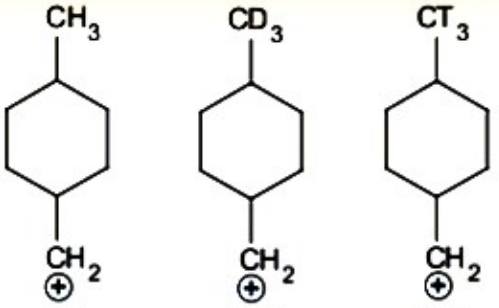
Minimum-I
Maximum distance

A5. Ans. 3 > 2 > 1, Distance dependence

Q6. $\underset{\underset{NO_2}{|}}{CH_2} - CH_2 - \overset{\oplus}{C}H_2, \quad \underset{\underset{F}{|}}{CH_2} - CH_2 - \overset{\oplus}{C}H_2, \quad \underset{\underset{OH}{|}}{CH_2} - CH_2 - \overset{\oplus}{C}H_2$

A6. 3 > 2 > 1, Reason = –I power of groups inversely proportional to stability of carbocation.

Q7.



A7. 3 > 2 > 1,
Reason – isotopic effect in inductive.

2. Stability of free Radical

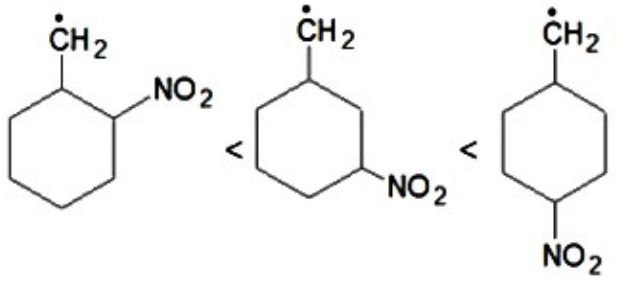
Stability of F.R. ∝ Presence of electron releasing group ∝ +I group ∝ $\frac{I}{-I \text{ group}}$

Stability of F.R. decline in the following order

a) $\underset{3^\circ}{(CH_3)_3C^\bullet} > \underset{2^\circ}{(CH_3)_2CH^\bullet} > \underset{1^\circ}{CH_3CH_2^\bullet} > CH_3^\bullet$

b) $\dot{C}H_2 - CH_3 > Br - \dot{C}H_2 - CH_3 > Cl - \dot{C}H_2 - CH_3 > F - \dot{C}H_2 - CH_3$
Maximum – I of F

c) $CH_3 - \dot{C}H_2 > \underset{F}{CH_2} - \dot{C}H_2 > \underset{CN}{CH_2} - \dot{C}H_2 > \underset{NO_2}{CH_2} - \dot{C}H_2$

d) 

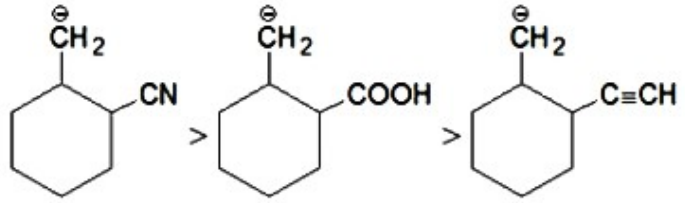
3. Stability of carbanion

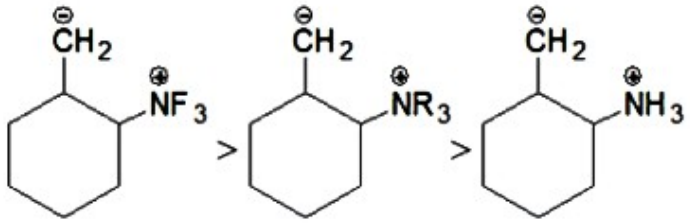
Stability of carbanion ∝ Presence of electron withdrawing group ∝ –I group ∝ $\frac{1}{+I \text{ group}}$

a) $\underset{3^\circ}{(CH_3)_3C^\ominus} < \underset{2^\circ}{(CH_3)_2CH^\ominus} < \underset{1^\circ}{CH_3CH_2^\ominus} < CH_3^\ominus$

b) $\overset{\ominus}{C}H_2 - CH_2 - NO_2 > \overset{\ominus}{C}H_2 - CH_2 - F > \overset{\ominus}{C}H_2 - CH_2 - OH > \overset{\ominus}{C}H_2 - CH_2 - NH_2$

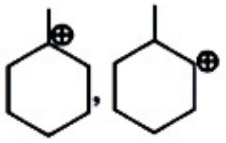
c) $\overset{\ominus}{C}H_2 - \underset{\underset{F}{|}}{CH} - CH_2 - CH_3 > \overset{\ominus}{C}H_2 - CH_2 - \underset{\underset{F}{|}}{CH} - CH_3 > \overset{\ominus}{C}H_2 - CH_2 - CH_2 - CH_2 - F$

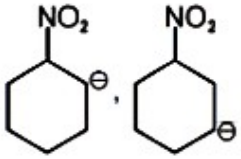
d) 

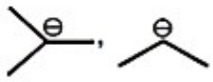
e) 

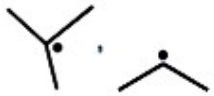
Q8. Compare Stability

(i) 

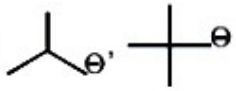
(ii) 

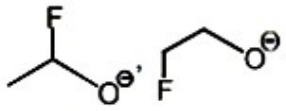
(iii) 

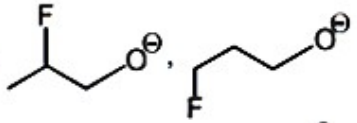
(iv) 

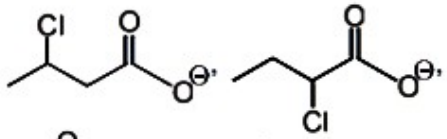
(v) 

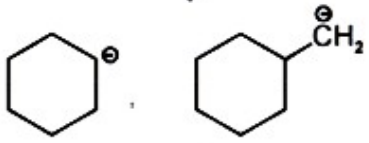
(vi) $F - CH_2 - \overset{\ominus}{C}H_2,\ Cl - CH_2 - \overset{\ominus}{C}H_2$

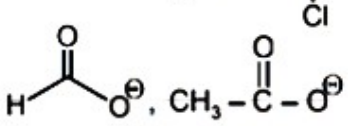
(vii) 

(viii) 

(ix) 

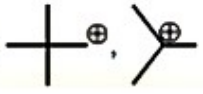
(x) 

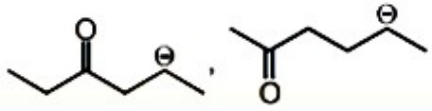
(xi) 

(xii) 

(xiii) $CF_3 - \overset{O}{\overset{\|}{C}} - O^{\ominus},\ CCl_3 - \overset{O}{\overset{\|}{C}} - O^{\ominus}$

(xiv) $CHCl_2 - \overset{O}{\overset{\|}{C}} - O^{\ominus},\ O_2N - CH_2 - \overset{O}{\overset{\|}{C}} - O^{\ominus}$

(xv) 

(xvi) 

A8.

(i) $1 > 2$	(ii) $1 > 2$	(iii) $1 > 2$
(iv) $2 > 1$	(v) $1 > 2$	(vi) $1 > 2$
(vii) $1 > 2$	(viii) $1 > 2$	(ix) $1 > 2$
(x) $2 > 1$	(xi) $2 > 1$	(xii) $1 > 2$
(xiii) $1 > 2$	(xiv) $1 > 2$	(xv) $2 > 1$
(xvi) $1 > 2$		

Introduction of Resonance & Conjugation condition

Consider structures of benzene:
(Draw Benzene Structure)

According to the given structure C_1–C_2 bond length should be smaller then C_2–C_3 bond length but C_1–C_2 and C_2–C_3 bond length must be equal. So this bond length data is not explained by single structure so we need two structures of benzene which is following

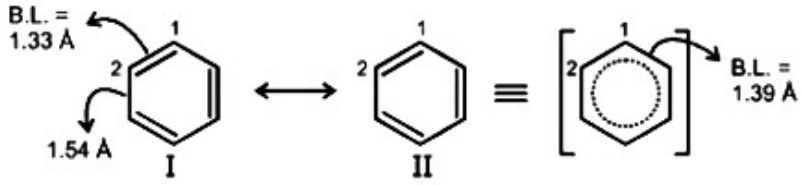


I and II are resonating structures of C_6H_6.

Resonating structure are hypothetical and actual structure is a resonance hybrid.
Resonance generally occurs when there is a conjugation of π-bonds (alternate double - single – double bond arrangement)

Delocalisation of lone pair and p electron in benzene.

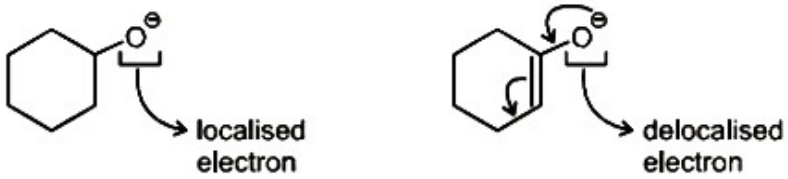
1.



Hypothetical structure
or
Resonating structure
or
Canonical structure
or
Uninterruped structure

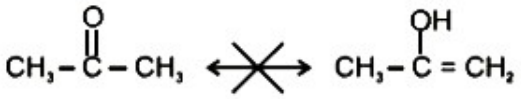
Real structure
or
Resonance hybrid

*** Localised and delocalised electron**



Important point regarding Resonance

1. Resonance exist only on paper and it is a hypothetical concept.
2. When property of molecule is not explained by one structure or two or more structure are needed to explain the property of the molecule then that structures are known as resonating structures and they are hypothetical in nature.
3. Combination of these resonating structures is known as resonance hybrid and it is a real structure.
4. Resonance is an intra-molecular process (i.e. it cannot take e^- from outside)
5. Resonance hybrid will be formed by contribution of resonating structures. More stable the resonating structure more will be the contribution in resonance hybrid.
6. No effect of pressure, temperature on resonating structures (because they are hypothetical)
7. In resonance we can only move electron, not atoms.
   Example :



   Not resonating structures.
8. Number of paired and unpaired electron should be same in each resonating structures.
   Example :



   no. of unpaired electron = 1 — no. of unpaired electron = 3
9. Total number of electron must be same in each resonating structure or charge should be conserved in all resonating structure because resonance is an intra-molecular process.
   Example :



   Total e^- = 18 — Total e^- = 20

   Total e^- = 18 — Total e^- = 20

Conditions for Resonance :

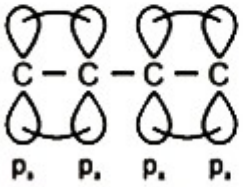
1. For resonance molecule or part of molecule where resonance take place should be planar (i.e. each atom having hybridisation → sp or sp^2)
2. For resonance molecule or part of molecule where resonance take place should have parallel p-orbitals.

Examples :

$CH_2 = CH - CH = CH_2$
$sp^2 \quad sp^2 \quad sp^2 \quad sp^2$

Buta-1,3-diene

* Buta-1,3-diene

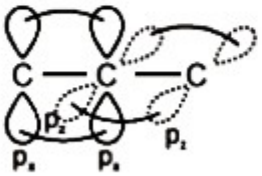


* Planar ✓
* Parallel p-orbital ✓
* Resonance ✓

$CH_2 = C = CH_2$
$sp^2 \quad sp \quad sp^2$

Propa-1,2-diene

* Propa-1,2-diene



* Planar ✓
* Parallel p-orbital ×
* Resonance ×

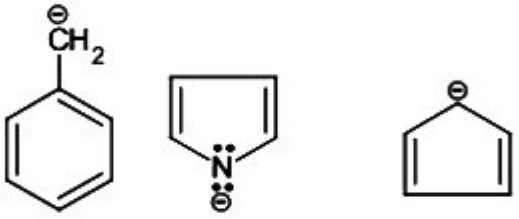
3. **Conjugation and Delocalization :**

Types of Conjugations

(a) Negative charge electron alternate to π electron conjugation :

If in a conjugated system π bond and Θ charge at alternate position so conjugation is known as negative charge electron alternate to π electron conjugation

$CH_2 = CH - \overset{\ominus}{C}H_2$ $\qquad$ $CH_2 = CH - CH = CH - \overset{\ominus}{C}H_2$



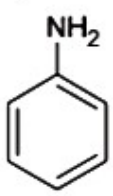
(b) Lone pair electron alternate to π electron conjugation :

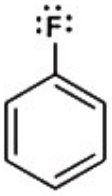
If in a conjugated system π bond and lone pair at alternate position so conjugation is known as lone pair electron alternate to π electron conjugation.

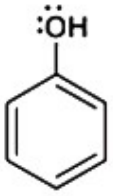
$CH_2 = CH - \overset{\cdot\cdot}{N}H_2$ $\qquad$ $CH_2 = CH - \overset{\cdot\cdot}{\underset{\cdot\cdot}{X}}:$

$CH_2 = CH - CH = CH - CH = CH - \ddot{N}H_2$

(1) (3) (5) (7)







(c) π electron alternate to π electron conjugation :

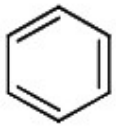
If π bond alternate with π bond so conjugation is known as π electron alternate to π electron conjugation

$CH_2 = CH - CH = CH_2$

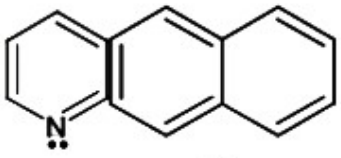
(A)

$CH_3 - CH_2 - CH = CH - CH = CH - CH = CH_2$

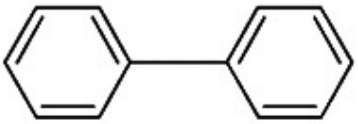
(B)



(C)



(D)



(E)

All of the above compounds have π, π conjugation.

(d) π electron alternate to vacant orbital (Positive charge) conjugation :

If π bond alternate with vacant orbital so conjugation is known as positive charge alternate to π electron conjugation.

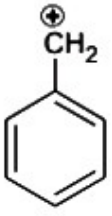
$CH_2 = CH - \overset{\oplus}{C}H_2$

$CH_2 = CH - CH = CH - \overset{\oplus}{C}H_2$

(B)



(C)



(D)



(E)

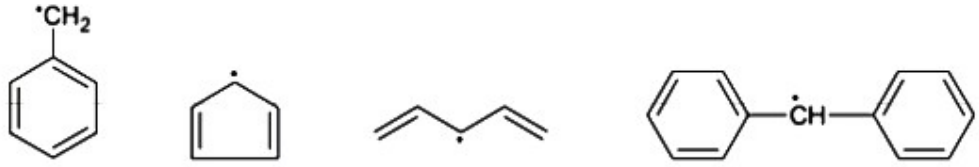
These species has two type of conjugation π electron alternate to π electron conjugation and positive charge alternate to π electron conjugation.

(e) π electron alternate to odd electron (free radical) conjugation :

If π bond and odd e⁻ are in alternate position.

$CH_2 = CH - \dot{C}H_2$

$CH_2 = CH - CH = CH - \dot{C}H_2$



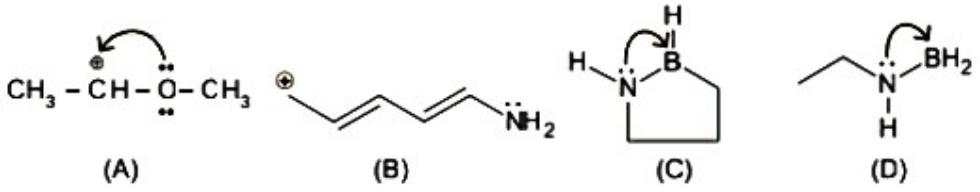
(f) Lone pair electron alternate to vacant orbital (Positive charge) conjugation :

If lone pair and vacant orbital are in alternate position

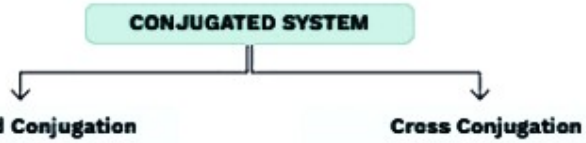
(i) $CH_3 - \overset{\oplus}{C}H - \ddot{N}H_2 \longleftrightarrow CH_3 - CH = \overset{\oplus}{N}H_2$

(ii) $CH_3 - \overset{\oplus}{C}H - \ddot{O}H \longleftrightarrow CH_3 - CH = \overset{\oplus}{O}H$

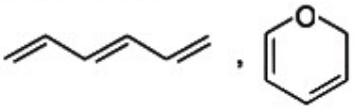
For example :



On the basis of direction conjugated system are of two types

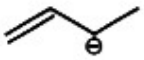
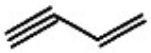
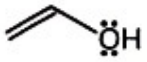
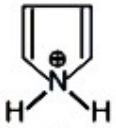
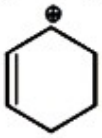
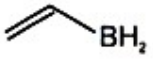
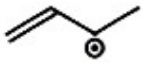
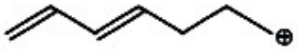
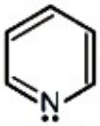
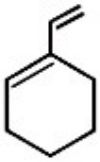
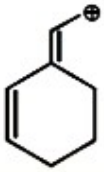
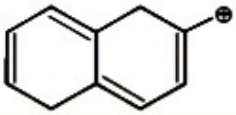


1. Conjugation: Conjugated system having at least 3 segments, also which has unidirectional flow of electrons are known as extended conjugation.

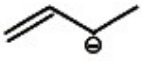
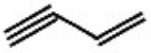
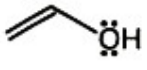
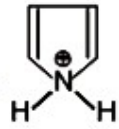
Ex: 

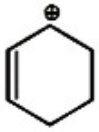
2. Cross Conjugation: Conjugated system having at least 3 segment and also having opposite direction of flow of electrons with no interaction between terminal segment is known as cross conjugation.

Q9. **In the following given examples find which are conjugated :**

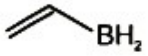
1. 
2. 
3. 
4. 
5. 
6. 
7. 
8. 
9. 
10. 
11. 
12. 
13. 
14. 
15. 
16. 
17. 
18. 
19. $CH_2 = C = C = CH_2$
20. 
21. 
22. 
23. 

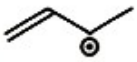
A9. (Star marked compounds have delocalization)

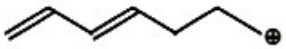
1. 
2*. 
3*. 
4. 
5*. 
6*. 
7*. 
8. 
9*. 
10*. 
11*. 
12. 

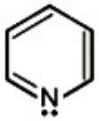
13*. 

14*. 

15*. 

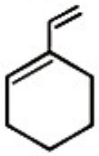
16*. 

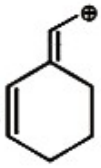
17*. 

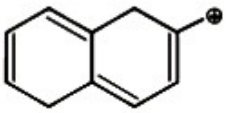
18*. 

19. $CH_2 = C = C = CH_2$

20*. 

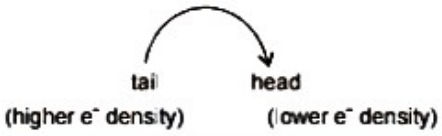
21*. 

22*. 

23*. 

How to draw Resonating Structures

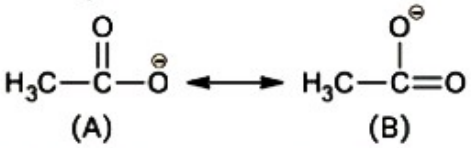
Symbols used in Resonance :

S.NO.	SYMBOL	NAME	USE
1.		Single curved arrow or single headed arrow	for transfer of two electrons
2.		fish hook arrow	for transfer of one electron
3.		double headed arrow	Used to separate resonating structures
4.			

Rules to draw valid resonating structures

1. Electron flow happens from high electron density to low electron density. Negative charge Lone pair electron → π-bond electron → δ+ (positive charge)
2. Position of atom cannot be change, only the π-electrons are delocalised.

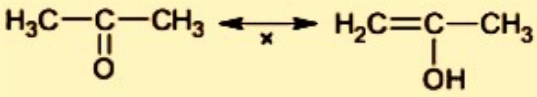
Example :



(A) (B)

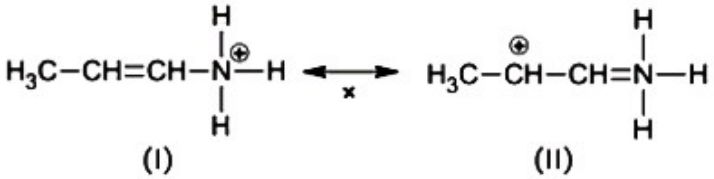
A & B are Resonance forms.

Note :



Above forms are not resonating forms because position of H is changed.

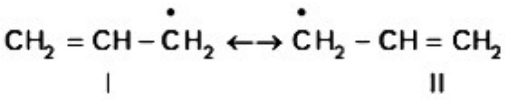
3. Second period elements should not violate the octet rule while drawing resonating structures



(I) (II)

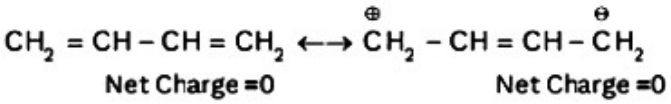
Above structures are not resonance form because in structure II, nitrogen has 10 valence electrons which violates octet rule.

4. Each Resonating contributor must have the similar number of unpaired and paired electrons.



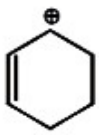
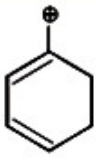
I & II are contributing structure because both have 8 paired electrons and 1 unpaired electron

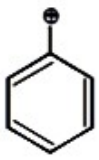
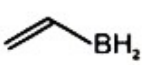
5. Net charge should be conserved in all resonating structures.



Contributing structures have same net charge so both are resonance forms.

Case-I : Double Bond Conjugated with Carbocation / Vacant p-orbital.

Q10. 1.  2.  3. 

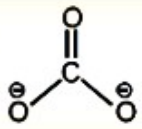
4.  5.  6.  7. 

Case-II : Double Bond Conjugated with Lone Pair / Negative Charge.

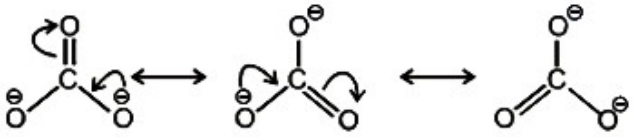
Note:

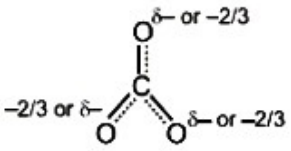
When double bond is conjugated with lone pair or (–ve) charge then (–ve) charge & lone pair will convert into π-bond & π-bond will convert into (–ve) charge & lone pair.

Q11.

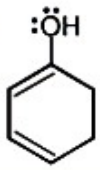


A11.

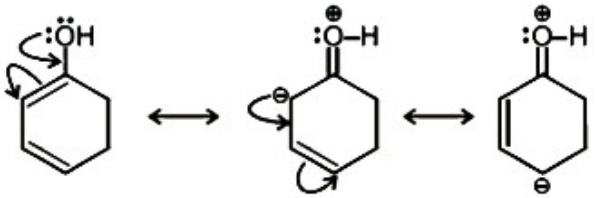


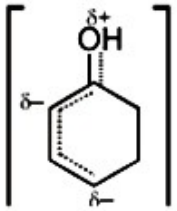


Q12.



A12.



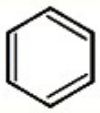


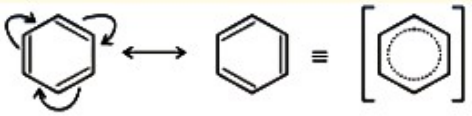
Q13. $CH_2 = CH - \ddot{N}H_2$

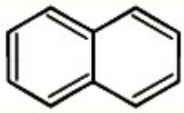
A13. $CH_2 = CH - \ddot{N}H_2 \longleftrightarrow \overset{\ominus}{C}H_2 - CH = \overset{\oplus}{N}H_2$

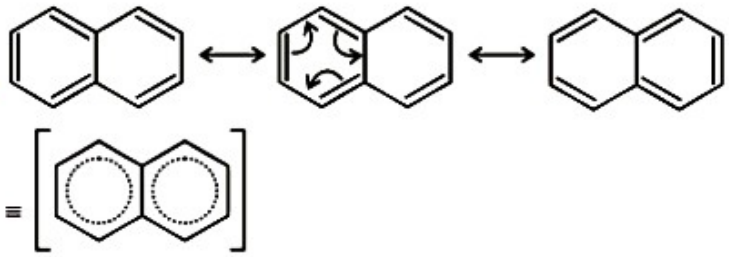
$$\left[\overset{\delta-}{CH_2} \cdots\cdots CH \cdots\cdots \overset{\delta+}{NH_2} \right]$$

Case-III : Double Bond is Conjugated with Double Bond :

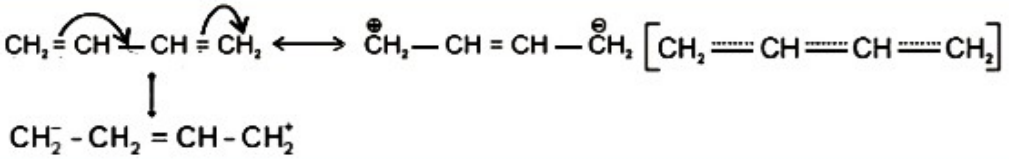
Q14. 

A14. 

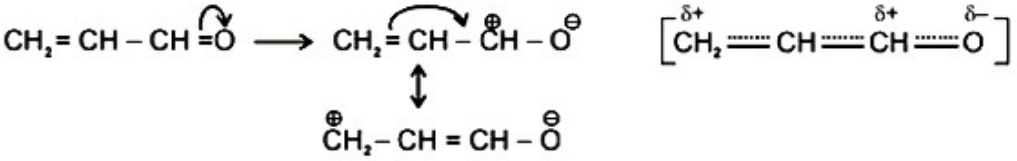
Q15. 

A15. 

Q16. $CH_2 = CH - CH = CH_2$

A16. 

Q17. $CH_2 = CH - CH = O$

A17. 

Case-IV : Double bond is conjugated with free radical.

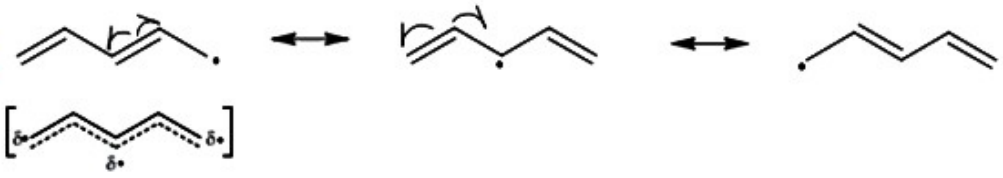
Q18. $CH_2 = CH_2 - \overset{\oplus}{C}H_2$

A18. $\overset{\curvearrowright}{CH_2} = \overset{\curvearrowright}{CH_2} - \overset{\oplus}{C}H_2 \longrightarrow \overset{\oplus}{C}H_2 - CH = CH_2 \equiv \left[\overset{\delta+}{C}H_2 \cdots CH \cdots \overset{\delta+}{C}H_2\right]$

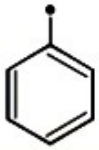
Q19.



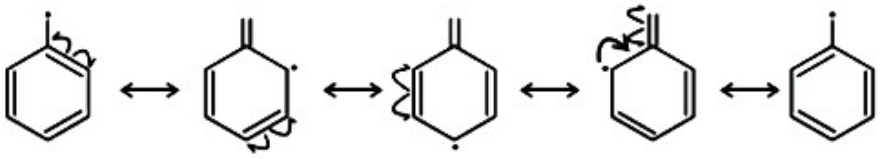
A19.

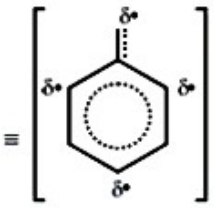


Q20.

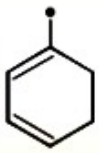


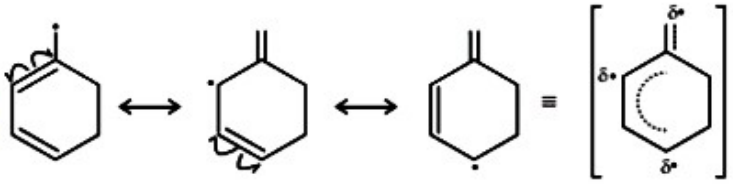
A20.





Q21.



A21. 

Case-V : d-orbital resonance — (–ve) charge or l.p. conjugated with vacant d-orbital.

Note :
When (–ve) charge is at adjacent position to vacant d-orbital then this is known as d-orbital resonance.

Concept Building Exercise

Q22. 1. $\overset{\ominus}{C}H_2-\ddot{S}H$ 2. $\overset{\ominus}{C}H_2-\ddot{C}l:$

3. $\overset{\ominus}{C}H_2-PH_2$ 4. $\overset{\ominus}{C}H_2-\ddot{O}H$

5. $\overset{\ominus}{C}H_2-\ddot{F}:$

A22. (Star marked compounds have delocalization)

1.* $\overset{\ominus}{C}H_2-\ddot{S}H \longleftrightarrow CH_2=\overset{\ominus}{S}H \equiv \left[\overset{\delta-}{C}H_2\cdots\overset{\delta-}{S}H\right]$

2.* $\overset{\ominus}{C}H_2-\ddot{C}l: \longleftrightarrow CH_2=Cl^{\ominus} \left[\overset{\delta-}{C}H_2\cdots\overset{\delta-}{C}l\right]$

3.* $\overset{\ominus}{C}H_2-\ddot{P}H_2 \longleftrightarrow CH_2=\overset{\ominus}{P}H_2 \left[\overset{\delta-}{C}H_2\cdots\overset{\delta-}{P}H_2\right]$

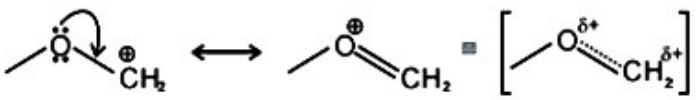
4. $\overset{\ominus}{C}H_2-\ddot{O}H \longleftrightarrow \times$

beause O do not have vacant d-orbital.

5. $\overset{\ominus}{C}H_2-\ddot{F}: \longleftrightarrow \times$

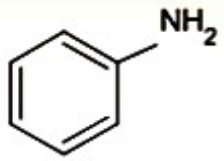
because F does not have vacant d-orbital.

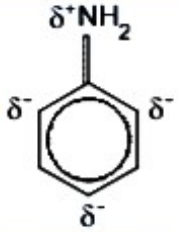
Case-VI : Lone pair conjugated with vacant orbital.

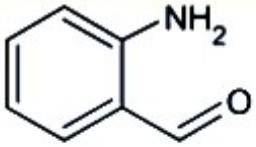


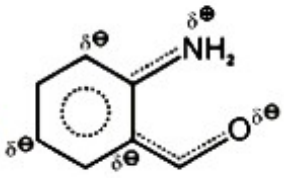
Draw the Resonance Hybrid of following species ?

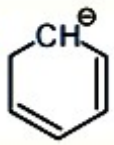
Concept Building Exercise

Q23. 

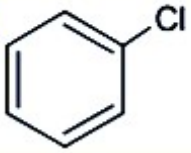
A23. 

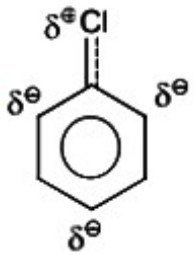
Q24. 

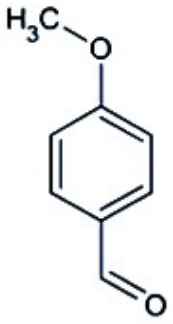
A24. 

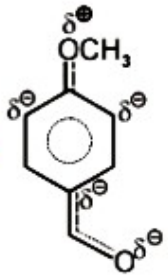
Q25. 

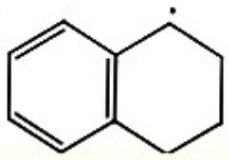
A25. 

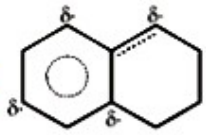
Q26. 

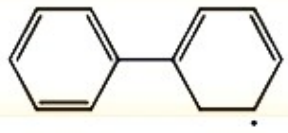
A26. 

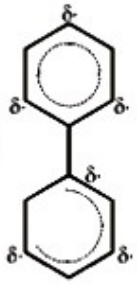
Q27. 

A27. 

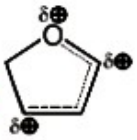
Q28. 

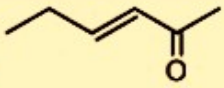
A28. 

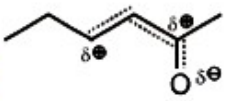
Q29. 

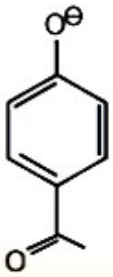
A29. 

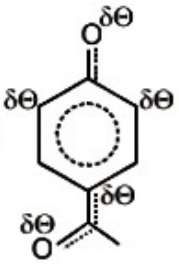
Q30. 

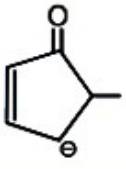
A30. 

Q31. 

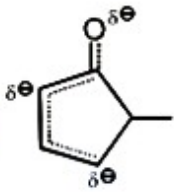
A31. 

Q32. 

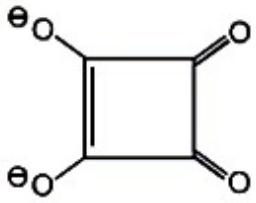
A32. 

Q33. 

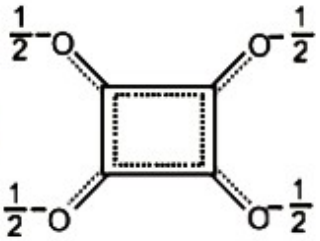
A33.



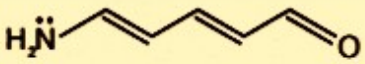
Q34.



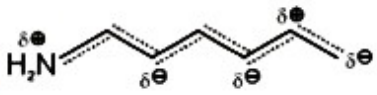
A34.



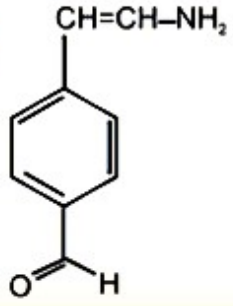
Q35.



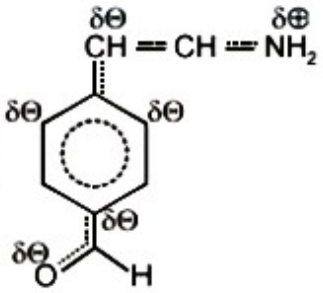
A35.

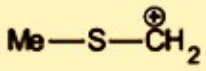


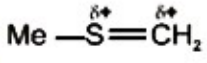
Q36.



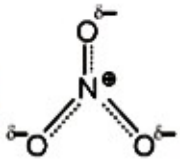
A36.

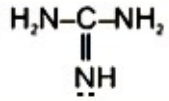


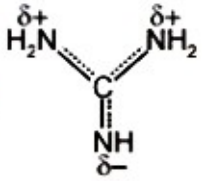
Q37. 

A37. 

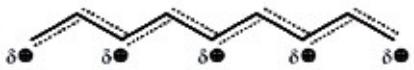
Q38. 

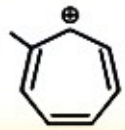
A38. 

Q39. 

A39. 

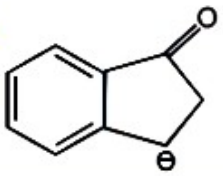
Q40. 

A40. 

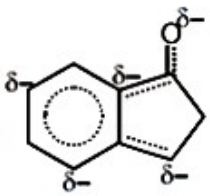
Q41. 

A41. 

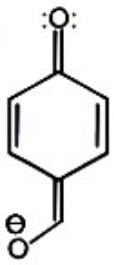
Q42.



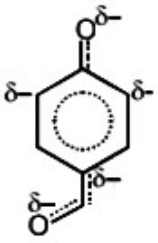
A42.



Q43.



A43.



Rules for Determining Stability Order of Resonating Structures

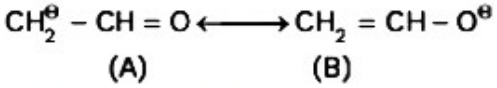
1. Among various resonating structure non polar contributors are more stable than polar resonating structure more number of covalent bonds is considered to be more contributing than others.



2. If two resonance structures have same number of bonds then that structure in which octet of every atom is complete is more resonating than structure having incomplete octet.



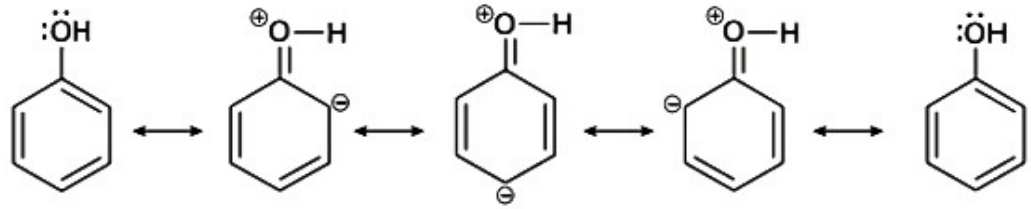
3. If resonating structures have same number of bonds then the neutral structure is extra stable than charged structure and if given both are charged Resonating structure then fewer charged contributor structure is more stable.
4. In case of charged contributing structures, the most stable one is that in which the positive and negative charge present on the most electropositive and most electronegative atoms of the structure.



(A) (B)

Stability of B > A

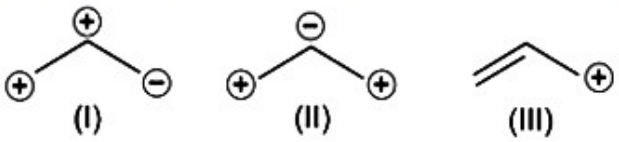
5. Resonating structure in which unlike charges are closer is more contributing than that structures in which unlike charges are far away.



(I) (II) (III) (IV) (V)

I ≈ V > II ≈ IV > III

6. Resonating structure in which like charges are farther away is more contributing as compared to structure in which like charges are closer.

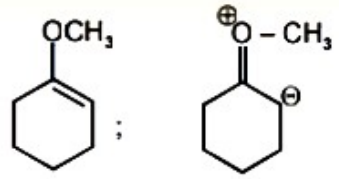


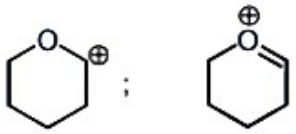
(I) (II) (III)

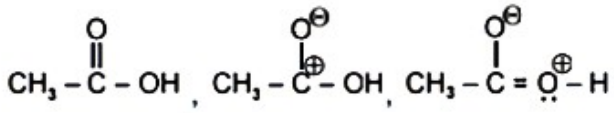
Order = III > II > I

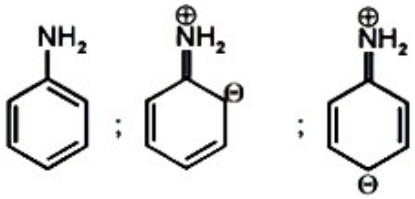
Concept Building Exercise

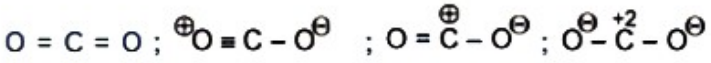
Q44.

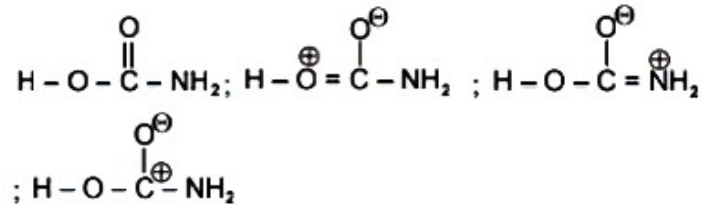
1.  I > II

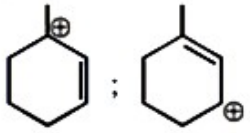
2.  II > I

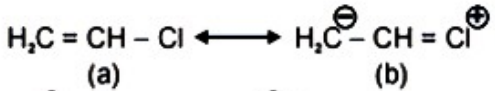
3.  I > III > II

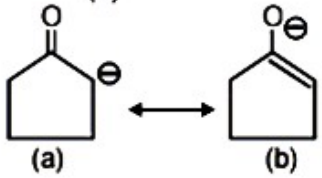
4.  I > II > III

5.  I > II > III > IV

6.  I > III > II > IV

7.  I > II

8.  a > b

9.  b > a

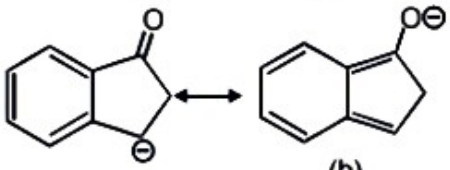
10.  b > a

11. $CH_3 - O - \overset{\oplus}{C}H_2 \longleftrightarrow CH_3 - \overset{\oplus}{O} = CH_2$
(a) (b)

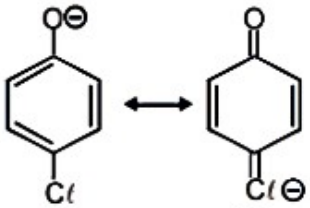
b > a

12. $H - N = C = O \longleftrightarrow H - \overset{\oplus}{N} \equiv C - \overset{\ominus}{O} \longleftrightarrow H - \overset{\ominus}{N} - C \equiv \overset{\oplus}{O}$
(a) (b) (c)

a > b > c

13. 
(a) (b)

b > a

14. 
(a) (b)

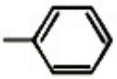
b > a

15. $Me - CH = C = \overset{\ominus}{C}H \longleftrightarrow Me - \overset{\ominus}{C}H - C \equiv CH$
(a) (b)

a > b

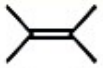
Hyperconjugation

α-Hydrogen

In general Carbon directly attached to ⊕/⊙/═/≡/  systems are known as α carbon & hydrogen attached to α-carbon is called α-hydrogen.

Concept Building Exercise

Q45. 1.  6 α H

2.  12 α H

3.  3 α H

4.  0 α H

5.  9 α H

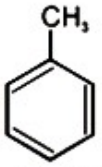
6.  6 α H

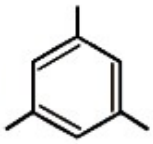
7.  0 α H

8.  6 α H

9.  6 α H

10. $CH_3 - C \equiv C - CH_3$ 6 α H

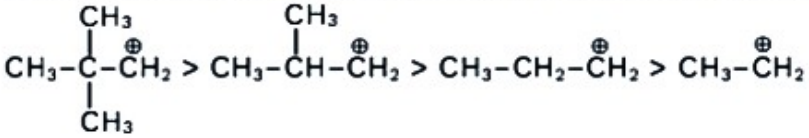
11.  3 α H

12.  9 α H

13.  4 α H

- **Definition** :
  Hyperconjugation or H-effect is delocalization of conjugated (C–H) sigma electrons with π-bond or positive charge or free radical.
  i) Hyperconjugating structures may be written involving "no bond" between the alpha carbon and hydrogen atoms so known as "no bond resonance."
  ii) Also called as "Nathan baker effect"."
  iii) It is also ever lasting effect.

- **Introduction :**
  Inductive effect direction of stability of cations expected to be

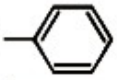


But actual order should be reversed which is explained on the basis of hyper conjugation.
IV > III > II > I

Terms for hyperconjugation:

i) Compound must have at least one sp^2-hybrid carbon of either alkene, alkyl carbocation or alkyl free radical compounds.

ii) α-carbon with respect to sp^2 hybrid carbon must have at minimum one hydrogen. If both these situations are fulfilled then hyperconjugation will take place in the molecule.

iii) This effect is observed for ⊕/⊙/═/≡/  system

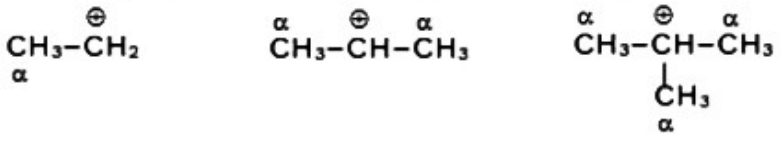
iv) Carbon-hydrogen σ bond conjugation is known as hyper-conjugation. In this effect complete transfer of σ-electron.

v) This effect also not depends on distance like resonance.

vi) For hyperconjugation at least one α-H is needed.

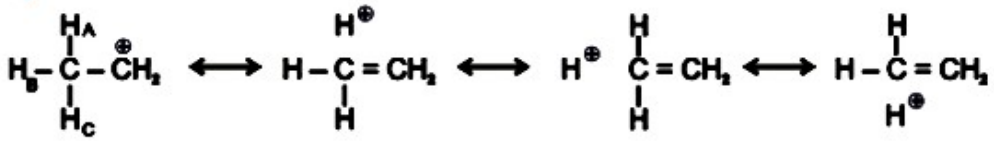
TYPES OF HYPER CONJUGATION :

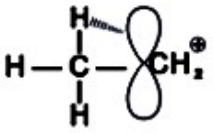
Case-I σ (C–H), vacant Orbital (positive charge) Hyperconjugation : This type of conjugation occurs in alkyl carbocation.



No. of α H = 3 No. of α H = 6 No. of α H = 9

Example :



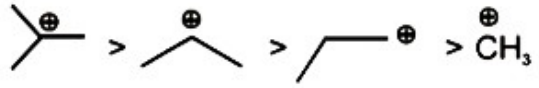


Orbital digram

Note : If number of α–H = n

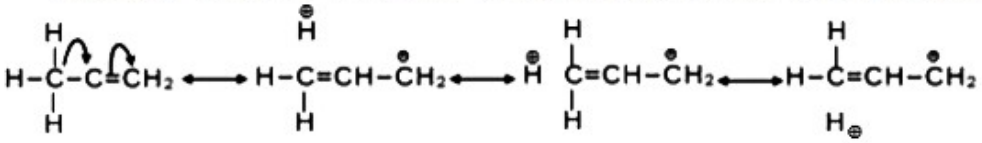
⇒ Number of hyperconjugating structure = n
⇒ Stability of carbocation ∝ Number of α-hydrogen

Example :



9αH 6αH 2αH 0αH

Hyperconjugating structures due to hyperconjugation may be written involving "no bond" between the alpha carbon and hydrogen atoms.

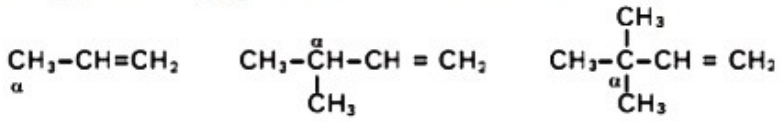


In the above hyperconjugating structures there is no covalent bond between carbon and hydrogen.

From this point of view, hyperconjugation may be regarded as "no bond resonance".

Case-II : σ (C–H), π hyperconjugation :

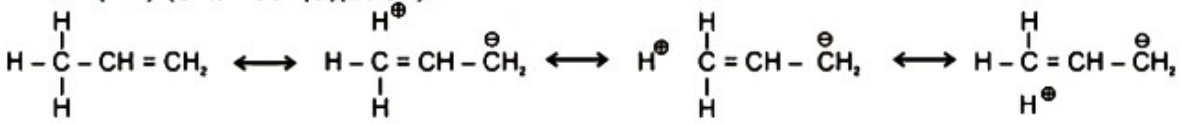
This type of conjugation occurs in alkenes.



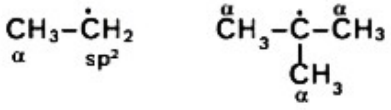
no. of α H = 3 no. of α H = 1 no. of α H = 0

Example : Hyperconjugation in double bond :

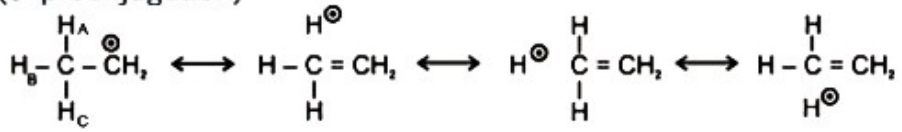
(+H) (σ-π* conjugation):

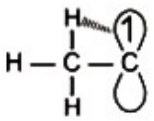


Case-III : σ (C –H), odd electron hyperconjugation : This type of conjugation occurs in alkyl free radicals –



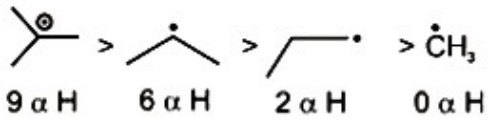
(σ-p conjugation)





If Number of α-H = n
Number of Hyper Conjugating Structure = n
Stability of carbon free radical ∝ Number of α-hydrogen

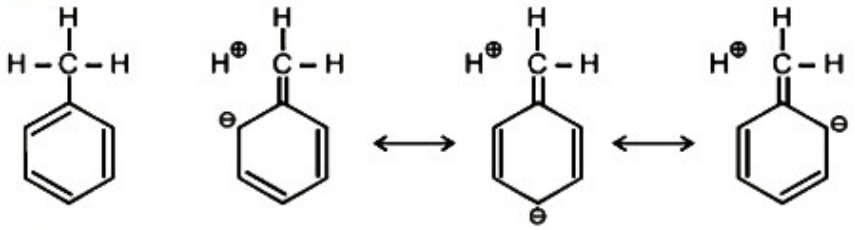
Example :



Note:

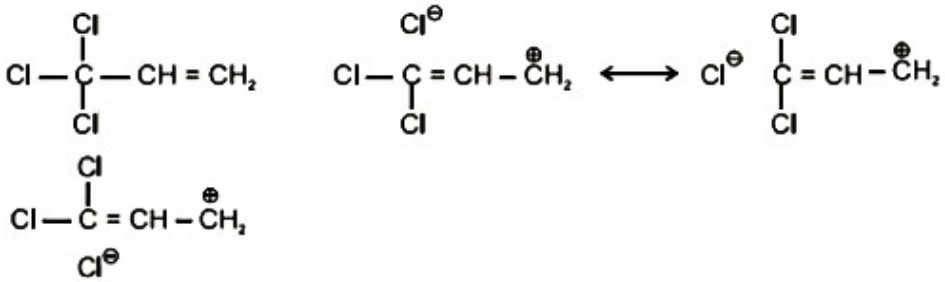
Carbanion never shows hyperconjugation, if there is one –C–H sigma bond alternate with negative charge then there will be no H-effect

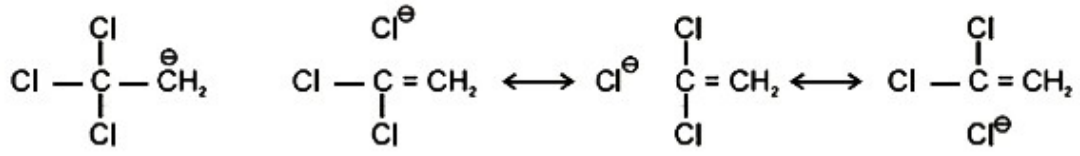
Case-IV : Hyperconjugation in phenyl : (+H) (σ-π* conjugation) :

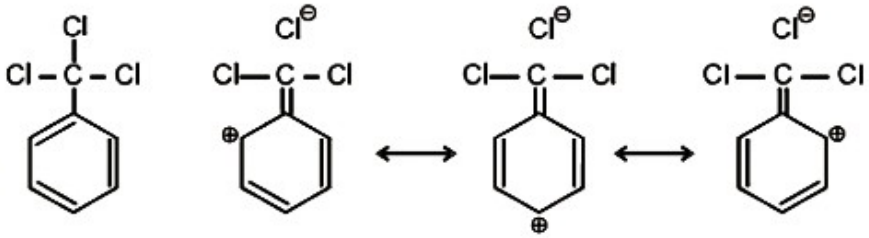


number of α-H = n
number of hyper conjugating structure = 3n

Case-V : Reverse Hyper Conjugation : (–H) (C – $\ddot{X}$:)

i) 

ii) 

iii) 

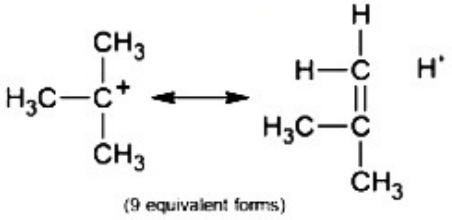
Order of effectiveness of hyperconjugation : α C–H bond > α C–D bond > α C–T bond $-CH_3 > -CD_3 > -CT_3$

Effects of Hyperconjugation

1. **Stability of Alkyl Carbocations :** Stability of alkyl carbocations $\propto$ Number of hyperconjugating structures $\propto$ number of α-hydrogens.

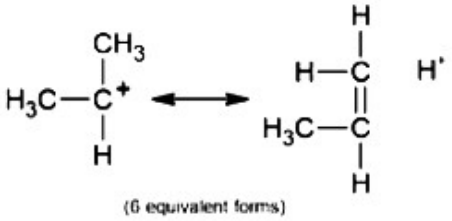
$$\boxed{(CH_3)_3C^+ > (CH_3)_2CH^+ > CH_3CH_2^+}$$

Above direction of stability can be explained by hyper conjugation.
In general higher the number of hydrogen atoms attached to carbon atoms, the more hyperconjugative forms can be drawn and thus greater will be the stability of carbonium ions.



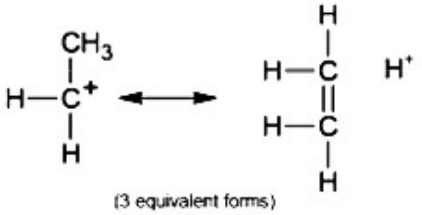
(9 equivalent forms)

3° carbonium ion



(6 equivalent forms)

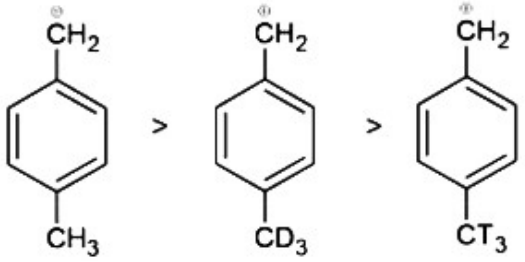
2° carbonium ion



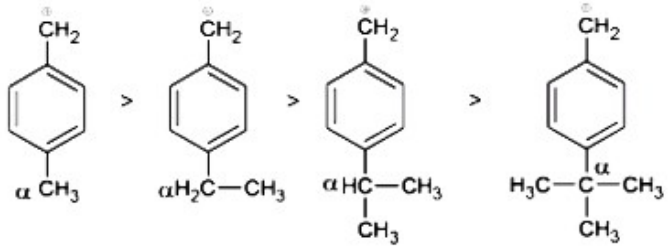
(3 equivalent forms)

1° carbonium ion

Example :

i) 

ii)



(3αH) (2αH) (1αH) (0αH)

(H effect dominates over I effect)

2. Stability of Alkenes : Stability can be explained by hyperconjugation.

Stability of alkenes ∝ Number of alpha hydrogens ∝ Number of hyperconjugating structures

Example :

i) $\overset{\alpha}{C}H_3 - CH = CH_2$ $\quad CH_3 - \overset{\alpha}{C}H_2 - CH = CH_2$ $\quad CH_3 - \overset{\alpha}{C}H(CH_3) - CH = CH_2$

(I) (II) (III)

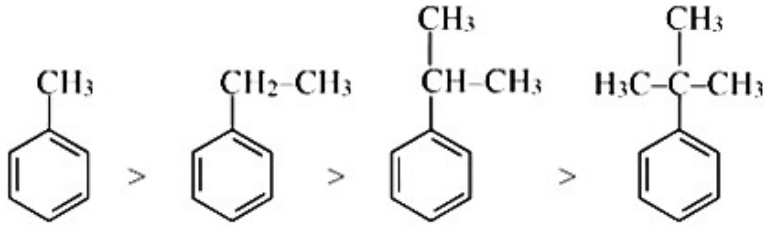
Stability order I > II > III

ii) $(CH_3)_2C = C(CH_3)_2$ $\quad (CH_3)_2C = CH - CH_3$ $\quad (CH_3)_2C = CH_2$

(I) (II) (III)

Stability order I > II > III

iii)



Stability order I > II > III > IV

Concept Building Exercise

Compare the stability of following alkenes :

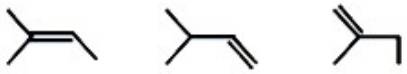
Q46. 

(a) (b) (c) (d) (e)

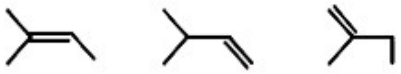
A46. 

12 α H (a) 9 α H (b) 6 α H (c) 3 α H (d) 0 α H (e)

a > b > c > d > e

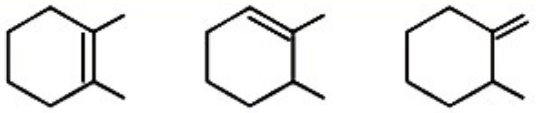
Q47. 

(a) (b) (c)

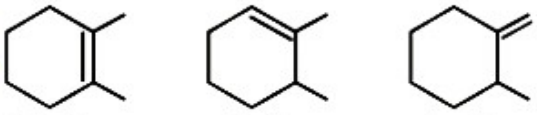
A47. 

10 α H (a) 6 α H (b) 5 α H (c)

a > c > b

Q48. 

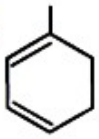
(a) (b) (c)

A48. 

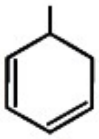
10 α H (a) 6 α H (b) 3 α H (c)

a > b > c

Q49.

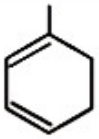


(a)



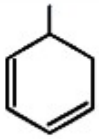
(b)

A49.



(a)

Resonance + 7α

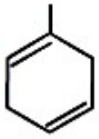


(b)

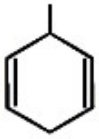
Resonance + 3α

a > b

Q50.

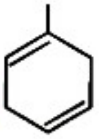


(a)



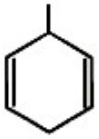
(b)

A50.



(a)

7α-H

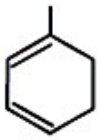


(b)

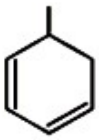
3α-H

a > b

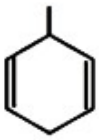
Q51.



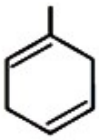
(a)



(b)

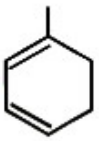


(c)



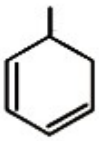
(d)

A51.



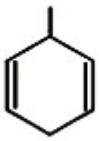
R+ 7α

(a)



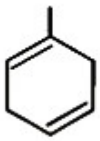
R+ 3α

(b)



3α

(c)



7α

(d)

a > b > d > c

Q52. 1-hexene (a) 2-hexene (b) 3-hexene (c)

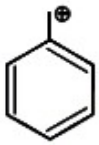
A52. 1-hexene  2α-H (a)

2-hexene  5α-H (b)

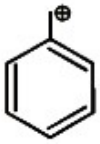
3-hexene  4α-H (c)

b > c > a

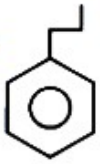
Q53. $CH_2 = CH - \overset{\oplus}{C}H_2$ (a)

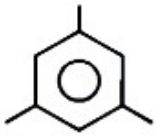
 (b)

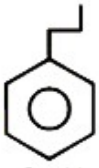
A53. $CH_2 = CH - \overset{\oplus}{C}H_2$ (a) 2R.S.

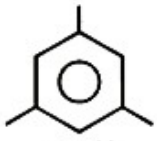
 (b) 5R.S.

b > a

Q54.  (a)

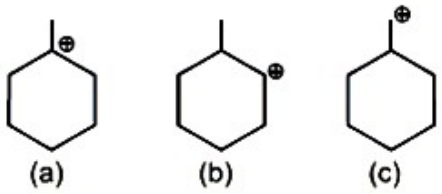
 (b)

A54.  2α-H (a)

 9α-H (b)

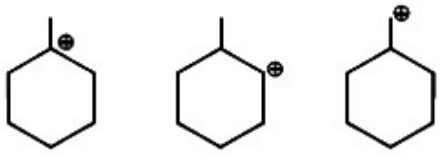
b > a

Q55.



(a) (b) (c)

A55.



7α-H 3α-H 1α-H

(a) (b) (c)

a > b > c

Q56. **(a) $CH_2 = CH - CH = CH - CH_3$**
(b) $CH_2 = CH - CH_2 - CH = CH_2$
(c) $CH_2 = C = CH - CH_2 - CH_3$

A56. (a) $CH_2 = CH - CH = CH - CH_3$ $(R + 3\alpha)$

(b) $CH_2 = CH - CH_2 - CH = CH_2$ 2α

(c) $CH_2 = C = CH - CH_2 - CH_3$ 2α

a > b > c

Stability of polyalkene :

Conjugated > Isolated > Cumulated

3. **Stability of free radicals :** Stability of free radicals can also be explained as that of carbonium ions.

Stability of Alkyl free radicals : $\propto$ +I, +R, - effect

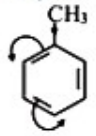
(a) $(CH_3)_3\dot{C} > (CH_3)_2\dot{C}H > CH_3\dot{C}H_2$

(b) $\underset{(i)}{\dot{C}H_3}$ $\underset{(ii)}{CH_3-\dot{C}H_2}$ $\underset{(iii)}{CH_3-\dot{C}H_2-CH_3}$ $\underset{(iv)}{CH_3-\overset{\dot{}}{\underset{\displaystyle CH_3}{\overset{|}{C}}}-CH_3}$

Stability is in growing order: IV > III > II > I

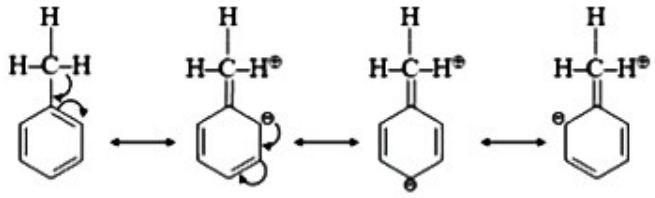
4. Orientation influence of methyl group :

The o, p directing influence of the methyl group in methyl benzenes is qualified partly to inductive and partly of hyperconjugation effect.



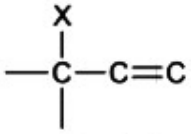
[Orientation encouragement of the methyl group due to +I effect]

CH_3 – (or alkyl group) is +H group, ortho-para directing group and activating group for electrophilic aromatic substitution reaction because of the hyperconjugation.



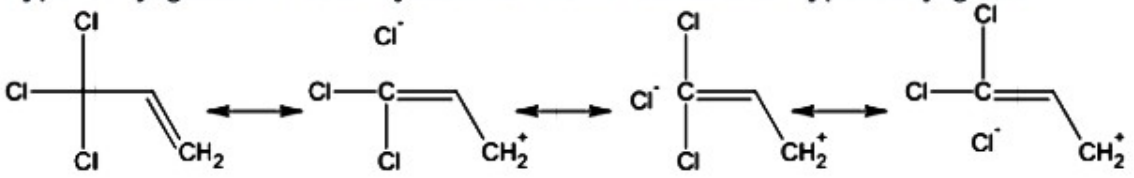
Reverse Hyperconjugation :

The phenomenon of hyperconjugation is also observed in the system given below:

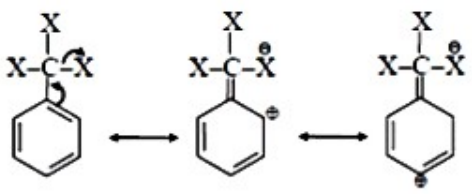


where X = halogen

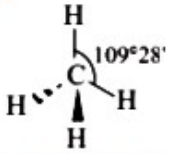
In such system the effect operates in the reverse direction. Hence the hyperconjugation in such system is known as reverse hyperconjugation.



- The meta directing influence and deactivating effect of CX_3 group for electrophilic aromatic substitution reaction can be explained by this effect.



BAYER'S STRAIN ANGLE THEORY :

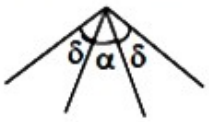


For any sp^3 atom angle must be 109° 28' or close to it for maximum stability.

For cycloalkanes



According to Bayer's cycloalkanes are strained.



$$\delta = \frac{1}{2}[109°28'-a]$$

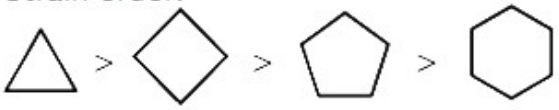
Hence for

3 carbon atoms →	$\alpha = 60°$,	$\delta = 24.5°$
4 carbon atoms →	$\alpha = 90°$,	$\delta = 9.5°$
5 carbon atoms →	$\alpha = 108°$,	$\delta = 0.5°$
6 carbon atoms →	$\alpha = 120°$,	$\delta = -5.5°$

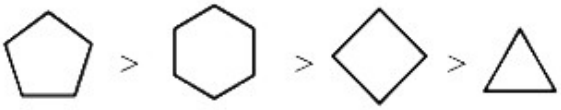
$$\text{Strain} \propto \frac{1}{\text{stability}}$$

According to Bayer's

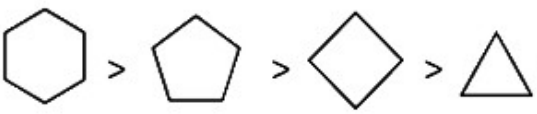
i) Strain order:



ii) Stability order:



But, actual stability order:


(Experimental evident)

Important exception in stability of alkene :

Concept Building Exercise

Q57.



(a) (b) (c)

A57. a > b > c (due to angle strain)

Q58.



(a) (b)

A58. b > a
(due to repulsion between 2 methyl group in (a))

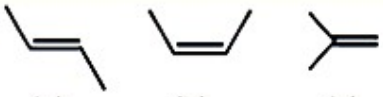
Q59.



(a) (b)

A59. b > a
By hyperconjugation (a) having 1° and 2°carbanion while (b) have only 1°carbanion therefore, (b) is more stable than (a).

Q60.



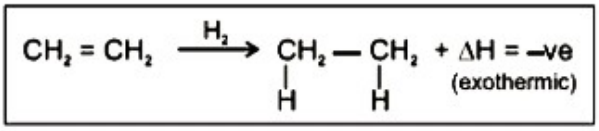
(a) (b) (c)

A60. c > a > b

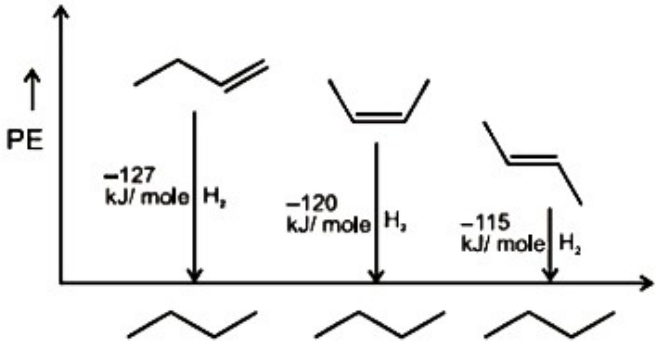
Application of Hyperconjugation

1. Heat of hydrogenation (HOH):

⇒ When 1 mole of alkene undergo hydrogenation then energy released in this process is heat of hydrogenation (HOH).



Case-I

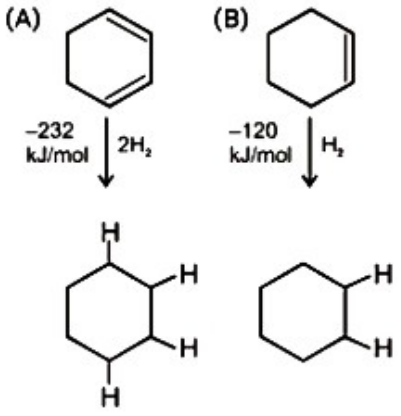


$$HOH \propto \frac{1}{\text{Stability of isomeric alkene}}$$

Case-II

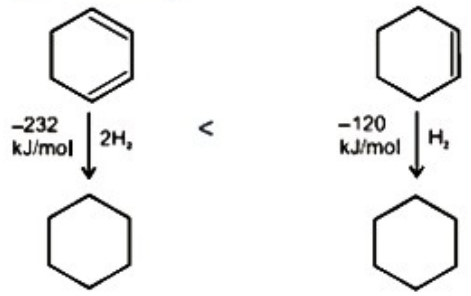
⇒ More is the number of π-bond, more is heat of hydrogenation

HOH α number of π – bond



Case-III

HOH per mole H_2 :



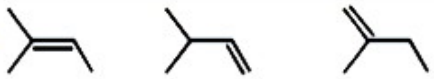
HOH/mole H_2

$\frac{-232}{2} = -116$ kJ/mol $\qquad\qquad \frac{-120}{1}$ kJ/mol

$$\text{HOH / mole } H_2 \propto \frac{1}{\text{Stability of alkene}}$$

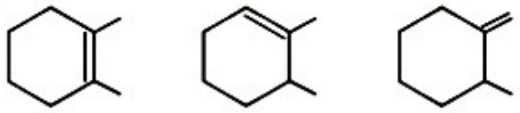
Compare HOH :

Concept Building Exercise

Q61. 

(a) (b) (c)

A61. Stability : a > c > b
HOH : b > c > a

Q62. 

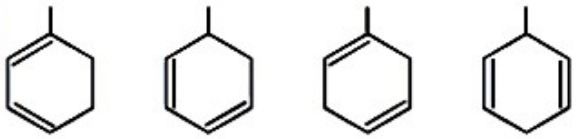
(a) (b) (c)

A62. Stability : a > b > c
HOH : c > b > a

Q63. $CH_3 - C \equiv C - CH_3$ (a) $\qquad$ $CH_3 - CH_2 - C \equiv CH$ (b)

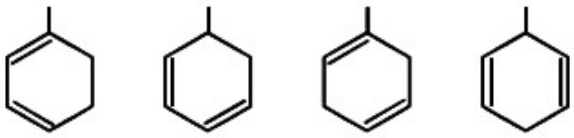
A63. Stability : a > b
HOH : b > a

Q64.



(a) (b) (c) (d)

A64.

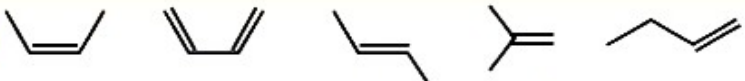


R + 7α R + 3α 7α 3α

(a) (b) (c) (d)

Stability : a > b > c >d
HOH : d > c > b > a

Q65.



(a) (b) (c) (d) (e)

A65. Stability : d > c > a > e
HOH : b >e > a > c > d

Q66. **(a) 1-hexene** **(b) 2-hexene** **(c) 3-hexene**

A66. Stability : b > c > a
HOH : a > c > b

Q67.



(a) (b)

A67.



R + 4α 4α

(a) (b)

Stability : a > b
HOH : b > a

Q68. 

(a) (b)

A68. a > b stability HOH → b > a

Q69. **(a) $CH_2 = CH - CH = CH - CH = CH_2$**
(b) $CH_3 - CH = CH - CH = CH - CH_3$
(c) $CH_2 = CH - CH = CH - CH_2 - CH_3$
(d) $CH_2 = CH - CH_2 - CH_2 - CH = CH_2$
(e) $CH_3 - CH = CH - CH_2 - CH = CH_2$
(f) $CH_2 = C = CH - CH_2 - CH_2 - CH_3$

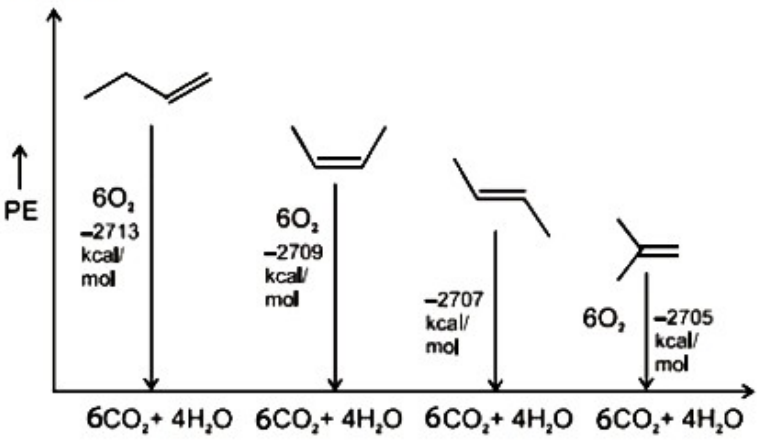
A69. (a) $CH_2 = CH - CH = CH - CH = CH_2$
(b) $CH_3 - CH = CH - CH = CH - CH_3$ R + 6α
(c) $CH_2 = CH - CH = CH - CH_2 - CH_3$ R + 2α
(d) $CH_2 = CH - CH_2 - CH_2 - CH = CH_2$ 4α
(e) $CH_3 - CH = CH - CH_2 - CH = CH_2$ 5α
(f) $CH_2 = C = CH - CH_2 - CH_2 - CH_3$ 2α
Stability : a > b > c > e > d > f
HOH : a > f > d > e > c > b

2. Heat of combustion (HOC):

Energy released during combustion is known as heat of combustion (HOC)..

$$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \xrightarrow{\Delta} xCO_2 + \frac{y}{2}H_2O$$

Case-I



$$\text{HOC} \propto \frac{1}{\text{Stability of compound}}$$

Case-II : More is the number of carbon more will be the HOC.
HOC α number of carbon



600 kJ/mol

>



350 kJ/mol

Case-III : HOC per mole CH_2



600 kJ/mol



350 kJ/mol

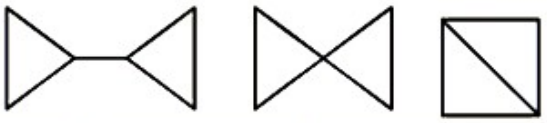
$\frac{600}{6}$ = 100 kJ/mol $<$ $\frac{350}{3}$ kJ/mol

$$\text{HOC / } CH_2 \propto \frac{1}{\text{Stability of compound}}$$

Compare HOC :

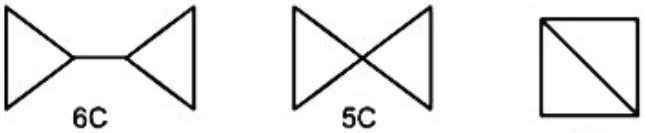
Concept Building Exercise

Q70.



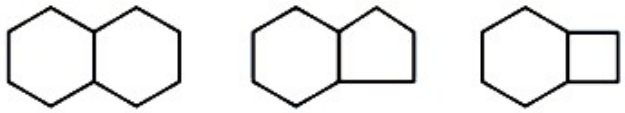
(a) (b) (c)

A70.

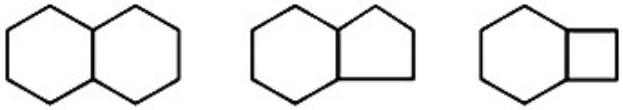


(a) (b) (c)

HOC : a > b > c

Q71. 

(a) (b) (c)

A71. 

10C 9C 8C

(a) (b) (c)

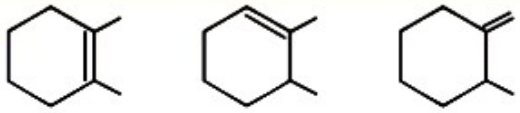
HOC : a > b > c

Q72. 

(a) (b) (c) (d) (e)

A72. Stability : e > a > d > b > c

HOC : e > c > b >d > a

Q73. 

(a) (b) (c)

A73. Stability : a > b > c

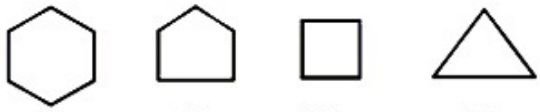
HOC : c > b > a

Q74. (a) $CH_3 - C \equiv C - CH_3$ (b) $CH_3 - CH_2 - C \equiv CH$

(a) (b)

A74. Stability : a > b

HOC : b > a

Q75. 

(a) (b) (c) (d)

A75. Stability : a > b > c > d

HOH : d > c > b > a

HOC : a > b > c > d

Q76.



(a) (b) (c) (d)

A76. Stability : a > b > c > d
HOC : d > c > b > a

3. Bond length in alkenes:

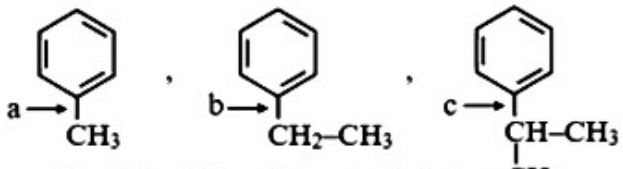
The bond length between carbon-carbon double bond ∝ number of hyperconjugating structures.

Example :

(i) $\overset{a}{\underset{\downarrow}{CH_3}}-CH=CH-CH_3, CH_3-\overset{b}{\underset{\downarrow}{CH}}=CH_2, \overset{c}{\underset{\downarrow}{CH_2}}=CH_2$

Bond length order ⇒ a > b > c

(ii)



Bond length order ⇒ a > b > c

4. Dipole Moment

Definition: The degree of polarity of a bond due to difference in electronegativity between two adjacent atoms in the molecule is called dipole moment.

Dipole moment is represented by μ.

μ = q × d where q = magnitude of separated charge d = The internuclear distance between two atoms

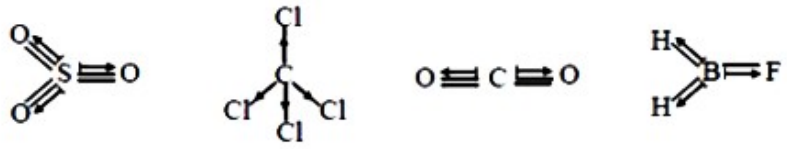
Dipole moment is represented by arrow head pointing towards more electronegative atom.

$$H \rightarrow Cl$$

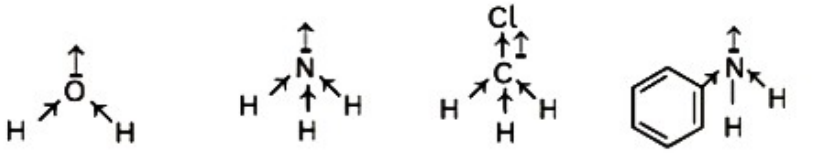
Dipole moment depends on shape of the molecule

Dipole moment of symmetrical compound is always zero.

Example :

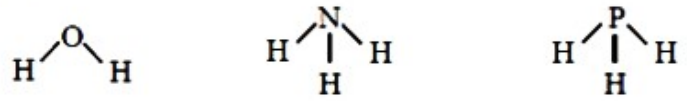


Dipole moment of unsymmetrical compound is always greater than zero



$\mu \neq 0$

More is the electronegativity of central atom of the molecule, more is the dipole moment

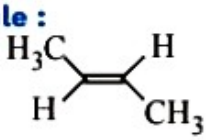


electronegativity of central atom is increasing hence μ is also increasing

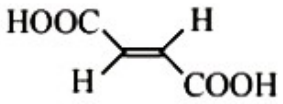
Dipole moment of cis isomers is more than that of trans isomer.

If group has linear moments, then the dipole moment of the trans isomer will be zero.

Example :



$\mu = 0$

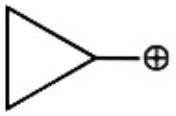


$\mu = 0$

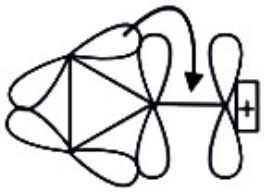
Hydrogen Bonding, Melting Point and Boiling Point

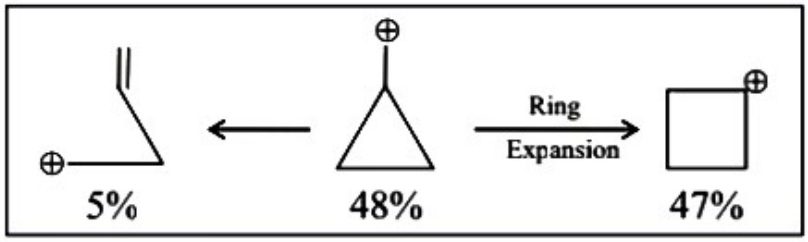
Sigma Bond Resonance (Dancing Resonance)

Sigma bond resonance is observed in cyclopropyl methyl carbocation.

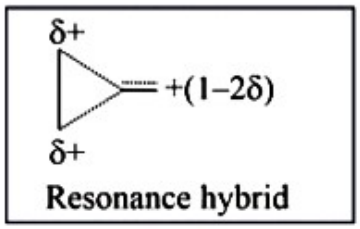


The Sigma bond of ring of cyclopropane (three carbon atoms) are very weak due to excess of angle strain. * σ bond in Resonance



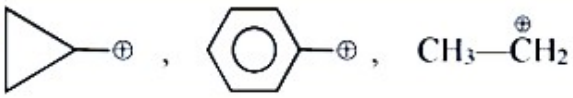


* The overlapping area between orbitals is so small that the C–C bond is extremely weak.
* The C–C bond is weaker then π-bond is ethene
* Due to this close overlapping the electrons delocalized to the p-orbital. The effect is observed to be very strong and is even stronger than hyper conjugation due to weakness of delocalized bond.

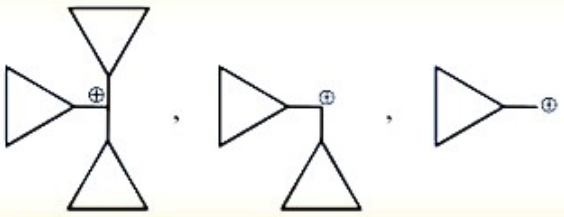


Compare stability of following carbocations :

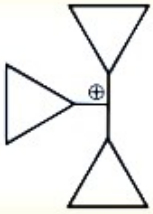
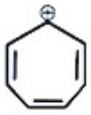
Concept Building Exercise

Q77. 

A77. 1 > 2 > 3

Q78.  , ,

A78. 1 > 2 > 3

Q79.  ,  , $Ph_3C^{\oplus}$, $Ph_2\overset{\oplus}{C}H$, $Ph-\overset{\oplus}{C}H-R$

A79. 2 > 1 > 3 > 4 > 5

Q80. $(CH_3)_3C^{\oplus}$, $Ph-CH_2^{\oplus}$, $CH_2=CH-CH_2^{\oplus}$, $CH_3-CH_2^{\oplus}$, $CH_3^{\oplus}$, $CH_2=CH^{\oplus}$, $CH\equiv C^{\oplus}$

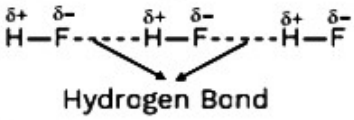
A80. 1 > 2 > 3 > 4 > 5 > 6 > 7

Inter Molecular Attraction Forces :

Definition : The hydrogen bond is an electrostatic force of attraction among covalently bonded hydrogen atom of one molecule and an electronegative atom (F, O, N) of another molecule.

Example :

In hydrogen flouride, hydrogen is attached to strongly electronegative atom flourine. Hydrogen acquires the positive charge which will be attracted electrostatically by the negative charge on flourine in another molecule of HF.



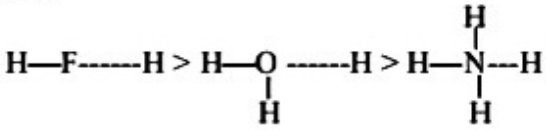
Condition for hydrogen bonding

a) The molecule must contain a highly electronegative atom linked to hydrogen atom.

b) The size of electronegative atom should be small. F, O and N atoms only form effective hydrogen bonding.

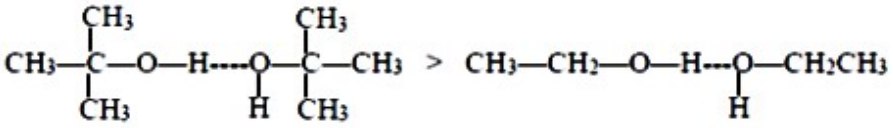
Factors affecting strength of hydrogen bonding

a) Greater electronegativity of atom attached to hydrogen, the stronger is the hydrogen bond



(strength of hydrogen bonding)

b) More is the crowding, strength of hydrogen bond is less

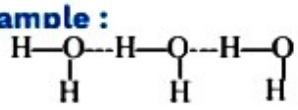


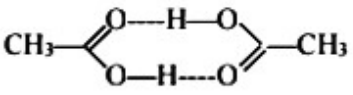
Types of hydrogen bonding

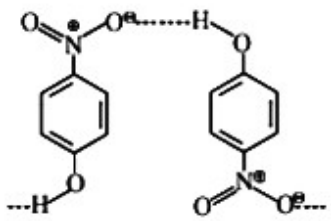
a) Intermolecular hydrogen bonding :

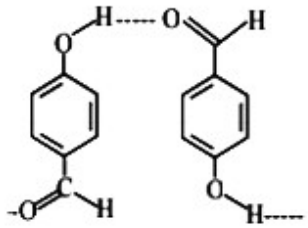
The hydrogen bonding between two molecules of same compound

Example :

i) 

ii) 

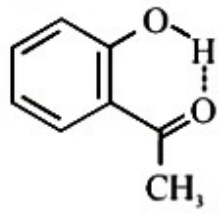
iii) 

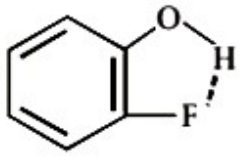
iv) 

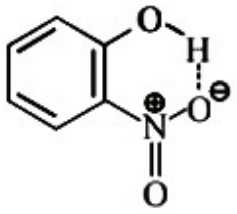
b) Intramolecular hydrogen bonding :

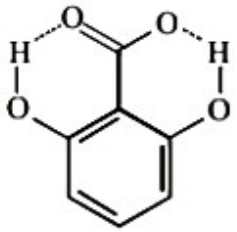
The hydrogen bonding between two atoms of same molecule. This type of electrostatic force of attraction is only possible if the ring size formed on attraction is 5 or 6.

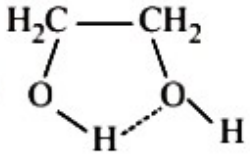
Example :

i) 

ii) 

iii) 

iv) 

v) 

Physical properties

a) Boiling point : The boiling point of a liquid is the temperature where its kinetic energy is sufficient to overcome the intermolecular attractive forces.

Boiling point $\propto$ Molecular weight

$\propto$ Intermolecular forces of attraction

In case of isomeric compounds, the boiling point depends on the surface area of the compound. More is the surface area of the compound; more will be boiling point of isomeric compounds.

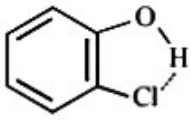
Example :

$CH_3-CH_2-CH_2-CH_2-CH_3$ B.pt. = 36.1°C

$CH_3-CH(CH_3)-CH_2-CH_3$ B.pt. = 27.9°C

$C(CH_3)_4$ B.pt. = 9.5°C

Boiling point decreases due to intramolecular hydrogen bonding



B. pt = 178°C ;



B.pt = 217°C

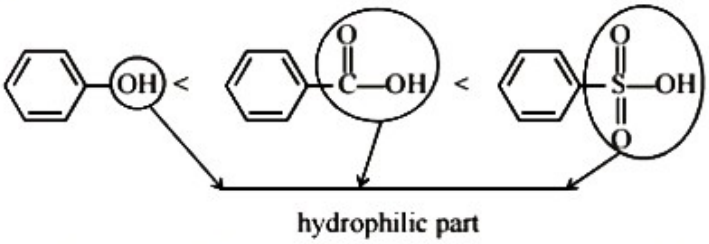
b) Solubilities :

Solubility $\propto$ polarity

Polarity = surface area of hydrophobic part

More is the surface area of hydrophillic part, more is the solubility

Example :



As the surface area of hydrophillic part increases, solubility increases.

c) Melting point

M.P $\propto$ molecular weight

$\propto$ Intermolecular forces of attraction

$\propto$ compactness

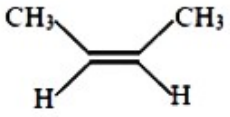
More is compactness, more energy is required to separate the molecules hence melting point is more.

⇒ Compactness of symmetrical compounds is more.

⇒ Trans isomer is more compact than cis

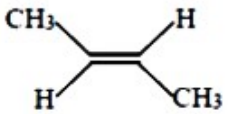
⇒ branched hydrocarbons are more compact than linear hydrocarbons.

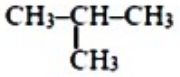
Example :



<

M.P.





<

M.P.

$CH_3–CH_2–CH_2–CH_3$



<

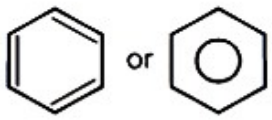
M.P.



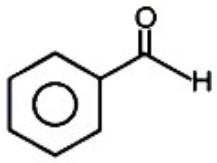
Aromatic and Aromaticity

Examples for aromatic compounds:

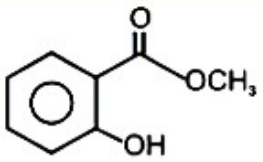
Q81.



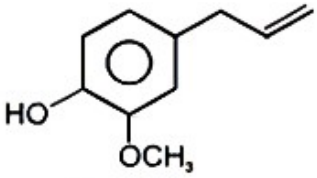
Benzene



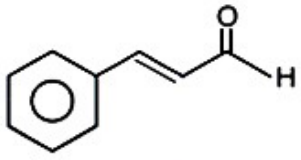
Benzaldehyde
(in oil of almonds)



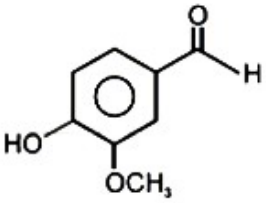
Methyl salicylate
(in oil of wintergreen)



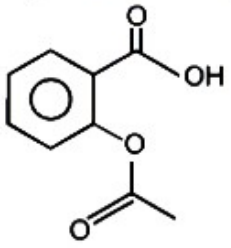
Eugenol
(in oil of cloves)



Cinnamaldehyde
(in oil of cinnamon)



Vanillin
(in oil of vanilla)



Acetylsalicylic acid
(Aspirin)

Task – pick any of two compounds and discuss with your classmates about uses

Definition :

Compound which are obtained from fractional distillation of coal tar and having characteristic aroma are known as aromatic compound.

Rules for compound to be aromatic :

1. Compound must be cyclic
2. Compound must be planar or nearly planar.
3. Compound must be conjugated throughout ring
4. Compound must have $(4n + 2)$ π conjugated or delocalised electrons where n is a whole number

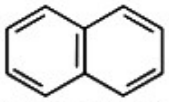
$n = 0, 1, 2, 3, 4, 5, 6, ...$

$n =$	0,	1,	2,	3,	4,	5.....
$(4n + 2)\pi$electron =	2	6	10	14	18	22....

Examples :



a) Aromatic compound



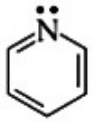
b) Aromatic compound



c) Aromatic compound



d) Aromatic compound



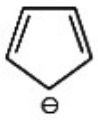
e) Aromatic compound



f) Aromatic compound



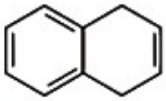
g) Aromatic compound



h) Aromatic compound



i) Aromatic compound



j) Aromatic compound

2. **Anti aromatic Compounds :** According to Huckel's rule, compound will be anti aromatic if it fulfills the following four conditions :

   a) Compound must be cyclic
   b) Compound must be planar.
   c) Compound must be conjugated through out ring
   d) Compound must have $(4n)\pi$ conjugated or delocalized electrons where n is a whole number

   n = 1, 2, 3, 4, 5, 6

n =	1,	2,	3,	4,	5.....
(4n) πelectron	= 4	8	12	16	20.....

Example :



a) Anti-Aromatic compound



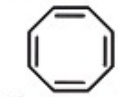
b) Anti-Aromati compound



c) Anti-Aromatic compound

3. **Non Aromatic Compounds:** The compounds having sp^3 hybridized atom is known as none aromatic compound.

Example:

a)  Non-Aromatic compound →  Tub-shape

Non-planar so non aromatic compound

b)  Non-Aromatic compound

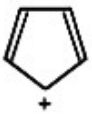
c)  Non-Aromatic compound

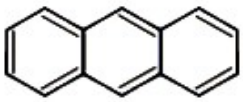
d)  Non-Aromatic compound

Although cyclooctatetraene has $(4n)\pi$ electrons but even then it is not an anti aromatic. Geometry of this compound is non planar. Thus it is non aromatic. IInd, IIIrd & IVth compounds are not completely conjugated.

Note:
Due to Aromaticity extra stability is achieved by aromatic molecule, than non-aromatic or anti aromatic molecule.

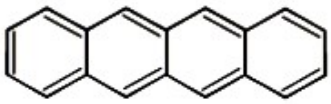
Q82. Which of the following compounds are aromatic?

a) 

b) 

c) cycloheptatrienyl cation

d) 

e) 

f) 

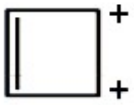
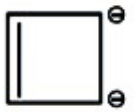
g) cyclononatetraenyl anion

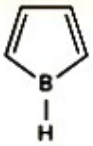
h) $CH_2 = CHCH = CHCH = CH_2$

Q83. Comment about following molecules

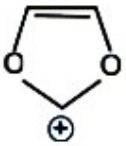
1. 

i) Aromatic
ii) Anti-aromatic
iii) Non-aromatic ✓

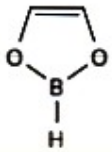
2.		i) ii) iii)	Aromatic Anti-aromatic Non-aromatic	✓
3.		i) ii) iii)	Aromatic Anti-aromatic Non-aromatic	✓
4.		i) ii) iii)	Aromatic Anti-aromatic Non-aromatic	✓
5.		i) ii) iii)	Aromatic Anti-aromatic Non-aromatic	✓
6.		i) ii) iii)	Aromatic Anti-aromatic Non-aromatic	✓
7.		i) ii) iii)	Aromatic Anti-aromatic Non-aromatic	✓
8.		i) ii) iii)	Aromatic Anti-aromatic Non-aromatic	✓
9.		i) ii) iii)	Aromatic Anti-aromatic Non-aromatic	✓
10.		i) ii) iii)	Aromatic Anti-aromatic Non-aromatic	✓

11. 

i) **Aromatic**
ii) **Anti-aromatic** ✓
iii) **Non-aromatic**

12. 

i) **Aromatic** ✓
ii) **Anti-aromatic**
iii) **Non-aromatic**

13. 

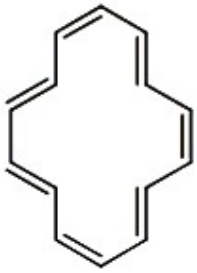
i) **Aromatic** ✓
ii) **Anti-aromatic**
iii) **Non-aromatic**

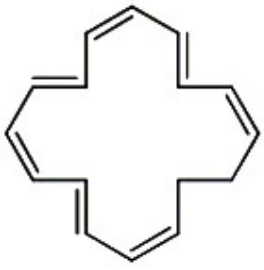
- **The Annulenes :** The word annulene is incorporated into the class name for monocyclic compounds that can be represented by structures having alternating single and double bonds. The ring size of an annulene is indicated by a number in brackets. Thus, benzene is [6] annulene and cyclooctatetraene is [8] annulene.

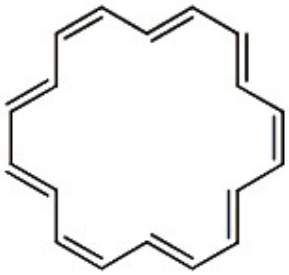
Q84.

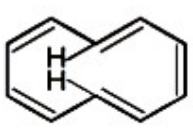

Benzene
([6] annulene)


Cyclooctatetraene
([8] annulene)

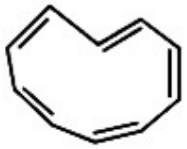

[14] Annulene
(aromatic)


[16] Annulene
(antiaromatic)


[18] Annulene
(aromatic)


I


II


III

[10] Annulenes
None is aromatic because none is planar

Definition : Those monocyclic compounds which are having alternate π-bond arrangement on its circumference are known as annulene.

Note :

Number of π-electrons should be written in [ ]

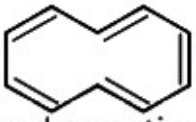
Name → [ ] annulene

Case-1 : Generally (4n+2) π e^- are considered as aromatic.

[6] annulene will be aromatic

[10] annulene will be non-aromatic

[14] annulene will be aromatic

[18] annulene will be aromatic  [10] annulene

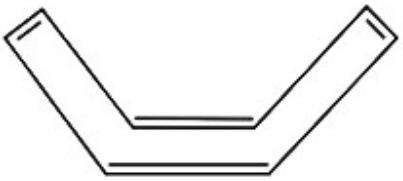
Case-2 : Generally 4nπ e^- are considered as anti-aromatic.

[4] annulene will be anti-aromatic.

[8] annulene will be non-aromatic

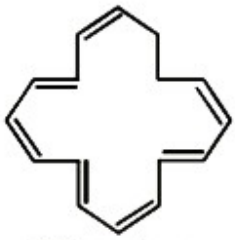
[12] annulene will be anti-aromatic

[16] annulene will be anti-aromatic

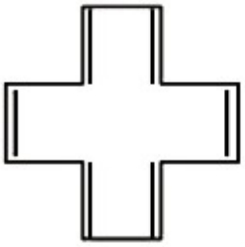


[8] annulene (Tub shape)

Examples :

1. 

[16] annulene
Anti-aromatic

2. 

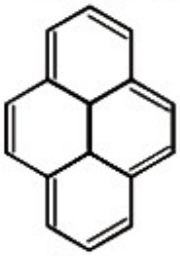
[12] annulene
Anti-aromatic

3. 

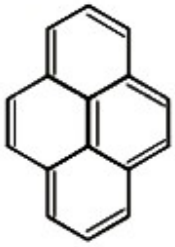
[14] annulene
Aromatic

4. 

[12] annulene
Non-aromatic

5. 

14π e^-
Aromatic

6. 

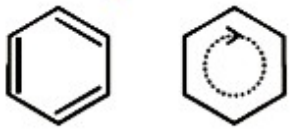
14π e^-
Aromatic

7. 

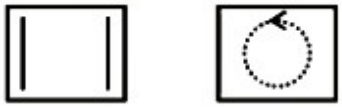
[16] annulene
Anti-aromatic
(due to repulsion
between Me)

Application of Resonance :

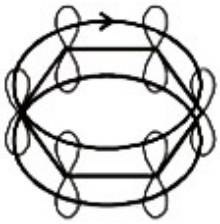
Aromaticity :

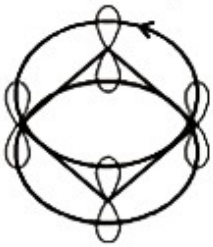


Aromatic
(clockwise ring
current)



Anti-aromatic
(anticlockwise ring
current)





Stability :

Aromatic compound > Non-aromatic > Anti-aromatic compound

To identify that the compound aromatic, non-aromatic or anti aromatic following rules are followed:

Rule-1 Compound should be cyclic.

Rule-2 Compound should be planar. Planarity is check by hybridisation (sp^2, sp)

Rule-3 Compound should have complete cyclic conjugation of π-electrons.

Rule-4 Huckel's rule :

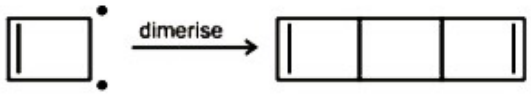
According to Huckel if cyclic conjugation of

$(4n + 2)\pi\ e^-$	$(4n)\pi\ e^-$
n = 0, 1, 2, 3,.....	n = 1, 2, 3, ...
$2\pi\ e^-$, $6\pi\ e^-$, $10\ \pi\ e^-$	$4\pi\ e^-$, $8\pi\ e^-$, $12\pi\ e^-$ ....
.....	.....

Aromatic compound	Anti-aromatic compound
	
Cyclic ✓	Cyclic ✓
Planar ✓	Planar ✓
Complete cyclic conjugation ✓	Complete cyclic conjugation ✓
$6\pi\ e^-$	$4\pi\ e^-$
Aromatic	Anti-aromatic

Important points for anti-aromatic compounds :

Anti-aromatic compounds are unstable in nature. Due to unpaired electron they undergo dimerization so they become stable.

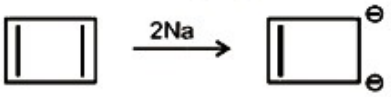


Anti-aromatic Non-aromatic



They are paramagnetic in nature.

Anti aromatic compounds readily reacts with active metal (Na, K) and behaves as oxidising agent.

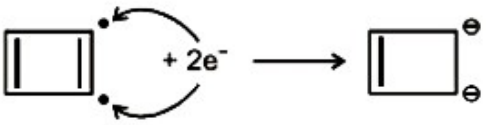


Anti-aromatic Aromatic

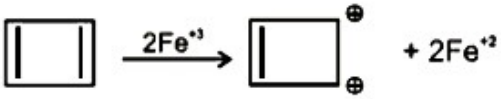
It is behaving like oxidising agent.

Mechanism:

$2Na \longrightarrow 2Na^{+} + 2e^-$

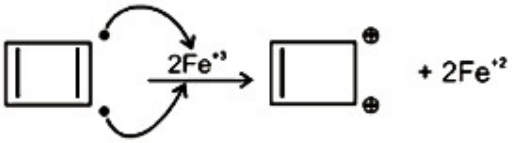


Anti-aromatic compound also behaves as reducing agent.
Reaction

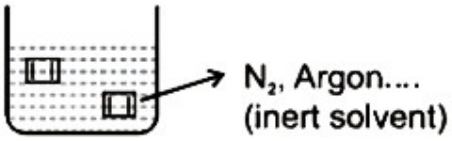


Anti-aromatic

Mechanism

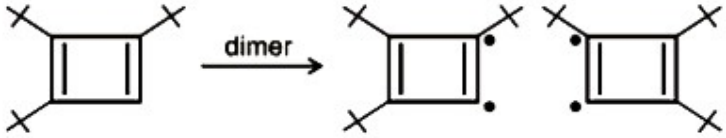


Anti-aromatic compounds can be isolated in inert solvent.



Anti-aromatic compound can be isolated at very low temperature (4K) so that kinetic energy minimises, but it is only possible at pluto.

Note :



no dimerisation

Because of repulsion between the two tertiary butyl groups.

Mesomeric Effect or Resonance Effect

Definition :

Permanent displacement of π electrons cloud of a conjugated system due to p-orbital of directly attached atoms or groups is known as mesomeric effect/resonance effect.

(a) This is a permanent effect where the displacement of conjugational π electrons are influenced by the group attached to carbon chain.

(b) There is a push or pull of π electrons of the conjugated system of molecule.

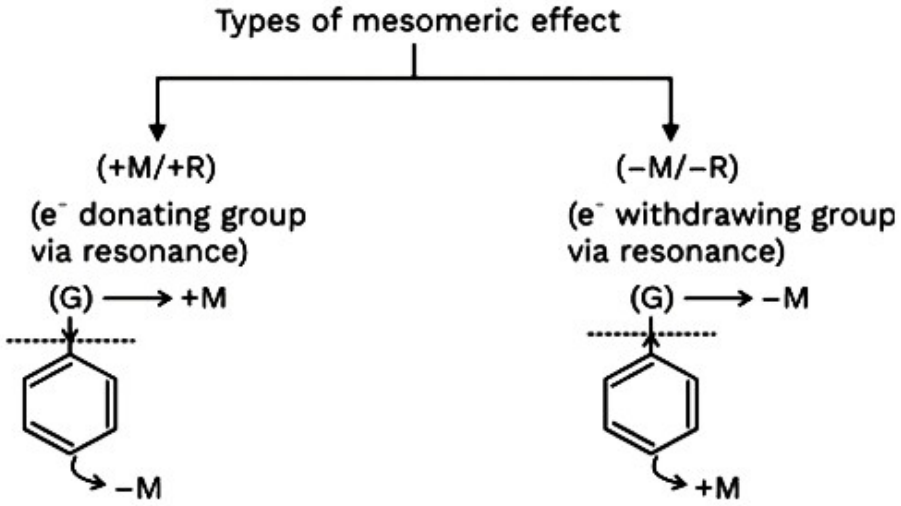
(c) Mesomeric effect operates only in the conjugated system.

(d) If the conjugation is not available, M effect does not propagate further.

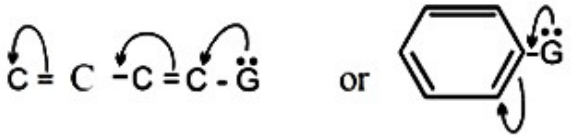
(e) M-effect has no correlation with the electro negativity of the connecting atom of the group

(f) M-effect is two type; +M and –M

There are two types of mesomeric effect



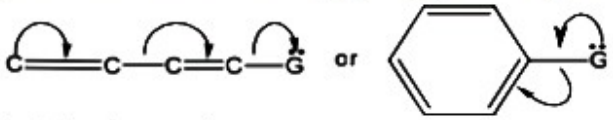
1. **+M effect (+R) :** If the group pushes the electrons into the π electron system, the effect is said as +M effect and the group is named Electron Releasing Group or +M group.

e.g.  or

+ M Effect is shown by : –Cl, –Br, $-NH_2$, –NHR, $-NR_2$, $-NHCOCH_3$, –OH , –OR

Condition : Electron pair must be present at first atom of group.

2. –M effect (–R) : If the π electrons are present system is pulled by the group then the effect is called –M effect and the group is called EWG group (electron with drawing group) or –M group.

e.g.  or

–M effect is shown by :

–CHO, C = O, –COOH, $-NO_2$, –CN, $-SO_3H$, $-SO_2Cl$

Condition : (i) Vacant orbital must be present

(ii) Multiple bond with more electron negative second atom.

Tips and Tricks

All electrons donating group (via conjugation / resonance) ⟶ +M

All (–ve) charge species ⟶ +M

All lone pair containing species ⟶ +M (via resonance)

All electrons withdrawing group ⟶ –M

All (+ve) charge species (vacant orbital) ⟶ –M

+ M effect order

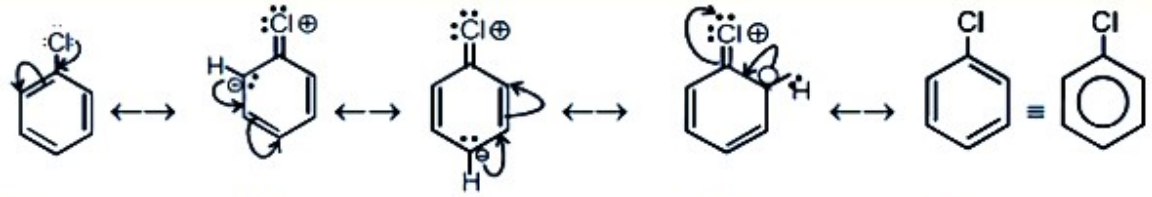
1. $-\overset{\ominus}{C}H_2 > -\overset{\ominus}{N}H > -O^{\ominus}$
2. $-\overset{\ominus}{N}H > -NH_2$
3. $-NR_2 > -NHR > -NH_2 > -NH-\overset{O}{\overset{\|}{C}}-R > -N\begin{matrix}\diagup \\ \diagdown\end{matrix}\begin{matrix}\overset{O}{\overset{\|}{C}}-R \\ \underset{O}{\underset{\|}{C}}-R\end{matrix}$
4. $-O^- > -OR$
5. $-NH_2 > -OH > -F$
6. $-F > -Cl > -Br > -I$

–M effect order

1. $-\overset{\ominus}{C}H_2 > -BH_2 > -PH_2$
2. $-N\begin{matrix}\diagup O \\ \diagdown O\end{matrix} > -CN > -\underset{O}{\underset{\|}{C}}-H > -\underset{O}{\underset{\|}{C}}-R > -\underset{O}{\underset{\|}{C}}-OR > -\underset{O}{\underset{\|}{C}}-NH_2$

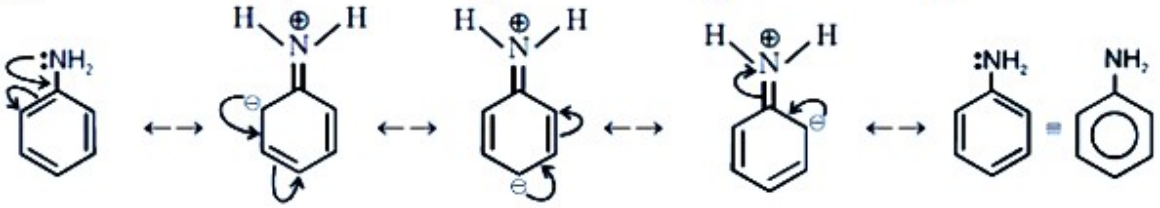
The various resonating structures of chlorobenzene, aniline and nitrobenzene are illustrated in following diagrams. Note that –Cl and $-NH_2$ show + M effect and $-NO_2$ show –M effect.

Q85. i)



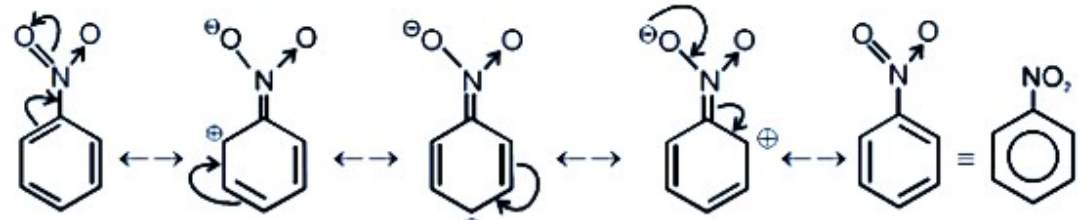
(a) (b) (c) (d) (e)

ii)



(a) (b) (c) (d) (e)

iii)



(a) (b) (c) (d) (e)

Note :

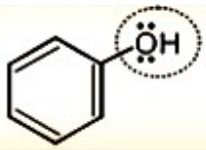
(i) Thus, mesomeric effect works at only ortho & para position, it is absent on meta position, while inductive effect works at all three-position o, m, p however intensity of effect decreases as the distance increases.

(ii) Mesomeric effect always dominates on inductive effect except halogen (Cl, Br, I, only).

Concept Building Exercise

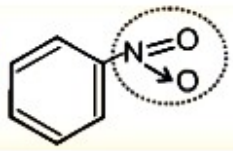
Identify circle group will be +M / −M / +I / −I

Q86.

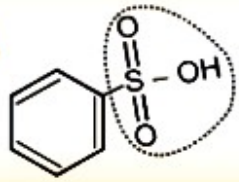


A86. −I, +M

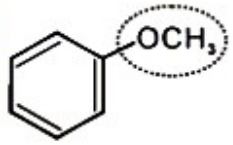
+M > −I

Q87. 

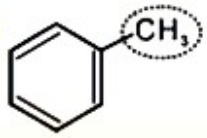
A87. −I, −M
−M > −I

Q88. 

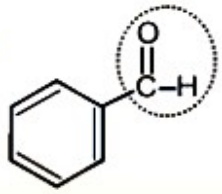
A88. −M > −I

Q89. 

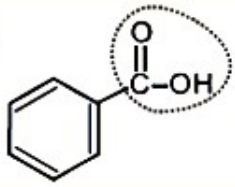
A89. −I, +M
+M > −I

Q90. 

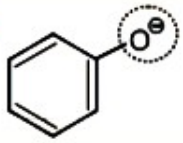
A90. +I
M = ×

Q91. 

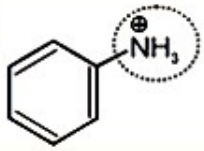
A91. −I, −M
−M > −I

Q92. 

A92. –M > –I

Q93. 

A93. +M > +I

Q94. 

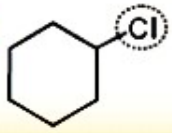
A94. –I

Q95. $CH_2 = CH - \ddot{O}CH_3$

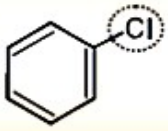
A95. +M > –I

Q96. $CH_2 = CH - CH = O$

A96. –M > –I

Q97. 

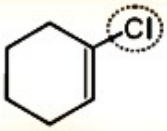
A97. – I

Q98. 

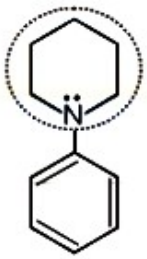
A98. –I >>> +M

Q99. CH_2= CH –Cl

A99. –I

Q100. 

A100. –I >> +M

Q101. 

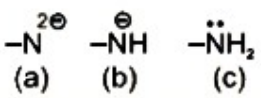
A101. +M > –I

Concept Building Exercise

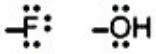
Identify +M order :

Q102. $\overset{\ominus}{C}H_2$ (a) $\overset{\ominus}{N}H_2$ (b) $\overset{\ominus}{O}H$ (c) $\overset{\ominus}{F}$ (d)

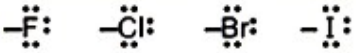
A102. a > b > c > d

Q103. 
(a) (b) (c)

A103. a > b > c

Q104. 
(a) (b)

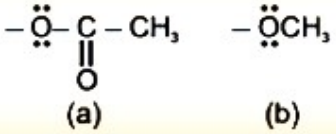
A104. b > a

Q105. 
(a) (b) (c) (d)

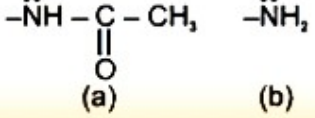
A105. a > b > c > d

Q106. 
(a) (b)

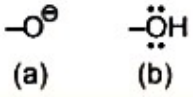
A106. b > a

Q107. 
(a) (b)

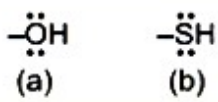
A107. b > a

Q108. 
(a) (b)

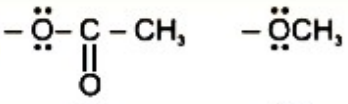
A108. b > a

Q109. 
(a) (b)

A109. a > b

Q110. 
(a) (b)

A110. a > b

Q111. 
(a) (b)

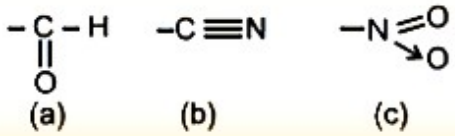
A111. b > a

Concept Building Exercise

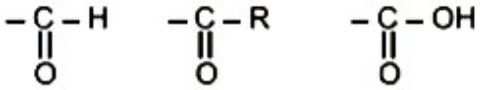
Identify –M order :

Q112. 
(a) (b) (c) (d)

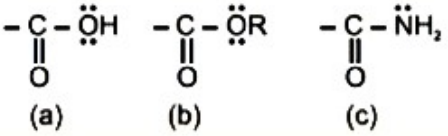
A112. d > b > c > a

Q113. 
(a) (b) (c)

A113. c > b > a

Q114. 
(a) (b) (c)

A114. a > b > c

Q115. 

A115. a > b > c

Comparison between electronic effects :

S.NO.	INDUCTIVE EFFECT	MESOMERIC EFFECT	HYPERCONJUGATION EFFECT
1	Saturated and unsaturated compounds	Unsaturated compounds especially having conjugated system.	Carbocation, carbon free radical and unsaturated compounds.
2	It involves partial shifting of sigma electrons	It involves complete shifting of π -electrons of π-bonds or lone pair of electrons	It contains partial shifting of sigma-electrons into adjacent π-orbital.
3	Pair is slightly displaced from its position and thus partial charges are developed.	Pair is completely transferred and thus full positive and negative charges are developed.	It is conveyed from one end to other end of the chain provided conjugated is present. It is distance independent.

Resonance Energy

The energy difference between most stable resonating structure and resonance hybrid is known as resonance energy.

It's the experimental value which is calculated by heat of hydrogenation (HOH)

Higher the value of resonance energy, greater is the resonance stabilization.

1. Which structure having more resonance energy A or B?



(A) (B)

Resonance energy and energy is different aspects in consideration.

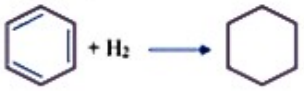
When we talk about energy of any compound that's generally potential energy of molecule.

Resonance energy only present when delocalization is possible in molecule.

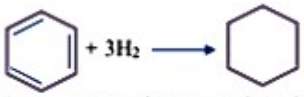
Resonance Energy of Benzene :



The resonance energy of benzene is calculated from the heat of hydrogenation as given below:



+ 29 Kcal./Mole



+ 3 × 29 Kcal. Mole = 87 Kcal./mole (Theoretical value)

but experimental value for benzene is 51 Kcal./mole so,

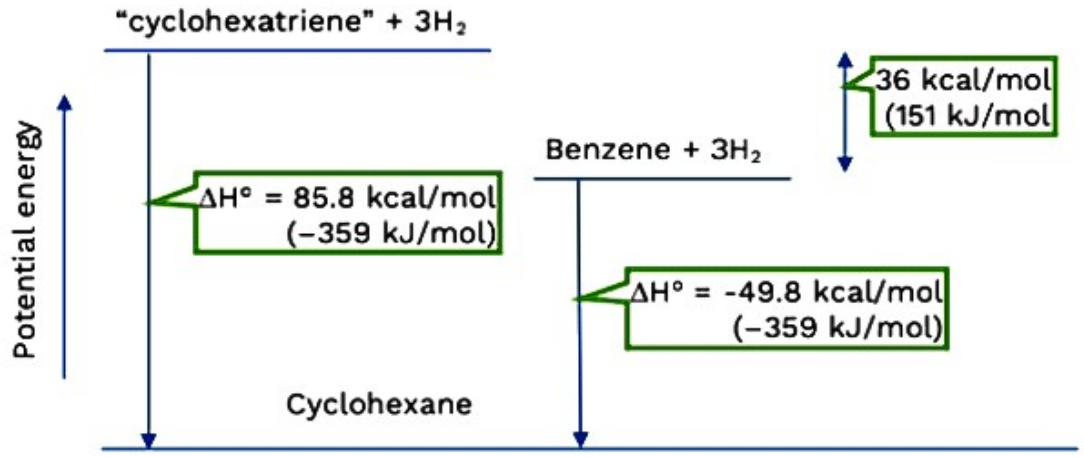
Resonance energy = Calculated value – Experimental value

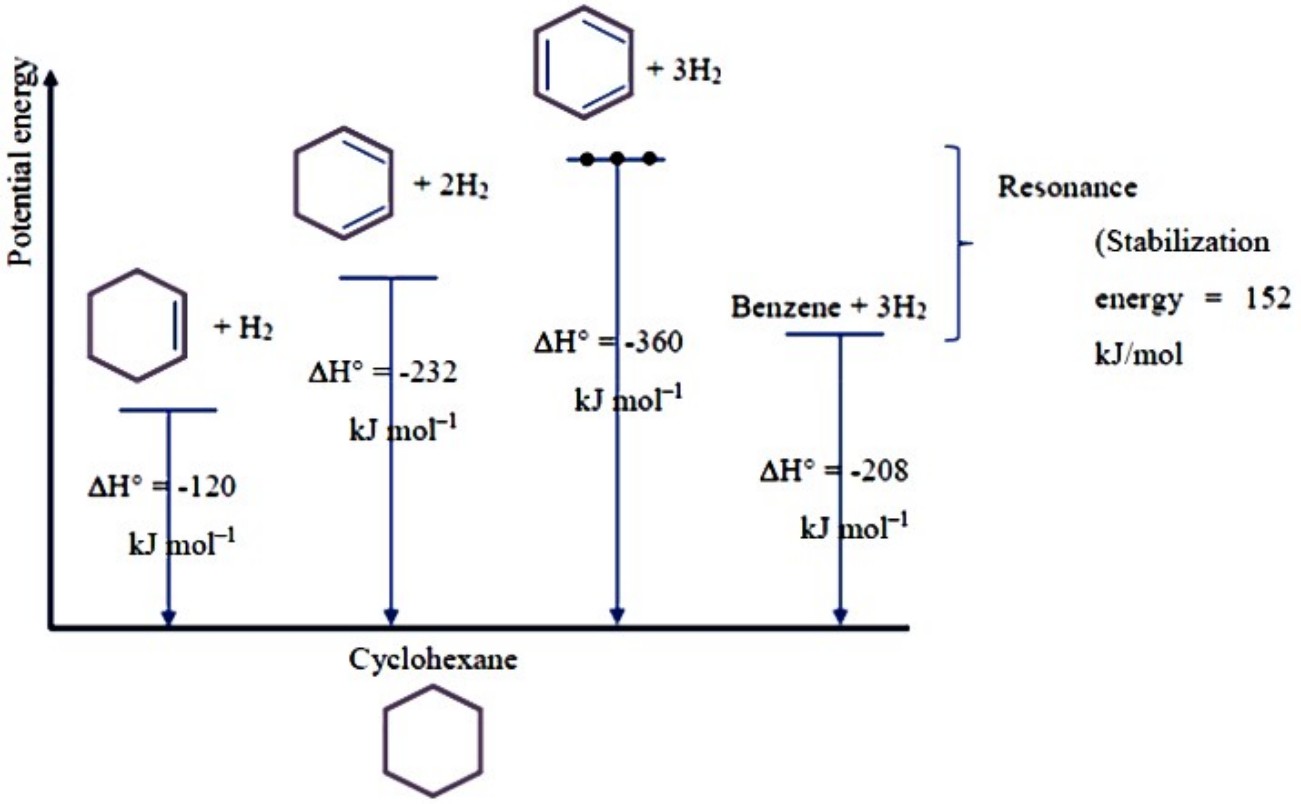
= 87 – 51 = 36 Kcal./mole

Resonance energy $\propto$ extent of resonance $\propto$ quality of resonance

Factors affecting resonance energy:

Case-1: On the basis of quantity :


"cyclohexatriene" + 3H₂
Potential energy
36 kcal/mol
(151 kJ/mol
Benzene + 3H₂
ΔH° = 85.8 kcal/mol
(–359 kJ/mol)
ΔH° = -49.8 kcal/mol
(–359 kJ/mol)
Cyclohexane


Potential energy
+ 3H₂
+ 2H₂
+ H₂
Resonance
(Stabilization
energy = 152
kJ/mol
Benzene + 3H₂
ΔH° = -360
kJ mol⁻¹
ΔH° = -232
kJ mol⁻¹
ΔH° = -120
kJ mol⁻¹
ΔH° = -208
kJ mol⁻¹
Cyclohexane

Concept Building Exercise

Q116.



(a)

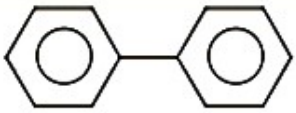


(b)

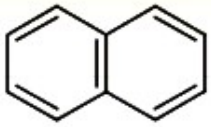
A116. R.E. b > a

More resonance, more will be resonance energy

Q117.

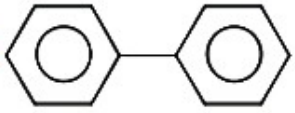


(a)

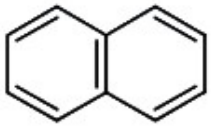


(b)

A117.



(a)
4 R.S.
R.E. a > b

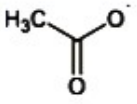


(b)
3 R.S.

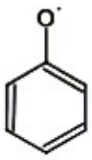
Case-2: On the basis of quality :

Concept Building Exercise

Q118.

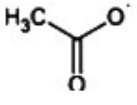


(a)

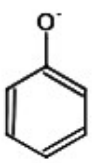


(b)

A118.



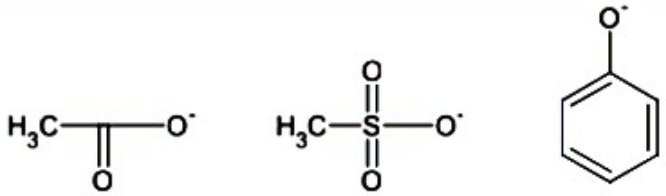
(a)
(O–O) (2eq. R.S.)



(b)
(O – C) (5 uneq. R.S.)

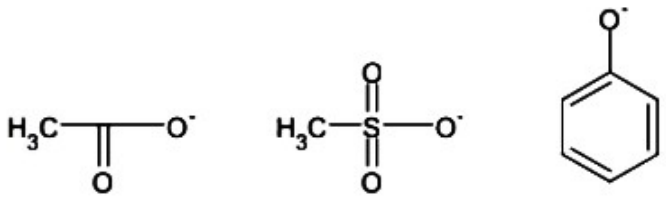
R.E. : a > b

Q119.



(a) (b) (c)

A119.



(a) (b) (c)

(O–O) (2eq. R.S.) (O–O) (3eq. R.S.) (5 R.S.)

R.E. : b > a > c

Case-3: On the basis of cross and extended resonance

Concept Building Exercise

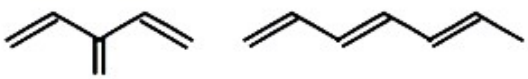
Q120.



(a) (b)

A120. R.E. a = b

Q121.

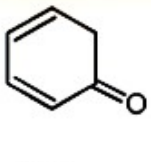


Cross (a) Extended (b)

A121. R.E. : b > a

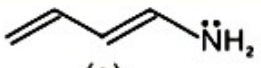
Q122.

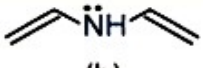

(a)


(b)

A122. R.E. : b > a

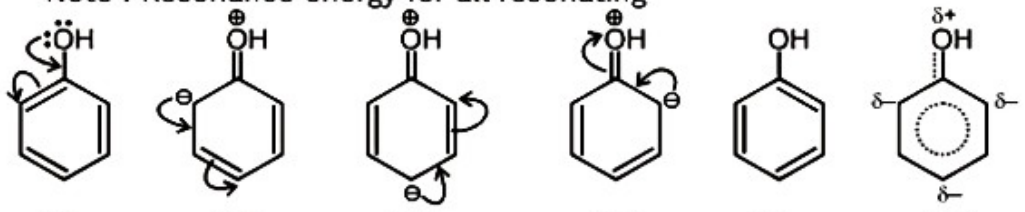
Q123.


(a)


(b)

A123. R.E. : a > b

Note : Resonance energy for all resonating


(I) (II) (III) (IV) (V) (VI)

Stablity : I = V > II = IV > III

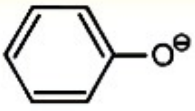
Case-4: On the basis of overlapping :

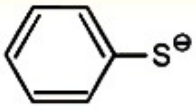
Concept Building Exercise

Q124. **(a) $CH_2 = CH\text{-}F$ (b) $CH_2 = CH - Cl$ (c) $CH_2 = CH - Br$ (d) $CH_2 = CH - I$**

A124. R.E. : a > b > c > d

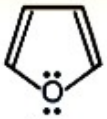
Q125.


(a)

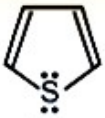

(b)

A125. R.E. : a > b

Q126.



(a)



(b)

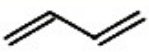
A126. R.E. : a > b

Case-5: On the basis of aromaticity :

Stability : Aromatic > Non-aromatic > Anti-aromatic

Concept Building Exercise

Q127.



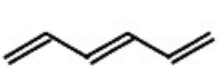
(a)



(b)

A127. a > b

Q128.



(a)



(b)

A128. b > a

Case-6: On the basis of number of benzenoid structure

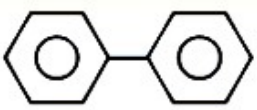
Fries rule:

More is the number of benzene structure in a compound more will be resonance.

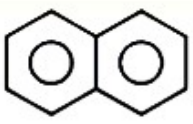
∴ more number in resonating energy.

Concept Building Exercise

Q129.



(a)



(b)

A129.

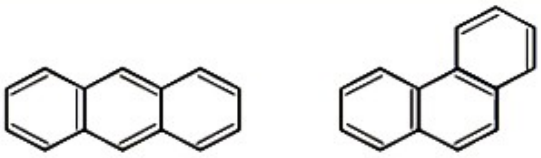


(a) (b)

4 R.S. 3 R.S.

a > b

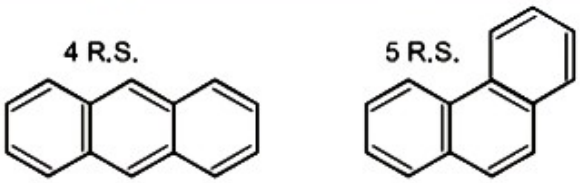
Q130.



(a) (b)

A130.

4 R.S. 5 R.S.



(a) (b)

b > a

Case-7: On the basis of electro negativity (E.N.)

Concept Building Exercise

Q131.



(a) (b)

A131. b > a

Q132.



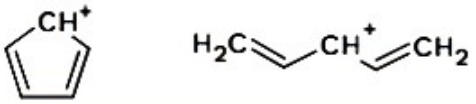
(a) (b)

A132. b > a

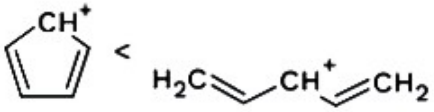
Concept Building Exercise

Compare Resonance energy between the given compounds:

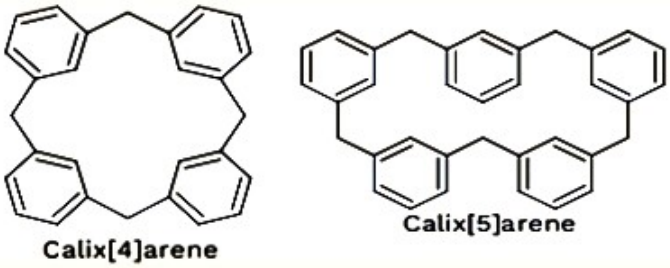
Q133.



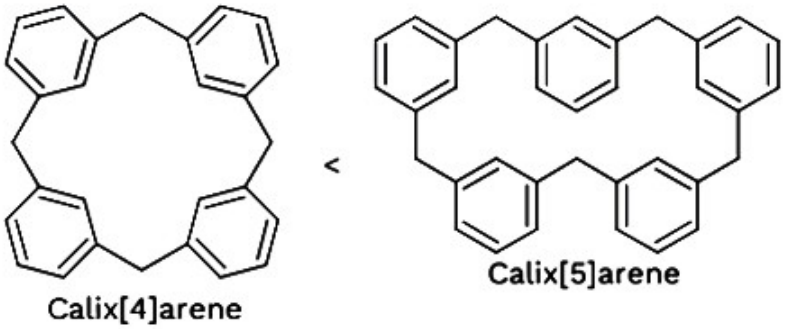
A133.



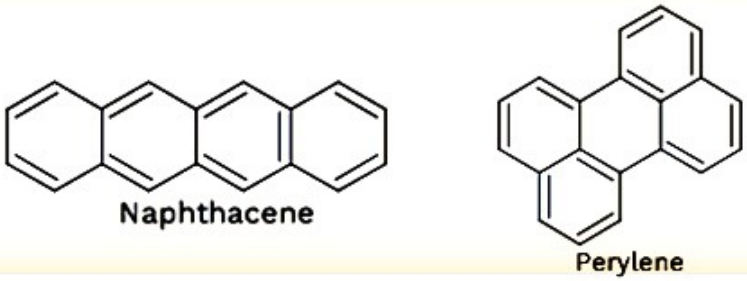
Q134.



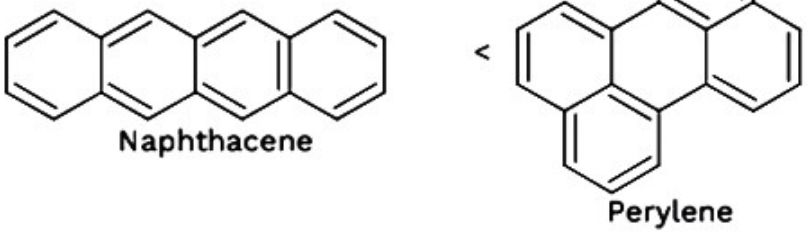
A134.



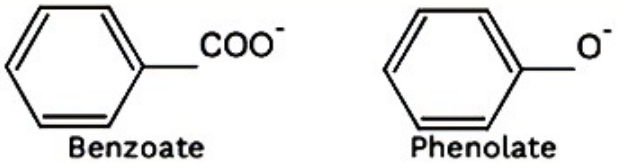
Q135.



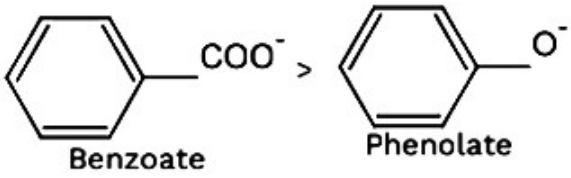
A135.



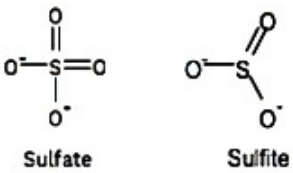
Q136.



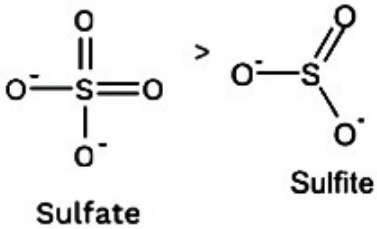
A136.



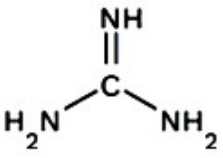
Q137.



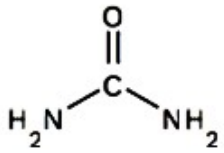
A137.



Q138.

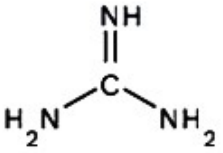


Guanidine



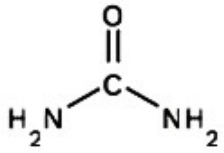
Urea

A138.



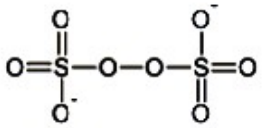
Guanidine

<

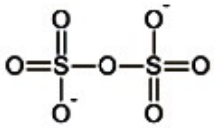


Urea

Q139.

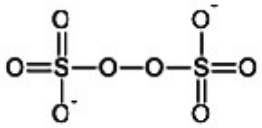


Peroxidisulfate



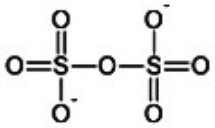
Disulfate

A139.



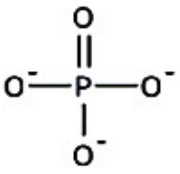
Peroxidisulfate

<



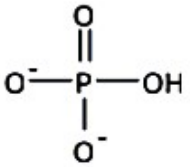
Disulfate

Q140.



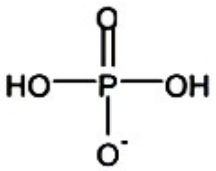
PO_4^{3-}

Phosphate



HPO_4^{2-}

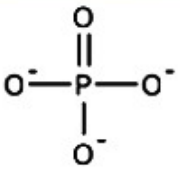
Hydrogen phosphate



$H_2PO_4^-$

Dihydrogen phosphate

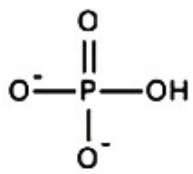
A140.



PO_4^{3-}

Phosphate

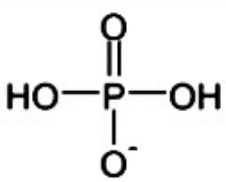
>



HPO_4^{2-}

Hydrogen phosphate

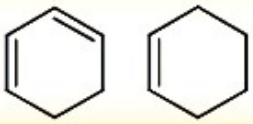
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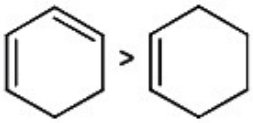
$H_2PO_4^-$

Dihydrogen phosphate

Q141.



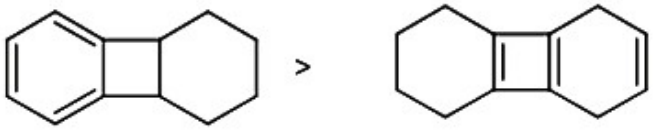
A141.



Q142.



A142.



Q143.

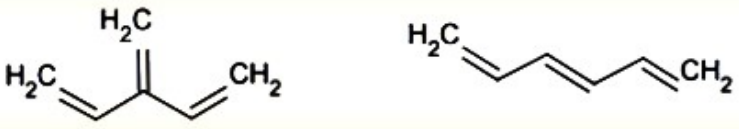


A143.

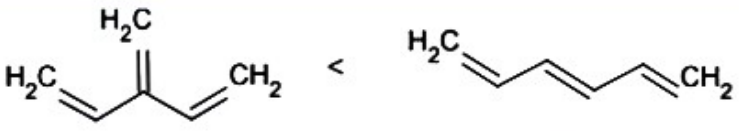


49 Kcal/mol 36Kcal/mol

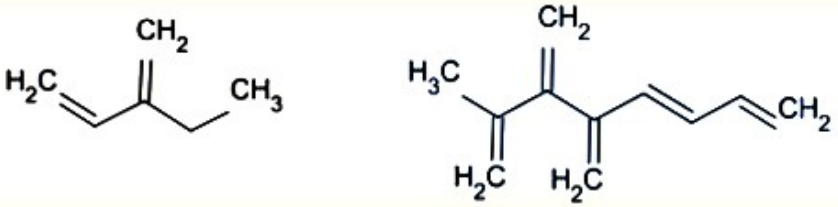
Q144.

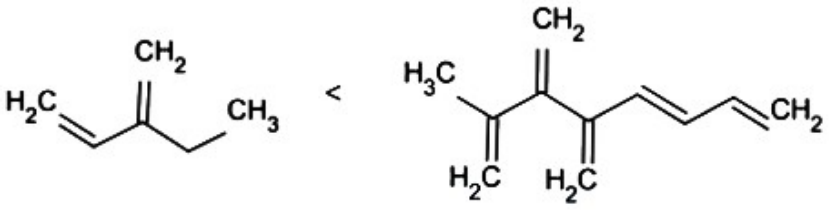


A144.

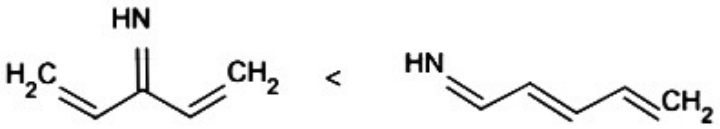


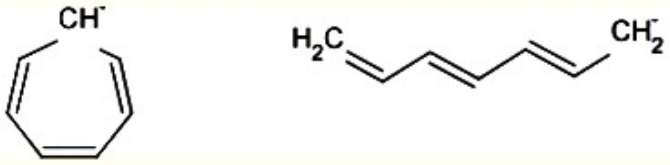
Q145.

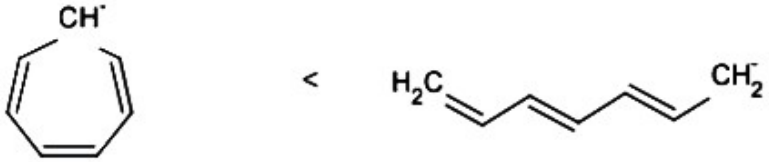


A145. 

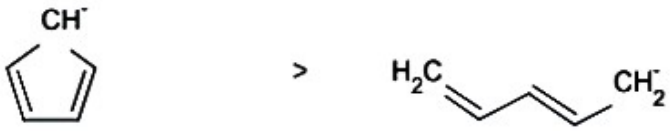
Q146. 

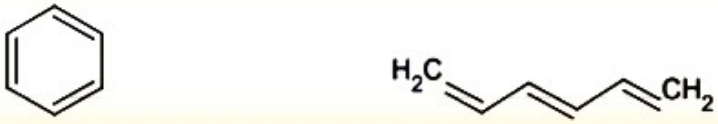
A146. 

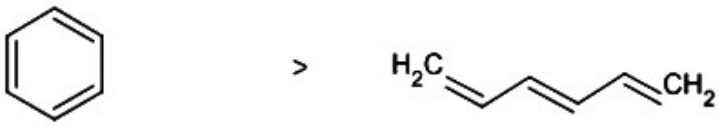
Q147. 

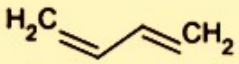
A147. 

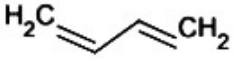
Q148. 

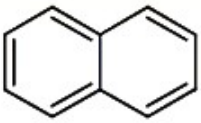
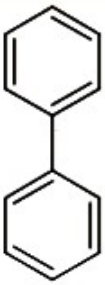
A148. 

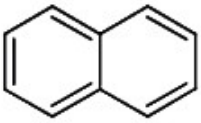
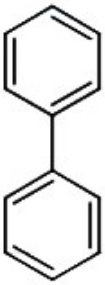
Q149. 

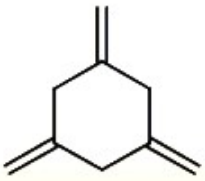
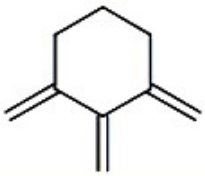
A149. 

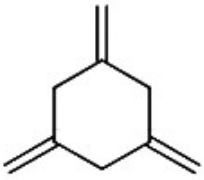
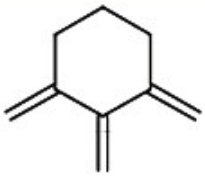
Q150.  

A150.  < 

Q151.  

A151.  < 

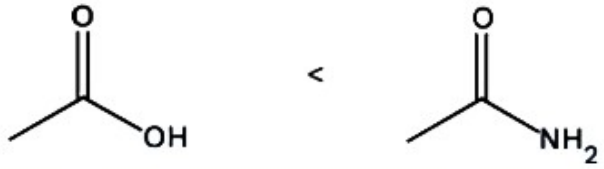
Q152.  

A152.  < 

Q153.



A153.



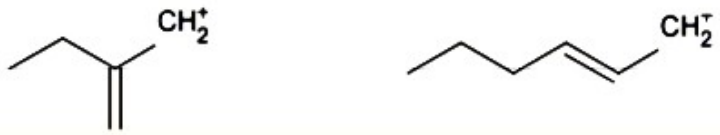
Q154.



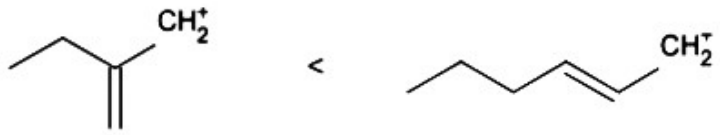
A154.



Q155.



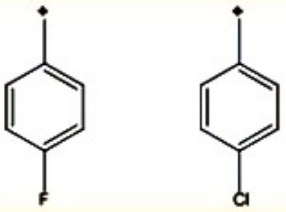
A155.

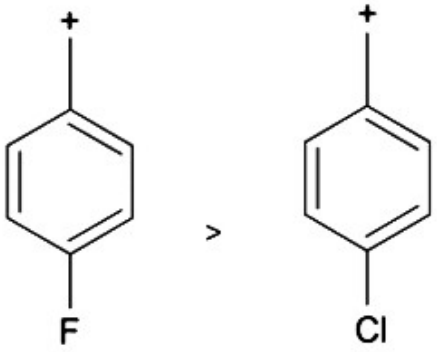


Q156.

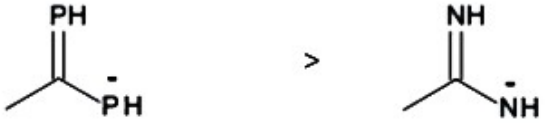


A156. 

Q157. 

A157 . 

Q158. 

A158. 

Q159. 

A159. 

Application of Mesomeric Effect

Stability of reaction intermediates

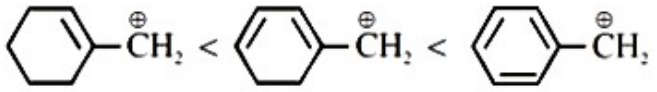
1. Stability of carbocation

More resonance in carbocation, more stability of carbocation.

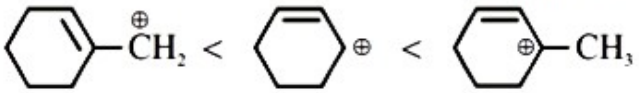
Aromatic cations are more stable than non aromatic cations.

Example:

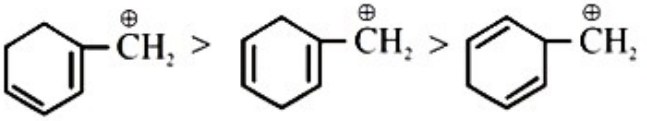
Stability order :

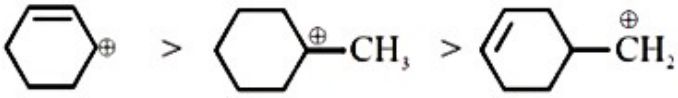
(a) 

Maximum resonance

(b) 

1° 2° 3°

(c) 

(d) 

Compare stability of following carbocations

(i) $CH_3 - \overset{\oplus}{\underset{\displaystyle CH_3}{\underset{|}{C}}} - CH = CH_2, CH_3 - \overset{\oplus}{C}H - CH = CH_2, \overset{\oplus}{C}H_2 - CH = CH_2$ (1 > 2 > 3)

(ii) $CH_2 = CH - \overset{\oplus}{C}H_2, CH_3 - CH_2 - \overset{\oplus}{C}H_2, CH_2 = CH - CH_2 - \overset{\oplus}{C}H_2$ (1 > 2 > 3)

(iii) $(C_6H_5)_3\overset{\oplus}{C}, (C_6H_5)_2\overset{\oplus}{C}H, C_6H_5\overset{\oplus}{C}H_2, CH_2 = CH - \overset{\oplus}{C}H_2, (CH_3)_3\overset{\oplus}{C}, (CH_3)_2\overset{\oplus}{C}H$

(1 > 2 > 5 > 3 > 4 > 6)

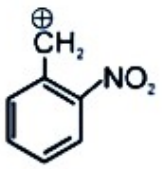
(vi) $CH_2 = CH - \overset{\oplus}{C}H_2$ $C_6H_5 - \overset{\oplus}{C}H_2$ (2 > 1)

allyl carbocation
Number of resonating
Structures = 2

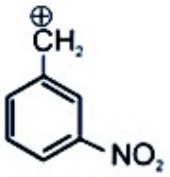
benzyl carbocation
Number of resonating
Structures = 5

Hence the benzyl carbocation is more stable than the allyl carbocation.

(v)



(I)



(II)



(III)

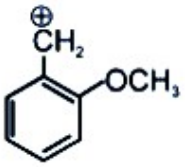
Increase in positive charge by −I and −R effect

Increase in positive charge only by −I effect

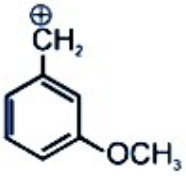
Increase in Positive charge by −I and −R effect

Stability order II > III > I

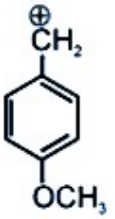
(vi)



(I)



(II)



(III)

Stabilised by + R effect destabilised by −I effect −I power is maximum (due to distance)

destabilised by −I effect only

Stabilised by + R effect destabilised by −I effect −I power is minimum (due to distance)

Stability order III > I > II

2. Stability of carbanion : More resonance in carbanion, more stability of carbanion
Example: stability order for following

(i) $CH_2 = CH - \overset{\ominus}{C}H_2, CH_2 = \overset{\ominus}{C}H, CH_3 - \overset{\ominus}{C}H$

I II III

stability order I > II > III

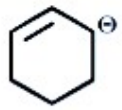
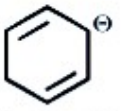
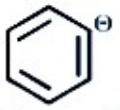
(ii) $\overset{\ominus}{C}H_2 - OH > \overset{\ominus}{C}H_2 - CH_2 - OH > \overset{\ominus}{C}H_2 - CH_2 - CH_2 - OH$

maximum(−I) Minimum(−I) of OH

(iii) $\overset{\ominus}{C}H_2 - NO_2 \quad \overset{\ominus}{C}H_2 - CH_2 - NO_2 \quad CH_3 - \overset{\ominus}{C}H - NO_2$

I II III

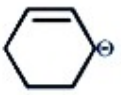
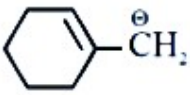
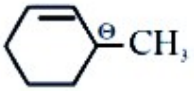
stability order I > III > II

(iv)  ,  , 

resonance more resonance localized ⊖ charge

stability order II > I > III

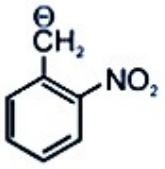
(v) $(C_6H_5)_3\overset{\ominus}{C} > (C_6H_5)_2\overset{\ominus}{C}H > C_6H_5\overset{\ominus}{C}H_2 > CH_2 = CH - \overset{\ominus}{C}H_2 > CH_2 = \overset{\ominus}{C}H$

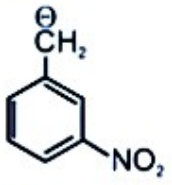
(vi)  >  > 

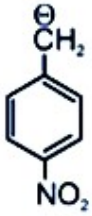
2° anion 1° anion 3° anion

I II III

stability order II > I > III

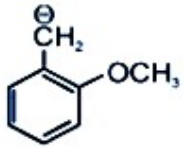
(vii) 

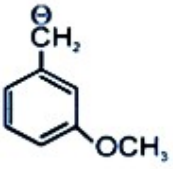


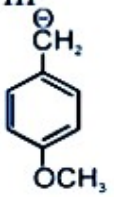


stability order I > III > II

I II III

(viii) 





I II III

stability order II > I > III

3. Stability of free radicals.

More resonance, more stability of free radical.

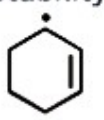
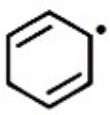
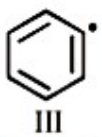
Example: Stability order

(i) $CH_2 = CH - \overset{\bullet}{C}H_2$ $CH_2 = \overset{\bullet}{C}H$ $CH_2 = CH - \overset{\bullet}{C}H = CH_3$

I II III

Less resonance No resonance More resonance

stability order III > I > II

(ii)  ,  , 

I II III

resonance more resonance localized

stability order II > I > III

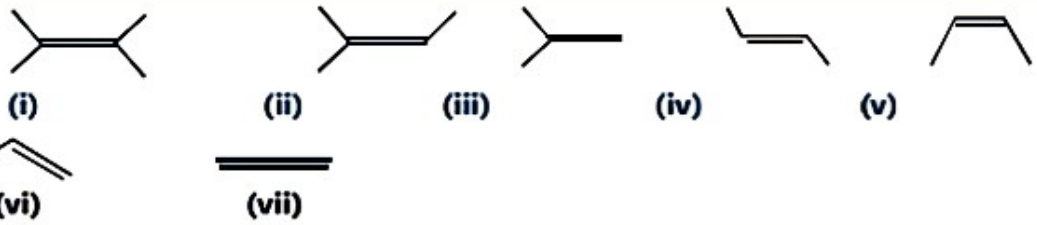
(iii) $(C_6H_5)_3\dot{C} > (C_6H_5)_2\dot{C}H > (C_6H_5)\dot{C}H_2 > CH_2 = CH - \dot{C}H_2 > (CH_3)_3\dot{C}$

$> (CH_3)_2\dot{C}H > CH_3\dot{C}H_2 > \dot{C}H_3$

Concept Building Exercise

Give the relative Stability of the following compound

Q160.



(i) (ii) (iii) (iv) (v) (vi) (vii)

A160. i > ii > iii > iv > v > vi > vii

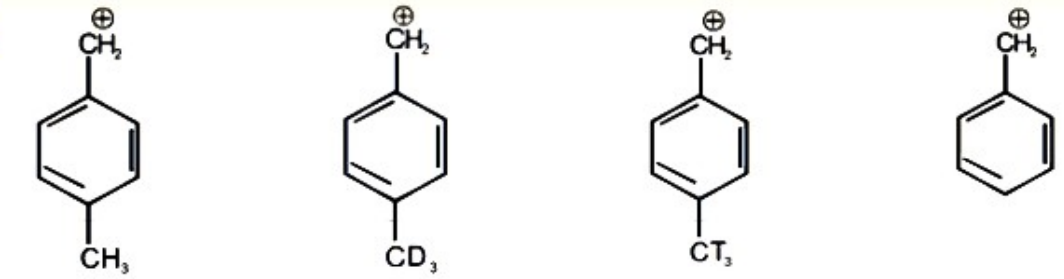
Q161.



(i) (ii) (iii) (iv)

A161. i > ii > iii > iv

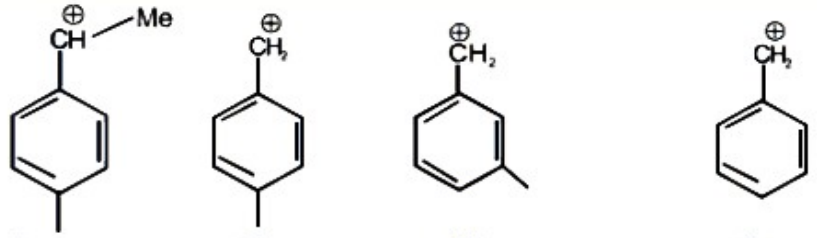
Q162.



(i) (ii) (iii) (iv)

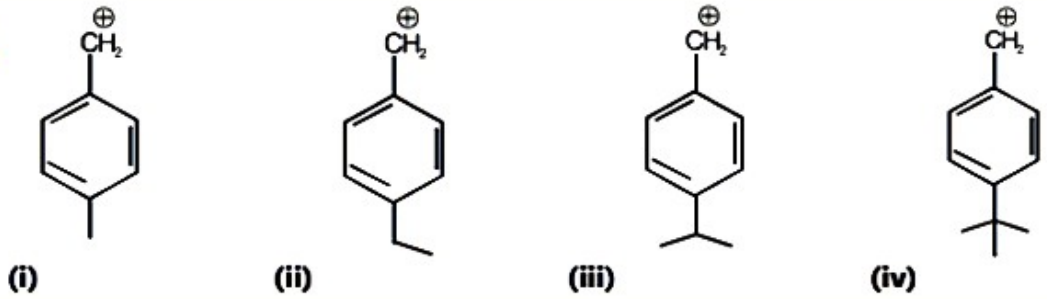
A162. i > ii > iii > iv

Q163.



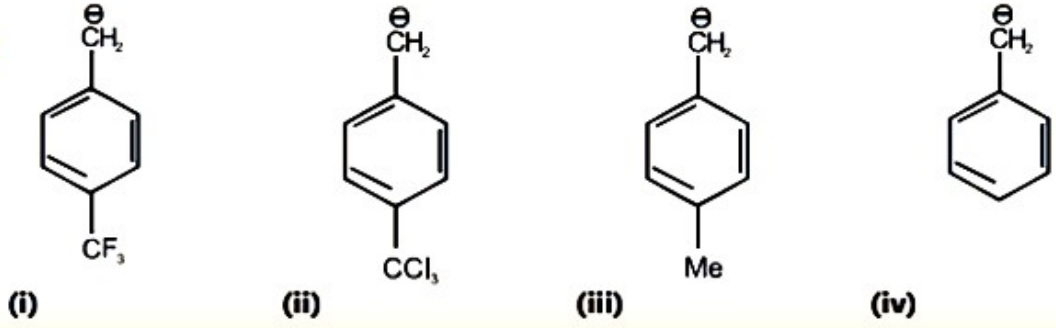
A163. i > ii > iii > iv

Q164.



A164. i > ii > iii > iv

Q165.



A165. i > ii > iv > iii

Tautomerism and Equilibrium Shift

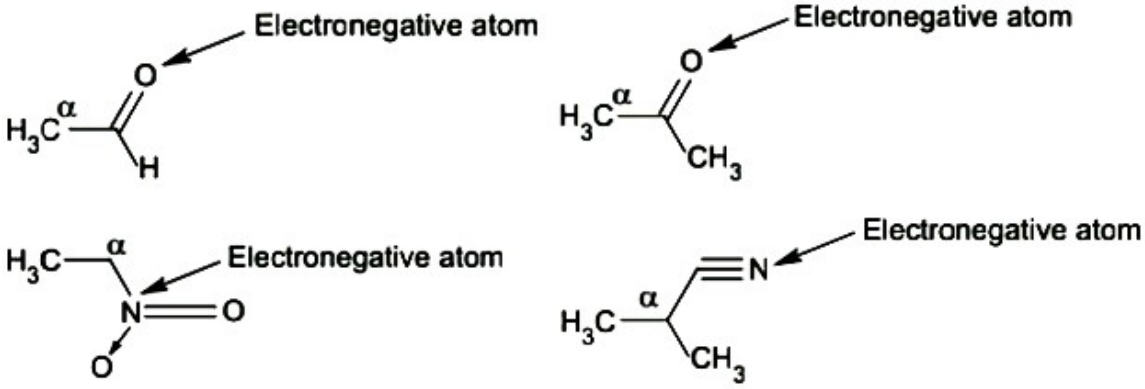
Various terms used for tautomerism

1. Desmotropism [desmos = bond ; tropos = turn , (From Greek)]
2. Pseudomerism
3. Kryptomerism
4. Allelotropism

Structural requirement for tautomerism

1. Compound should have electronegative atom bonded with multiple bond (i.e. N and O).
2. Compound should have atleast one acidic hydrogen present on α-carbon of the molecule.
   If the compound favour these two conditions it will show tautomerism.

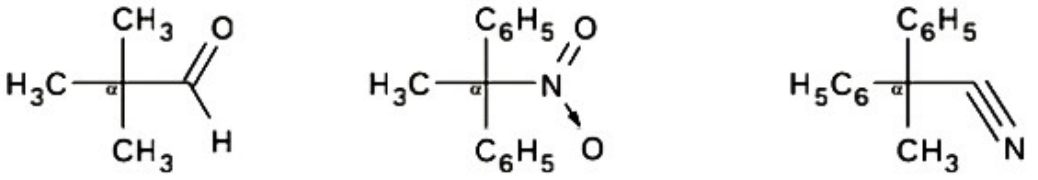
For example.



These compounds fulfill both the conditions, hence these will show tautomerism.

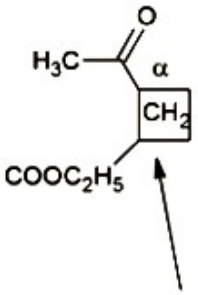
In these compounds electronegative atom is bonded with multiple bond but compound has no hydrogen on α-carbon hence these compounds will not show tautomerism.

For example.

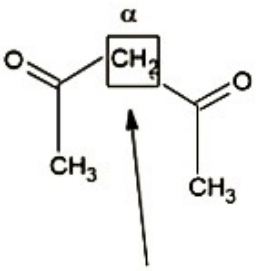


If compound has active methylene group (or) active methyne group then α-carbon for tautomerism is always carbon of active methylene (or) methyne group.

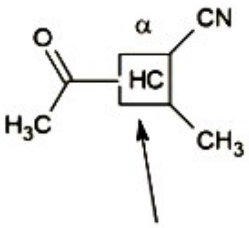
For example.



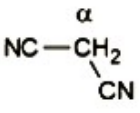
active methylene group
(1)



active methylene group
(2)



active methylene group
(3)

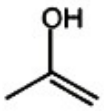
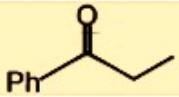
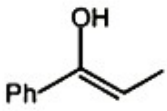


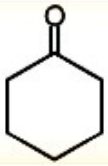
(4)

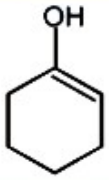
Note :
These compounds will also show tautomerism because these compounds fulfill both structural requirements.

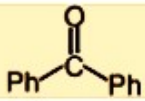
Concept Building Exercise

Fill up the following table for the compounds given below. Also state which Enol forms are capable of showing geometrical isomerism?

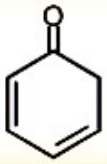
S.No.	Compound	Show tautomerism	Draw enol form (if show tautomerism)
Q166.			
A166.	✓		
Q167.			
A167.	✓		**Shows G.I.**

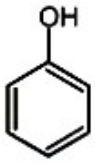
Q168. 

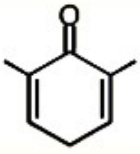
A168. ✓ 

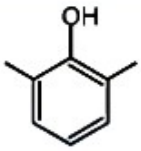
Q169. 

A169. × -

Q170. 

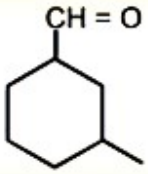
A170. ✓ 

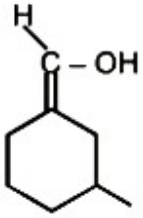
Q171. 

A171. ✓ 

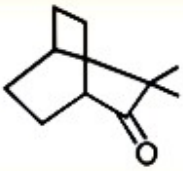
Q172. 

A172. × **Not available**

Q173. 

A173. ✓ 

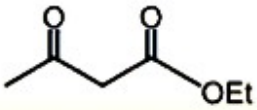
Shows G.I.

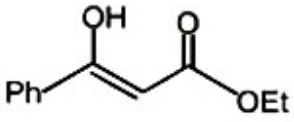
Q174. 

A174. × – **Bredt's rule voilation**

A175. 

A175. × – **No α-H available**

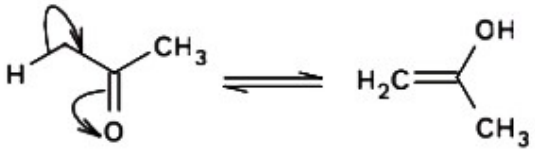
A176. 

A176. ✓ 

Shows G.I.

Cause of Tautomerism :

Migration of acidic hydrogen from α-carbon to electronegative atom which is bonded with multiple bond is the cause of tautomerism and this phenomenon is known as tautomerism.



(I) (II)

1. Both (I) and (II) are isomers known as tautomers.
2. Tautomers always be in equilibrium state.
3. Number of sigma, pi and lone pair of electrons in both tautomers are always same.
4. Tautomerism is a chemical phenomenon which takes place only in liquid state.
5. Tautomersim is catalysed by acid as well as base.
6. Tautomers are always reversible functional groups, functional isomers.

Difference between tautomerism and resonance

Tautomers of a compound actually exist and can be separated. On the other hand, resonating structures are imaginary and question of separation of such structure even does not arise.

Tautomers are different in the position of at least one atom relative to other atoms in the molecule.

Chief difference is the position of electrons and π-electrons or unshared e^- not the atom.

Position of ions in tautomers differ in bond length, i.e., migratory part has different bond lenghts in two cases of its attachment in appreciable difference. In resonating structure, difference is not more than 0.3Å, i.e., two extreme case in benzene are 1.54Å and 1.33Å.

Tautomers are in dynamic equilibrium ⇌ interconvertible and no such equilibrium in resonance.

Tautomerism has no contribution on stabilization of a molecule resonance has its effect on stabilization.

Tautomerism may occur in planar or non-planar while resonance only in plannar.

Tautomerism is of several types the two important types are dyad and triad systems :

1. **Diad System:** It involves the oscillation of H atom between two polyvalent atoms, e.g.., hydrogen cyanide (I) and hydrogen isocyanide (II).



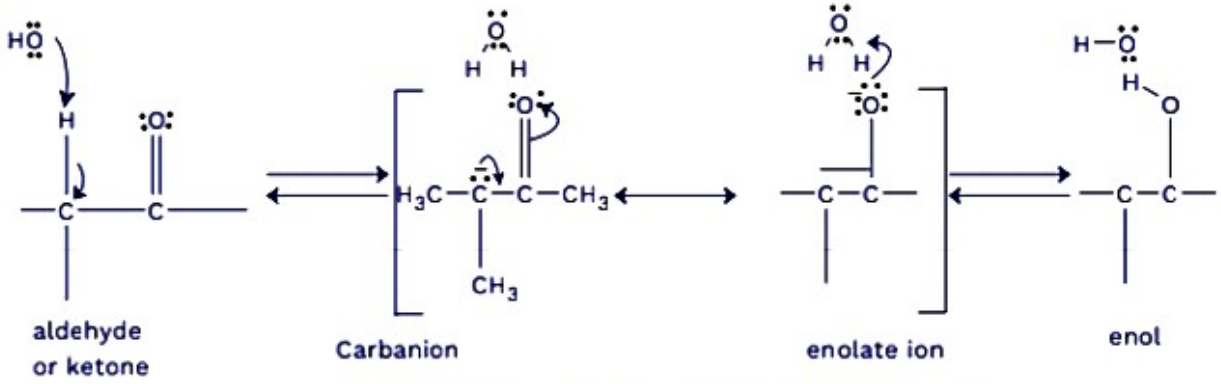
(I) (II)

In the above example the H atom vibrates between carbon and nitrogen atoms. The alkyl derivatives of (I) are called cyanides while those of (II) isocyanides.

2. **Triad System:** In it one H atom oscillates between three polyvalent atoms.

Keto-enol tautomerim : In this case the polyvalent atoms are one oxygen and two carbon atoms, e.g. aceto acetic ester, acetyl acetone, benzoyl acetone, acetaldehyde, acetone, phenol etc. The form containing keto (oxo) group is called keto while that having alkene and –ol functions is called an enol form.

a) Keto-enol Tautomerisation (Base promoted enol)



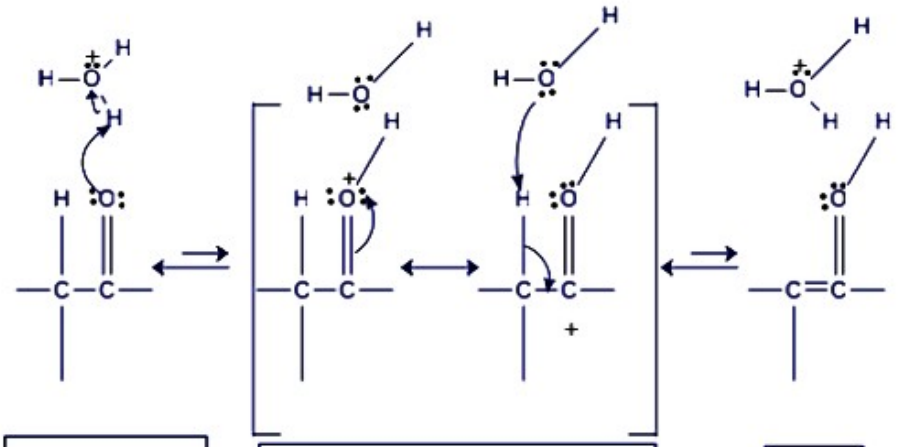
(conjugate base of both carbonyl compound and enol

Regeneration of hydrogen ion the carbanion by aq solution on the α-carbon provides back the carbonyl compound.Protonation on oxygen gives the enol.

b) Acid-catalyzed enolization:

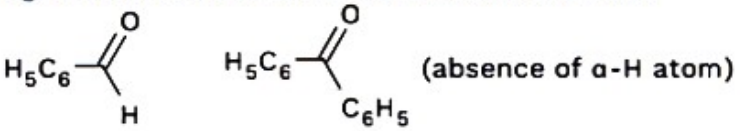
Includes the conjugate acid of the carbonyl compound. Recall that this ion has carbocation characteristics. Losing proton from oxygen gives back the starting carbonyl compound; loss of the proton from the α-carbon gives the enol.

enol and its carbonyl isomer have the similar conjugate acid.

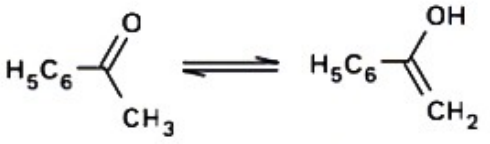


Note :

The conversion of a keto form into enol form is known as enolization. It is catalysed by the presence of acid (or) alkali. This dynamic isomerism is shown only by those aldehydes, ketones or esters which have at least one labile α-hydrogen atom. For example, benzaldehyde and benzophenone do not have labile α-hydrogen, hence these do not show tautomerism.

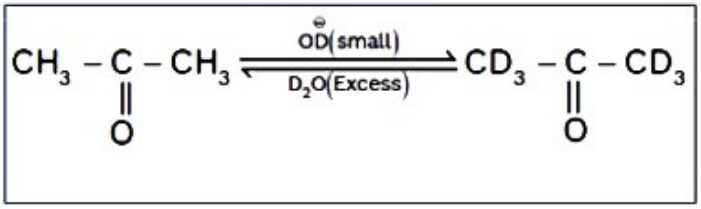


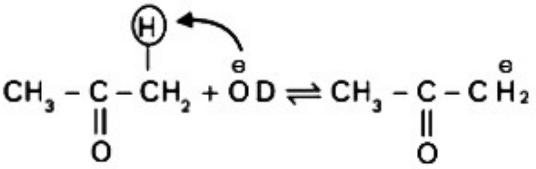
On other hand, acetophenone contains α-H atoms, hence exhibits tautomerism.

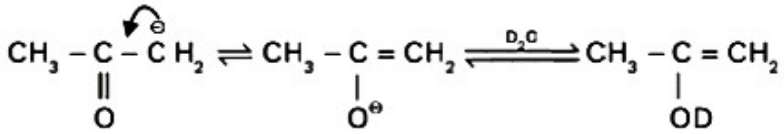


Keto form Enol form

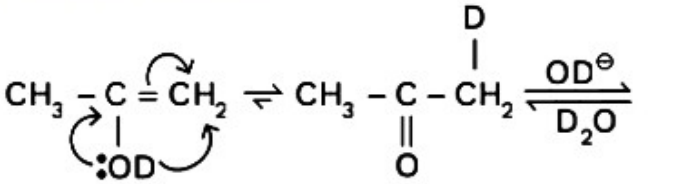
Deuterium Exchange Method

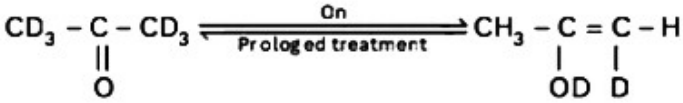


Step 1: 

Step 2: 

Now in reverse direction

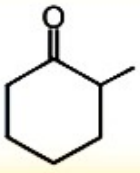




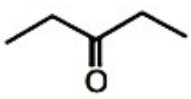
S.N.	Compound	Number of replaceable Hydrogen by Deuterium DO–/D2O (prolonged treatment)

Q177. 

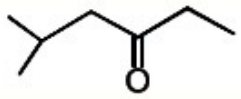
A177. 4

Q178. 

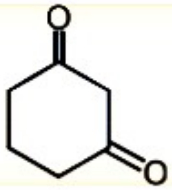
A178. 3

Q179. 

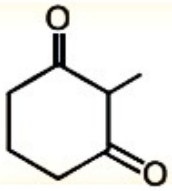
A179. 4

Q180. 

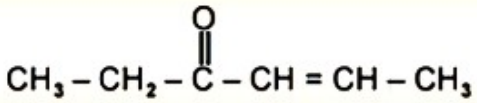
A180. 4

Q181. 

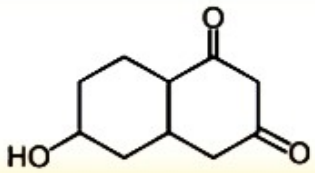
A181. 6

Q182. 

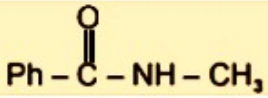
A182. 5

Q183. 

A183. 6

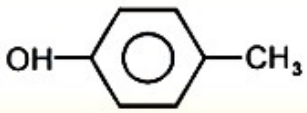
Q184. 

A184. 6

Q185. 

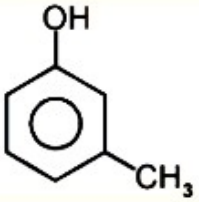
A185. 1

Q186.



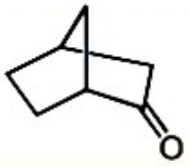
A186. 3

Q187.



A187. 4

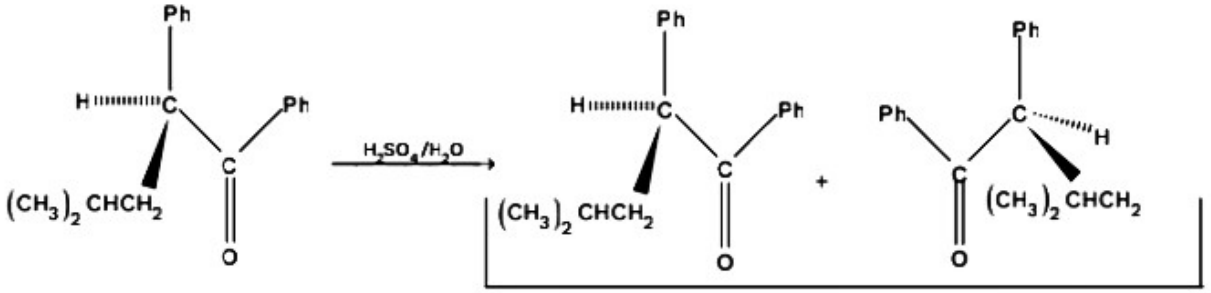
Q188.



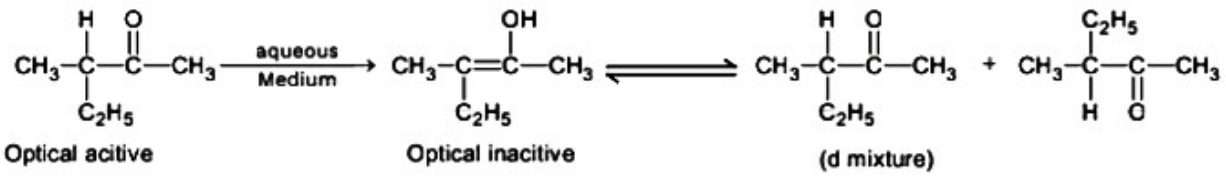
A188. 2

Racemization

1.



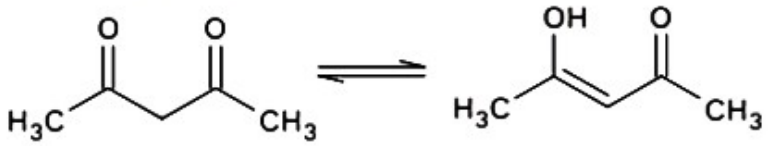
2.



Stability of Enol

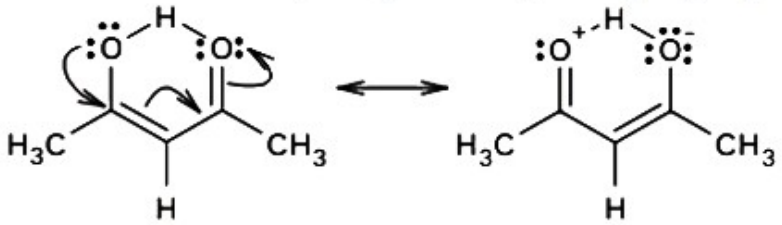
Stability and enol Content in Dicarbonyl:

Those molecules have two carbonyl groups separated by one $-CH_2-$ group (called β-dicarbonyl compounds), the quantity of enol present at equilibrium is far higher e.g.



(24%) (76%)

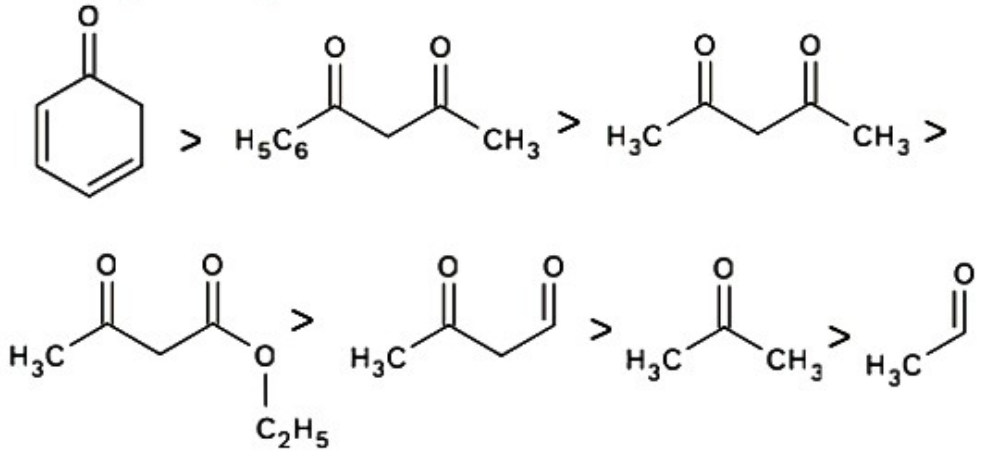
The superior stability of the enol form of β-dicarbonyl compounds can be attributed to stability gained through resonance steadying of the conjugated double bonds and (in a cyclic form) through hydrogen bonding.



Enolization is in the following order: CH_3COCH_3 < $CH_3COCH_2COOC_2H_5$ < $C_6H_5COCH_2COOC_2H_5$ < CH_3COCH_2CHO < $CH_3COCH_2COCH_3$

For example

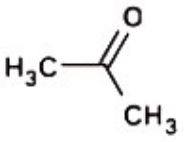
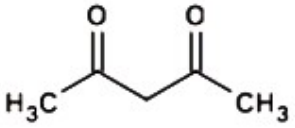
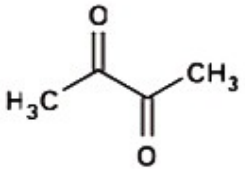
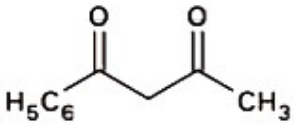
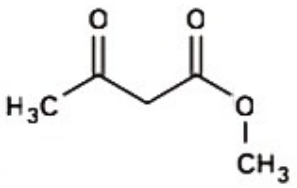
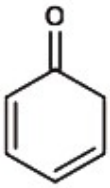
Percentage of enolic contents of some common compounds in decreasing order is given below:



Percentage Composition of Tautomeric Mixture : Percentage of enol contents of some compounds is given below in the table.

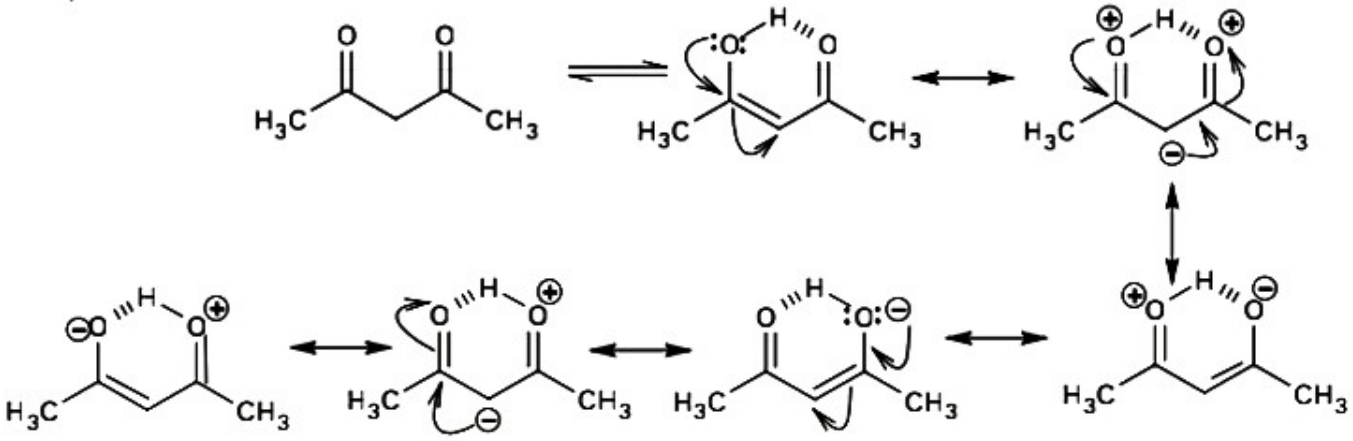
Percent enol content $\propto$ conjugation
$\propto$/ Temperature
$\propto$/ Hydrogen bonding
$\propto$/ Base strength

Enol content of some compounds

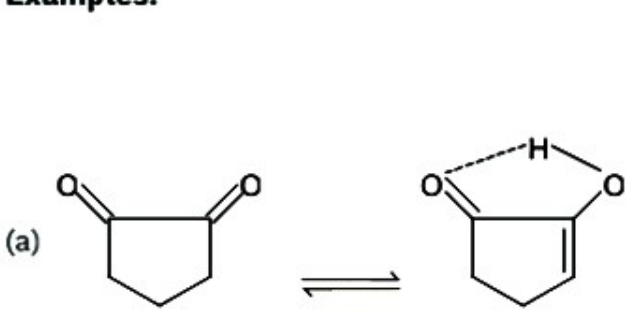
COMPOUND	ENOL PERCENTAGE	COMPOUND	ENOL PERCENTAGE
	0.00025		80.4
	0.0056		89.0
	4.8		99.99

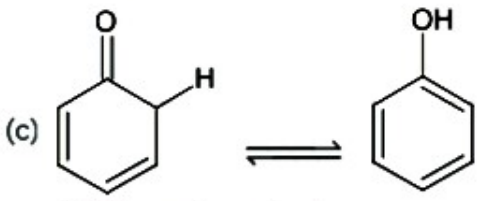
The conversion of a keto form into an enol form is known as enolisation and the enolisation of a compound has been found to depend upon various factors such as structural factor, temperature and nature of solvent. However, the most important is the structural factor (resonance and hydrogen bonding).

1. Ketonic form predominates in simple monocarbonyl compounds like acetaldehyde, acetone and cyclohexanone. This is due to the greater bond strength of C = O {>C = O, 365 kJ/mole) present in keto form than the carbon-carbon double bond (C = C, 250 kJ/mole) present in enolic form.
2. Enolic form predominates in β-di ketones due to intramolecular hydrogen bonding and resonance. Intramolecular hydrogen bonding stabilizes enol form by 7 kcal/mole and resonance stabilizes enol form by 15 kcal/mole. Thus enol form is more stable than keto form by 22 kcal/mole in 1,3-diketones.

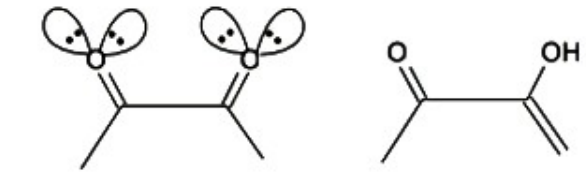


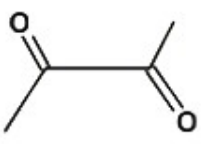
Examples:





100% enol content
* converted into anti form

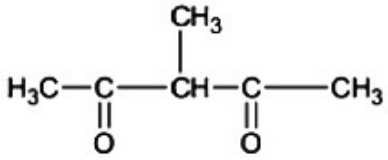


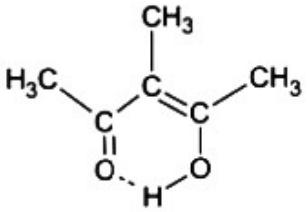


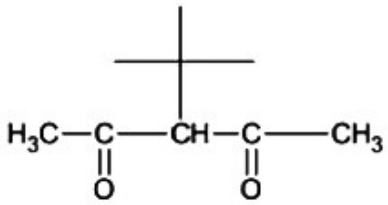
0% enol
* (Aromatic)
* dipole becomes zero
* keto is more stable
More stable conformer

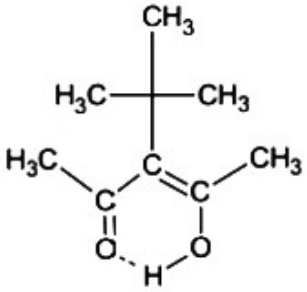
⇒ Release of energy due to stable ring formation

* Steric repulsion decreases enolic content due to repulsion in planer condition.

(a) 



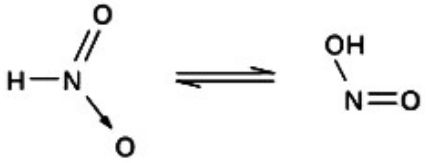
(b) 



* up to down repulsion increases
* Enolic form becomes more unstable
* Stability decreases, keto % increases.

(ii) *Triad System containing Nitrogen*

(a) Nitrous acid exists in two tautomeric forms:



Nitro form Nitrite form

The alkyl derivatives of nitro form and nitrite form are called as nitro alkanes and alkyl nitrites, respectively.

(b) Primary and secondary nitroalkanes have lable α-H atom, hence exhibit nitro-aci nitro type of tautomerism.

(c) Nitroso-oxime System: In this system one tautomer exits as nitroso while the other tautomer exists as an isonitroso (oxime) form.

(d) Imine-Enamine System

Other examples of Tautomeric

(a) Imine-Enamine system :

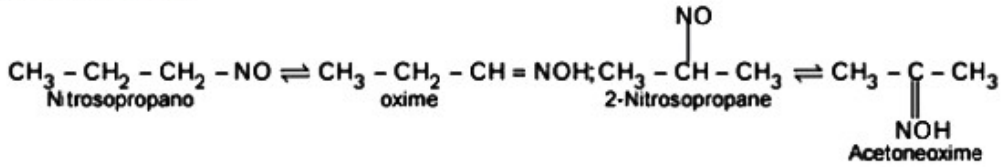
$R_2CH - CR = NR \rightleftharpoons R_2C = CR - NH - R$

Imine Enamine

Among these two tautomers, enamines are stable only when there is no hydrogen on the nitrogen, otherwise the imine form predominates.

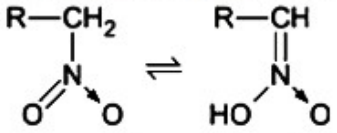
(b) Nitroso-oxime system

Primary and secondary nitro compounds, primary and secondary nitroso compounds too exhibit tautomerism with their extra stable isonitroso or oxime form.

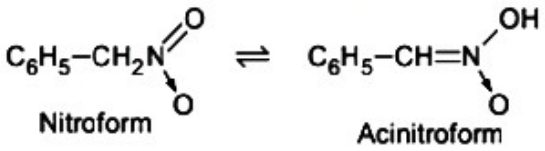


(c) Nitro-Acinitro system :

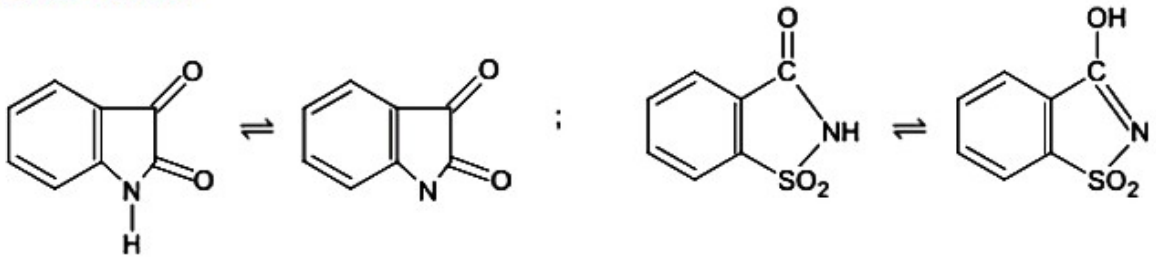
The acidic nature of the nitro compounds gives rise to the belief that the nitro compounds exist in two forms, a more stable or normal nitro form and the less stable acinitro form



The solidity of the nitro form is more as compared to the acinitro form because it is stabilised by resonance.



(d) Lactam-Lactim



Acid Base

Definition of Acid

(a) Arrhenius Acid: An acid is a matter that can contribute (or loose) a proton in H_2O.

$$HA \xrightleftharpoons{\text{Solution}} \overset{\oplus}{H} + \overset{\ominus}{A}$$

Arrhenious Acid Con Base

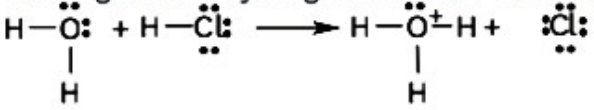
(b) The Bronsted Lowry Explanation of Acids and Bases: An acid is a substance that can contribute (or loose) a proton, and a base is a substance that can accept a proton.

If the conjugate Base of an acid is more stable then the acid is strong and if the conjugate base of an acid is less stable the acid is weak.

$HCl \rightarrow Cl^-$ Chloride ion is stable and it is weak conjugate base

$NH_3 \rightarrow NH_2^-$ Amide ion is unstable and it is strong conjugate base.

When gaseous hydrogen chloride dissolves in water:

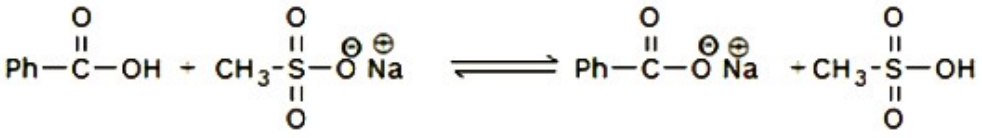


Base (proton acceptor	Acid (proton donor)	Conjugate acid of H_2O	Conjugate base of HCl

Example

1. $\underset{\text{Acid}}{Ph-COOH} + \underset{\text{Base}}{Ph-O^-Na^+} \rightleftharpoons \underset{\text{Conjugate Base}}{Ph-COO^-Na^+} + \underset{\text{Conjugate Base}}{Ph-OH}$

2. $\underset{\text{Acid}}{R-SO_3H} + \underset{\text{Base}}{RO^-Na^+} \rightleftharpoons \underset{\text{Conjugate Base}}{R-SO_3^-Na^+} + \underset{\text{Conjugate Acid}}{R-OH}$

3. 

Acid Base Conjugate Base Conjugate Acid

4. $\underset{\text{Acid}}{Ph-OH} + CH_3-O^-Na^+ \rightleftharpoons \underset{\text{Conjugate Base}}{Ph-O^-Na^+} + \underset{\text{Conjugate Acid}}{CH_3-OH}$

5. $\underset{\text{Base}}{H_2O} + \underset{\text{Acid}}{HNO_2} \rightleftharpoons \underset{\text{Conjugate Acid}}{H_3O^+} + \underset{\text{Conjugate Base}}{NO_2^-}$

Q189. Draw conjugate Base

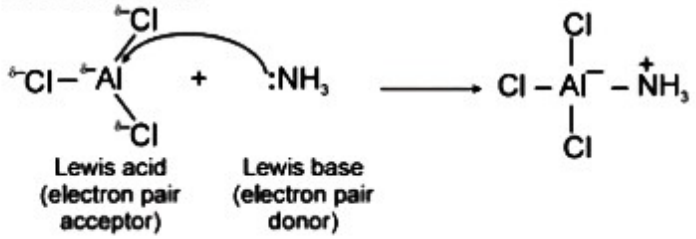
1. $H-F \rightarrow \overset{\ominus}{F}$
2. $H_2O \rightarrow \overset{\ominus}{O}H$
3. $NH_3 \rightarrow \overset{\ominus}{N}H_2$
4. $H_3O^+ \rightarrow H_2O$
5. $\overset{\oplus}{N}H_4 \rightarrow NH_3$
6. $CH_3-OH \rightarrow CH_3-\overset{\ominus}{O}$
7. $CH_3-SH \rightarrow CH_3-\overset{\ominus}{S}$
8. $Ph-OH \rightarrow Ph-\overset{\ominus}{O}$
9. $CH_3-\overset{\overset{O}{\|}}{C}-OH \longrightarrow CH_3-\overset{\overset{O}{\|}}{C}-O\ominus$

Q190. Draw conjugate Acid

1. $\overset{\ominus}{F} \rightarrow HF$
2. $\overset{\ominus}{O}H \rightarrow H_2O$
3. $\overset{\ominus}{N}H_2 \rightarrow NH_3$
4. $NH_3 \rightarrow \overset{\oplus}{N}H_4$
5. $H_2O \rightarrow H_3\overset{\oplus}{O}$
6. $CH_3-\overset{\ominus}{O} \rightarrow CH_3-OH$
7. $CH_3-OH \rightarrow CH_3-\overset{\oplus}{O}H_2$
8. $Ph-\overset{\ominus}{O} \rightarrow Ph-OH$
9. $CH_3-\overset{\overset{O}{\|}}{C}-O\ominus \longrightarrow CH_3-\overset{\overset{O}{\|}}{C}-OH$

(c) The Lewis meaning of acids and bases

According to Lewis acids are electron pair acceptors and bases are electron pair contributors.



CONCEPT	ACID	BASE
• Arihenius	H^+ donor	$\overset{\ominus}{O}H$ donor
• Bronsted & Lowry	H^+ donor	H^+ acceptor
• Lewis	e^- pair acceptor	e^- pair donor

(d) Acid dissociation constant (K_a)

The strength of an acid is spoken in terms of acid dissociation constant K_a or pK_a A strong acid has large K_a value or small pK_a value.

$$HA \rightleftharpoons H^+ + A^-$$

$$K_a = \frac{[H^+][A^-]}{[HA]} \text{ and } P_{K_a} = -\log K_a$$

Ex. $HCOOH > CH_3COOH > H_2CO_3 > HOCl > H_2O$ (Acidic strength)

Ka $2 \times 10^{-4} > 1.8 \times 10^{-5} > 4 \times 10^{-7} > 1 \times 10^{-8} > 1.8 \times 10^{-16}$

P_{ka} $3.7 < 4.74 < 6.4 < 8.0 < 15.74$

How to Compare Acidic Strength

Acidic Strength of any acid depends upon different factors like inductive, Resonance, Mesomeric, Electronegativity, Size of Atom, Ortho effect, Steric effect etc.

We will discuss here different pattern of questions expected in JEE Advance and Mains

Periodic trends in acid base

In same period

In the hydrides of different elements acidic nature increases from left to right along a period.

$CH_4 < NH_3 < H_2O < HF \rightarrow$ Acidic nature

$CH_3^- < NH_2^- < OH^- < F^- \rightarrow$ Stablitiy of conjugate base

$SiH_4 < PH_3 < H_2S < HCl \rightarrow$ Acidic nature

$SiH_3^- < PH_2^- < HS^- < Cl^- \rightarrow$ Stablitiy of conjugate base

In same group

In a group the acidic nature of hydrides increases from the top to bottom. In a group stability of conjugate base depends on size. Increasing size of ion, the stability increases

$HF < HCl < HBr < HI \rightarrow$ Acidic nature

$F^- < Cl^- < Br^- < I^- \rightarrow$ Stability of conjugate base

Acidic Strength of carbon compounds:

The Strength of an acid depends on stability of conjugate base. If conjugate base is more stable, the acid is more acidic and vice versa. The Stability of a conjugate base depends on.

i) Electronegativity of atom with-ve charge

ii) I–effect

iii) R-effect

iv) Aromaticity

Among hydrocarbons the order of acidic strength is, Alkanes < alkenes < alkynes

$CH_3 - CH_3 < CH_2 = CH_2 < CH \equiv CH \left(\text{Acidic strength}\right)$

$CH_3 - CH_2^- < CH_2 = CH^- < CH \equiv C^- \left(\text{Stability}\right)$

Within alkanes increase in size/ branches decreases acidic nature.

$CH_4 > CH_3 - CH_3 > CH_3 - CH_2 - CH_3 \left(\text{Acidic Strength}\right)$

$CH_3 > CH_3 - CH_2^- > CH_3 - CH_2 - CH_2^- \left(\text{Stability}\right)$

Acidic Strength based on inductive effect

Case 1: Inductive effects of substituent groups of some substituted acetic acids.

ACID	pK_a	I EFFECT OF THE SUBSTITUENT
$\boxed{H}-CH_2-CO_2H$	4.75	H = reference point
$\boxed{CH_3}-CH_2-CO_2H$	4.90	+I
$\boxed{CH_3-O}-CH_2-CO_2H$	4.30	−I
$\boxed{I}-CH_2-CO_2H$	3.00	−I
$\boxed{Br}-CH_2-CO_2H$	2.90	−I
$\boxed{Cl}-CH_2-CO_2H$	2.80	−I
$\boxed{(CH_3)_3\overset{+}{N}}-CH_2-CO_2H$	1.80	−I

Case-2: Position of the substituent and inductive effect

ACID	pK_a	I EFFECT OF THE SUBSTITUENT
$CH_3-CH_2-\overset{\alpha}{C}H(\boxed{Cl})-CO_2H$	2.84	−I
$CH_3-\overset{\beta}{C}H_2(\boxed{Cl})-\overset{\alpha}{C}H-CO_2H$	4.06	−I
$\overset{\gamma}{C}H_3(\boxed{Cl})-CH_2-\overset{\alpha}{C}H-CO_2H$	4.52	−I

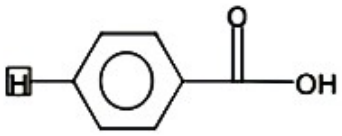
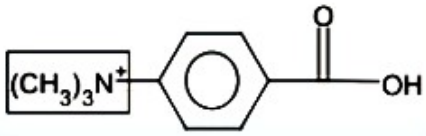
Case-3: Distance between the groups and the extent of inductive effect.

ACID	pK_a	I EFFECT OF THE SUBSTITUENT
$\boxed{HO_2C} - CO_2H$	1.23	–I
$\boxed{HO_2C} - CH_2 - CO_2H$	2.28	–I
$\boxed{HO_2C} - CH_2 - CH_2 - CO_2H$	4.19	–I

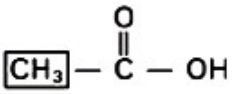
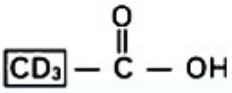
Case-4:

ACID	pK_a	I EFFECT OF THE SUBSTITUENT
$\boxed{H} - CH_2 - CO_2H$	4.75	H = reference point
$\boxed{Cl} - CH_2 - CO_2H$	2.86	One (–I) effect
$\boxed{Cl} - \underset{\boxed{Cl}}{CH} - CO_2H$	1.25	Two (–I) effects
$\boxed{Cl} - \underset{\boxed{Cl}}{\overset{\boxed{Cl}}{C}} - CO_2H$	0.65	Three (–I) effects

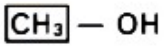
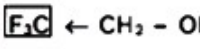
Case-5: Aromatic acids and inductive effect

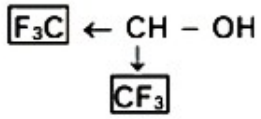
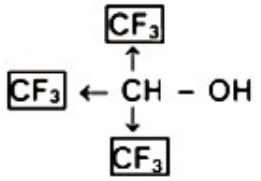
ACID	pK_a	I EFFECT OF THE SUBSTITUENT
	4.2	H = reference point
	3.9	–I effect

Case-6: Isotopic substituents and inductive effect

ACID	pK_a	I EFFECT OF THE SUBSTITUENT
	4.75	+I
	4.6	+I

Case-7: Electron withdrawing groups and inductive effects.

ACID	pK_a	I EFFECT OF THE SUBSTITUENT
	15.5	+I
	12.4	–I

ACID	pK_a	I EFFECT OF THE SUBSTITUENT
	9.3	–I
	5.4	–I

Case-8: Hybridization state of the carbon atom in the substituent and inductive effect

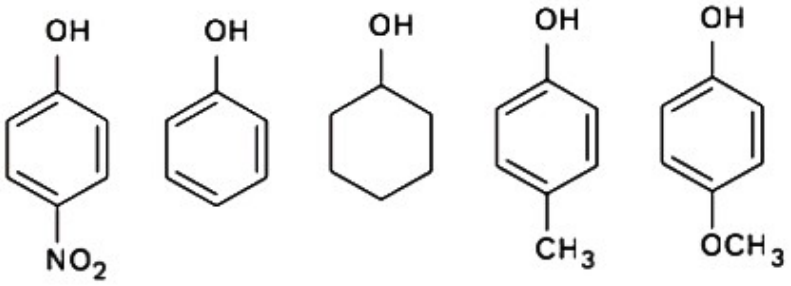
ACID	pK_a	I EFFECT OF THE SUBSTITUENT
	16	+I effect
	15.5	–I effect
	13.5	–I effect

Acidity of Phenols :

The phenoxide ion is more stabilized by resonance than the unionized phenol.
Groups which are –I, –M increases acidic character of phenol because effectively dispersing the negative charge of phenoxide ion.

+I and +M groups decrease acid strength.

Ex. (a)



I II III IV V

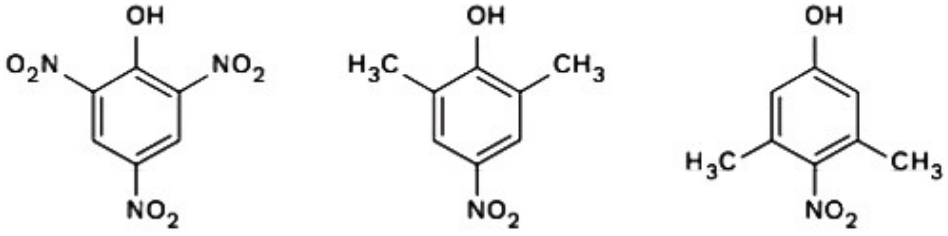
Ans. Acid strength order : I > II > IV > V > III

Sol. **Step-1:** III will be least acidic as it has no dispersion of negative charge (No delocalisation of negative charge).

Step-2: since –I, –M group will increase acid strength, Nitrophenol will be most acidic followed by phenol,

Step-3: Amongst cresol and methoxyphenol, methoxyphenol has +M effect of $-OCH_3$ which increases e– density hence decrease acidic strength

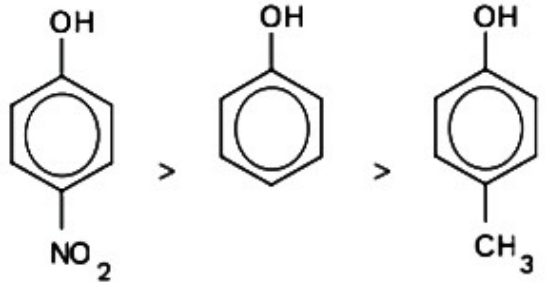
Example :



pKa = 0.71 pKa = 7.22 pKa = 8.25

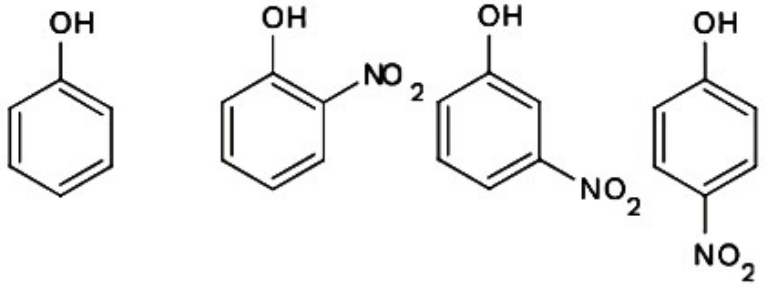
Examples related to phenol

(a)



p – nitrophenol phenol p – cresol

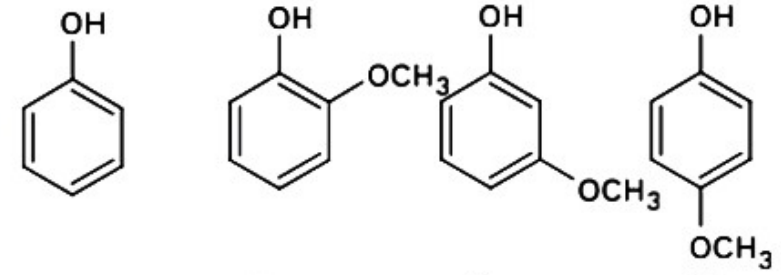
(b)



I II III IV

Order of acidic Strength is IV > II > III > I

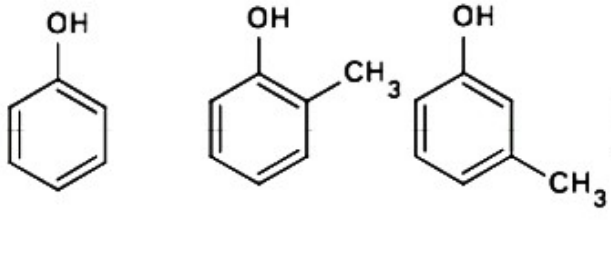
(c)



III > I > IV > II

I II III IV

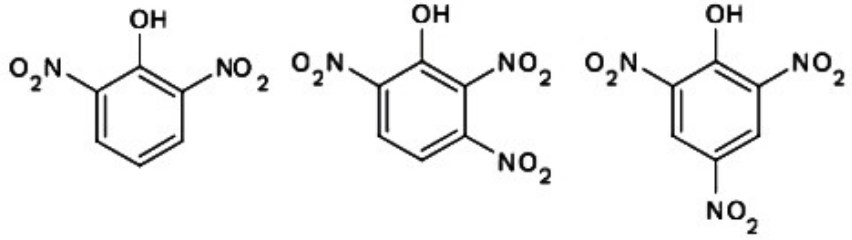
(d)



I > III > IV > II

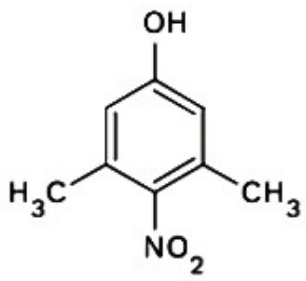
I II III IV

(e)

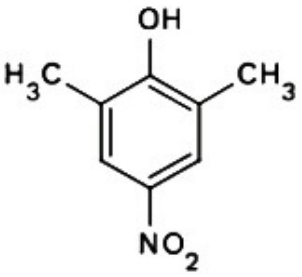


III > II > I

I II III

(f) 

I



II

Acidity order : II > I

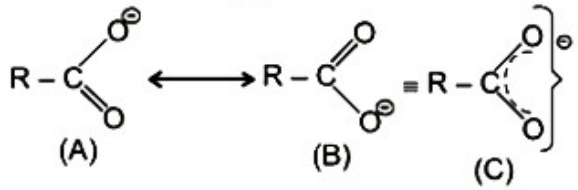
In (I) the nitro group turns away from the plane of benzene ring due to steric hindrance caused by methyl groups and resonance is inhibited. But in (II) nitro can show – M effect on – OH group.

Table : pK_a values of some phenols and Ethanol. (Ref. NCERT)

COMPOUND	FORMULA	pK_a
o-Nitrophenol	$o - O_2N - C_6H_4 - OH$	7.2
m-Nitrophenol	$m - O_2N - C_6H_4 - OH$	8.3
p-Nitrophenol	$p - O_2N - C_6H_4 - OH$	8.1
Phenol	$C_6H_5 - OH$	10
o-Cresol	$o - CH_3 - C_6H_4 - OH$	10.2
m-Cresol	$m - CH_3C_6H_4 - OH$	10.1
p-Cresol	$p - CH_3 - C_6H_4 - OH$	10.2
Ethanol	C_2H_5OH	15.9

Acidity of carboxylic acids :

Conjugate base of carboxylic acid exists as two equivalent canonical structures (A) and (B). This ion is resonance stabilized and resonance hybrid structure is (C).



Electron withdrawing group (–M, –I effect) increases acidic nature.

Electron releasing group (+M, +I effect) decreases acidic nature.

Example :

(a) $F - CH_2 - COOH > Cl - CH_2COOH > Br - CH_2COOH > I - CH_2COOH$

(b) $Cl - \underset{\underset{Cl}{|}}{\overset{\overset{Cl}{|}}{C}} - COOH > Cl - \overset{\overset{Cl}{|}}{CH} - COOH > Cl - CH_2COOH > CH_3COOH$

(c) $HCOOH > CH_3COOH > CH_3 - CH_2 - COOH$

(d) $\begin{matrix} COOH \\ | \\ COOH \end{matrix} > H_2C \begin{matrix} COOH \\ \diagdown \\ \diagup \\ COOH \end{matrix} > \begin{matrix} H_2C—COOH \\ | \\ H_2C—COOH \end{matrix}$

The effect of the following groups in increasing acidity order is

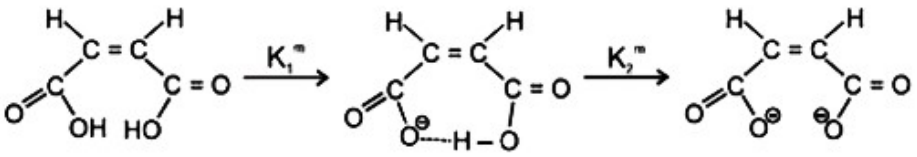
$Ph < I < Br < Cl < F < CN < NO_2 < CF_3$

These acids are arranged in direction of decreasing acidic strength (based on pK_a values) :

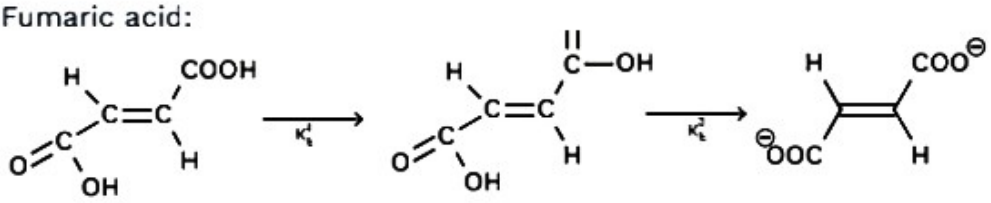
$CF_3COOH > CCl_3COOH > CHCl_2COOH > NO_2CH_2COOH > NC - CH_2COOH > FCH_2COOH > ClCH_2COOH > BrCH_2COOH > HCOOH > ClCH_2CH_2COOH > C_6H_5COOH > C_6H_5CH_2COOH > CH_3COOH > CH_3CH_2COOH$

Comparison between two geometrical isomers :

Maleic acid:



Fumaric acid:



Now, $k_1^m > K_1^f$

Since the conjugate base is stabilized by intramolecular H bonding.

But $K_2^f > K_2^m$ Since in maleate ion, after donation of H⊕ two –COOΘ groups faces each other and makes system unstable. In fumarate ion this repulsion is minimized.

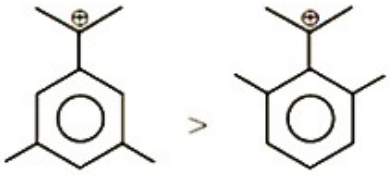
Steric Inhibition of Resonance (SIR Effect)

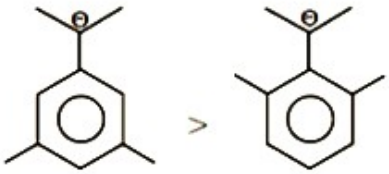
1. A steric inhibition of resonance occurs due to a steric repulsion of group/atom present at ortho position in known as **SIR EFFECT**.
2. On following atoms/groups SIR EFFECT is not observed.

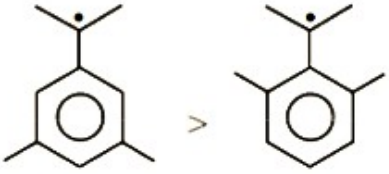
ATOMS	LINEAR GROUPS	NON PLANAR GROUPS
–I	–CN	$-SO_3H$
–Cl	–C≡C–H	
–Br	$-NH_2$	
–F	–OH	

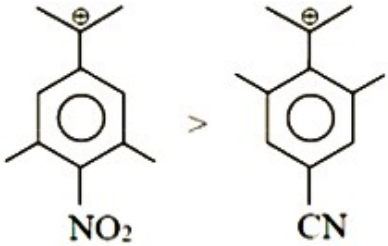
3. On following atoms/groups SIR EFFECT is observed.
   i) $-NMe_2$ ii) –NHMe iii) –COOH iv) $-NO_2$
   v) $-\overset{\oplus}{C}Me_2$ vi) $-\overset{\ominus}{C}Me_2$ vii) $-\overset{\bullet}{C}Me_2$

Example: Stability order of following compound:

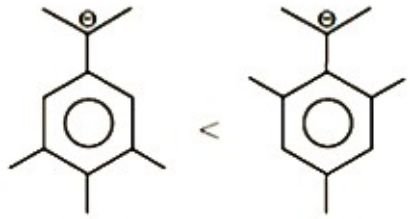
1.  (SIR EFFECT)

2.  (SIR EFFECT)

3.  (SIR EFFECT)

4. 

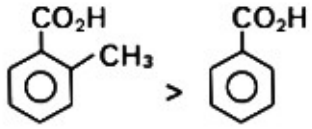
(SIR EFFECT)

5. 

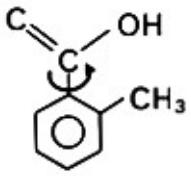
(SIR EFFECT)

Ortho-substituted Benzoic Acid is stronger than benzoic acid, why ?

It is common observation that commonly ortho substituted benzoic acids are more acidic as compared to their isomers and benzoic acids itself. This is called ortho effect (which is combined effect of steric hindrance, crowding & electronic effect) in benzoic acid. However exceptions are seen



o-Toluic acid Benzoic acid



In case of benzoic acid the methyl group (+M effect will not operate), the bond rotation across the phenyl ring and carboxylic group is checked, and the two groups are held perpendicular to each other; as such, resonance does not take place between them. The inhibition of +M effect in o-toluic acid, makes it a stronger acid compared to benzoic acid.

Conclusion of ortho effect (sir effect example)

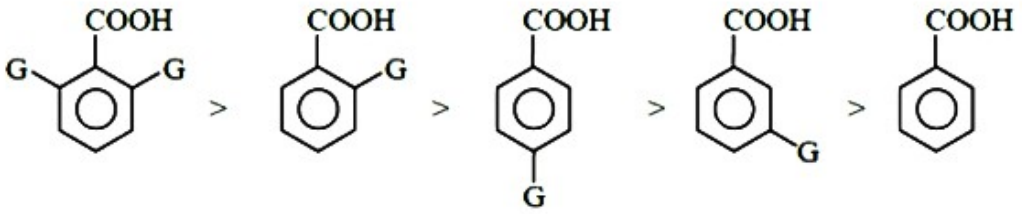
i) Ortho substituted benzoic acid is always as stronger acid than m- and p-derivative. This is known as ortho effect.

ii) A large group brings the COOH group out of plane and resist resonance through benzene ring.

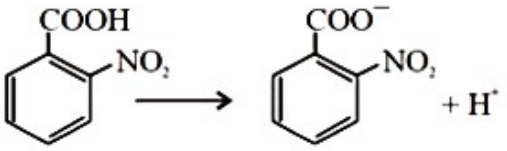
iii) This is the SIR EFFECT caused by the groups.

iv) This occurs only when the groups are at ortho position.

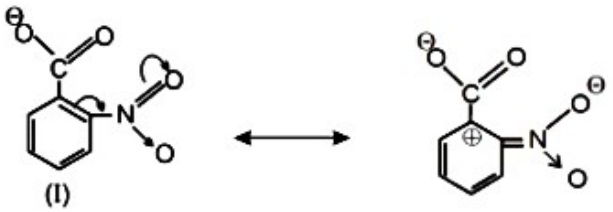
Acidity order of substituted benzoic acids



NOTE : G = EWG

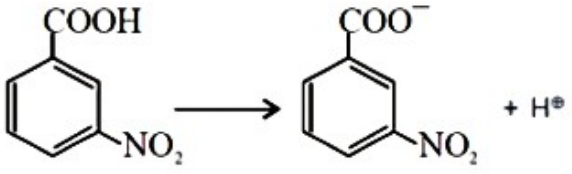


I



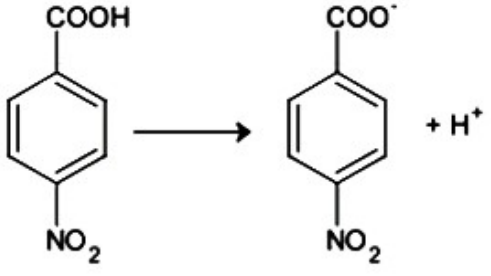
(I)

Thus, anion is stabilised by –R and –I effect and –I power is maximum.

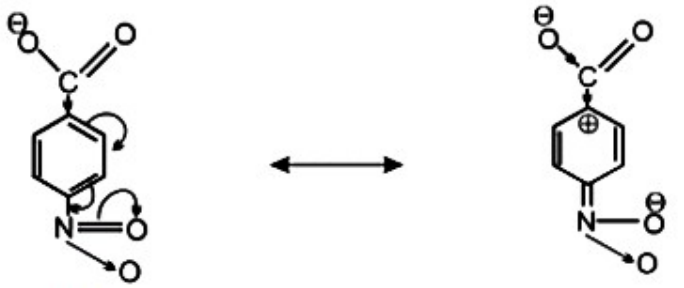


II

Anion is stabilised only by –I effect of NO_2 group

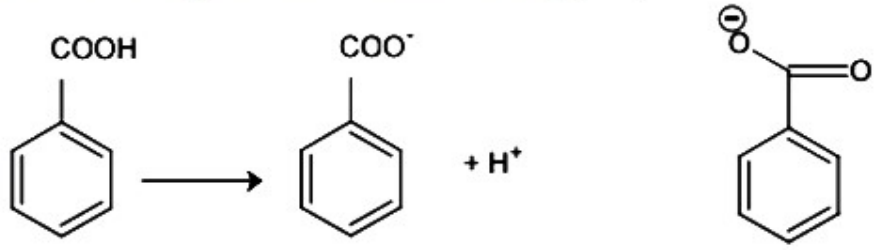


III



(III)

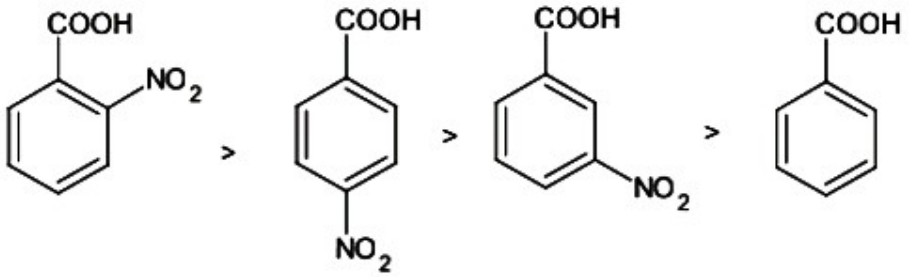
Anion is stabilised by -R and –I effect of NO_2 group



IV (IV)

Thus decreasing order of the stability of these anions is follows : I > III > II > IV

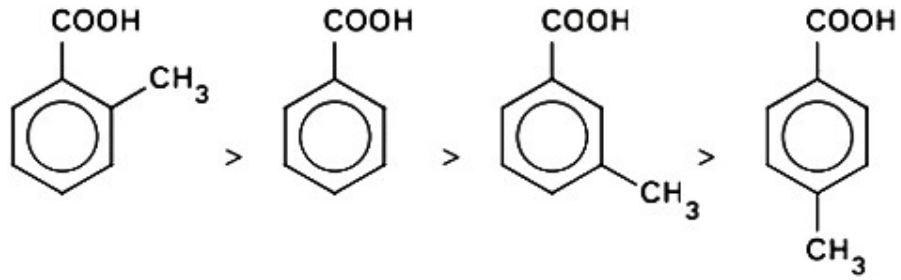
We know that ortho derivative is the most acidic therefore decreasing order of acidity of these acids is as follows :

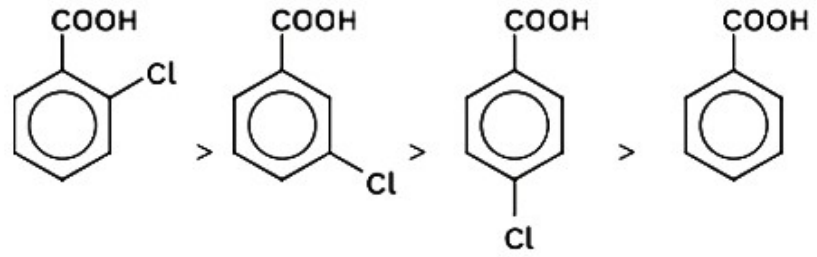


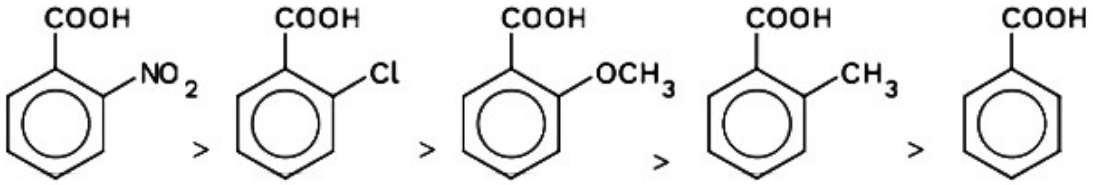
Acidity in decreasing order

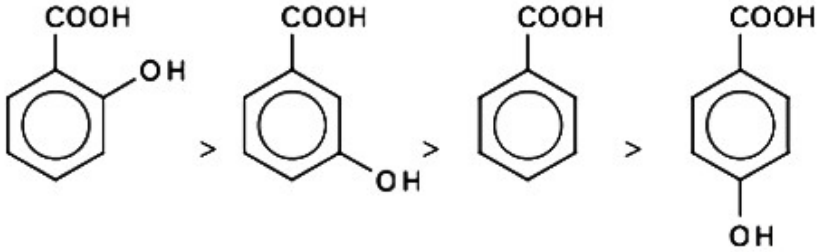
Other examples of ortho effects

Ex. (a)



(b) 
COOH
Cl
>
COOH
Cl
>
COOH
Cl
>
COOH

(c) 
COOH
NO2
>
COOH
Cl
>
COOH
OCH3
>
COOH
CH3
>
COOH

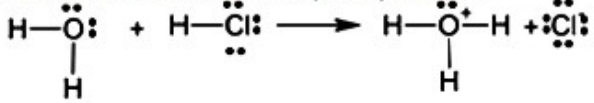
(d) 
COOH
OH
>
COOH
OH
>
COOH
>
COOH
OH

Base Concepts and Theory

Theories of Base

1. **Arrhenius base** : Base is a substance that can donate (or loose) OH^- ions in H_2O.
2. **The Bronsted** : Lowry definition of acids and bases

   Substance which can donate (or loose) a proton, and a base is a substance that can accept a proton.

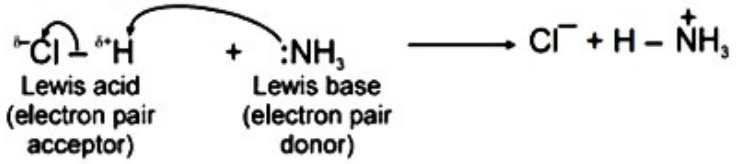


Base (proton acceptor) | Acid (proton donor) | acid of H_2O | Conjugate base of HCl | Conjugate

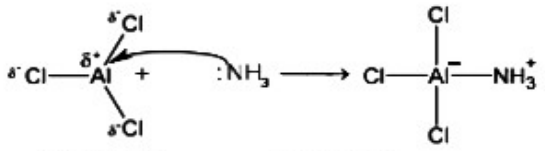
3. **The Lewis definition of acids and bases :**

   Lewis proposed that acids are electron pair acceptors and bases are electron pair donors.

   Lets Understand : aluminium chloride, reacts with ammonia in the same way that a proton donor does. Using curved arrows to show the donation of the electron pair of ammonia (the Lewis base), we have the following examples:

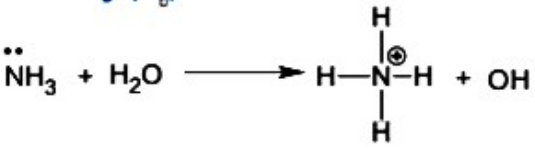


Lewis acid (electron pair acceptor) | Lewis base (electron pair donor)



Lewis acid (electron pair acceptor) | Lewis base (electron pair donor)

4. **Basicity (K_b) :**



$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$$

$$pK_b = -\log K_b$$

Basicity order in periodic table:

1. Basic strength decreases down the group,
2. Basic strength decreases along the period because electronegativity increases so electron donor tendency decreases.

Aliphatic bases :

1. On the basis of + I effect basic strength of amines should be 3° > 2° > 1° > NH_3 But this order is applicable only when the amines are in gaseous state or in case of non-polar aprotic solvent.
2. Basic strength of nitrogeous compound depends upon the hybridised state of nitrogen

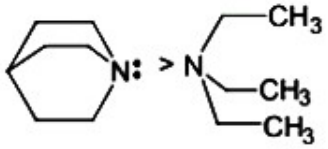
$$R - CH_2 - CH_2 - \underset{\downarrow}{NH_2} > R - CH_2 - CH = \underset{\downarrow}{NH} > R - CH \equiv CH - \underset{\downarrow}{NH_2} > R - C = \underset{\downarrow}{N}$$

sp^3(localised) sp^2(localised) sp^2(delocalised) sp

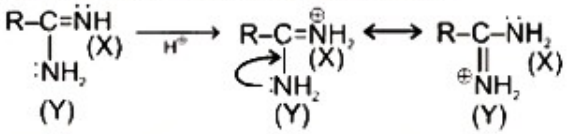
3. More electronegative atom (–I group) will decrease the basic strength

$$CH_3 - CH_2 - CH_2 - NH_2 > CH_2 = CH - CH_2 - NH_2 > H - C \equiv C - CH_2 - NH_2$$

4. Cyclic amines are more basic than acyclic amines of same nature



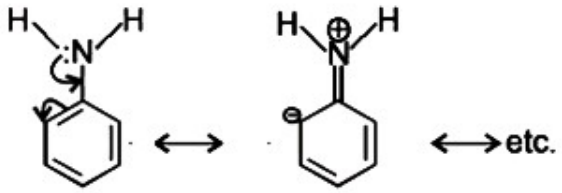
5. Amidines are more basic in nature because their conjugate acid are more stable due to resonance.



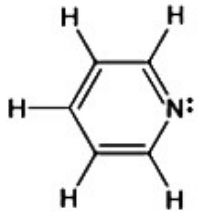
Nitrogen (X) is more basic than nitrogen (Y).

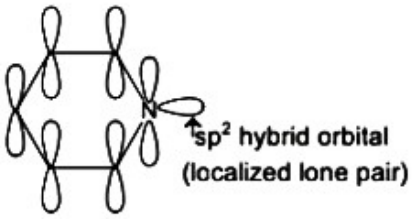
Basic Strength of Aromatic Amines and substituted Anilines:

a) Aniline : Lone pair of aniline lies in conjugation with a multiple bond, it resides in '2p' atomic orbital, so that it can get resonance stabilisation and hence, basic strength decreases. So, Aniline is a weaker base than NH_3



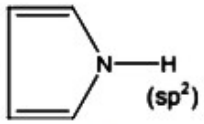
b) Pyridine (C_5H_5N) : 6-π electrons, aromatic

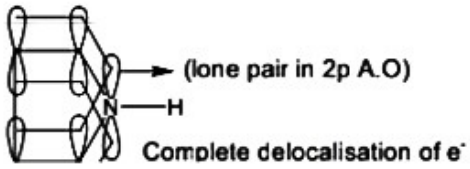
S.F:- 

A.O. Diagram : 

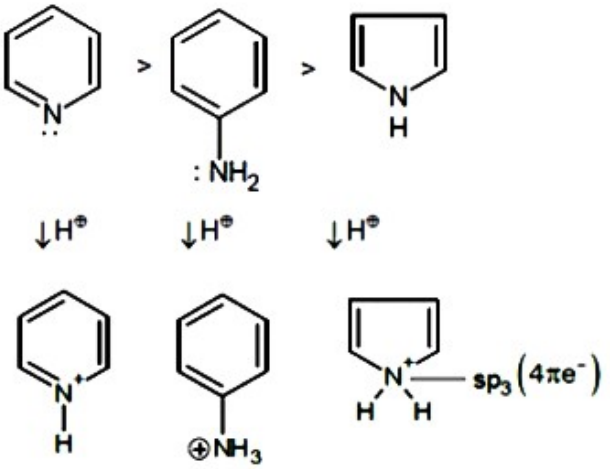
Lone pair of N in pyridine is localised so it is more basic than aniline.

c) Pyrrole (C_4H_5N:) : 6-π electrons, aromatic

S.F:- 

A.O. Diagram : 

Lone pair of N in pyrrole is delocalised in the aromaticity so it is very less basic than aniline.



Aromatic Aromatic Non-aromatic

General basicity order :

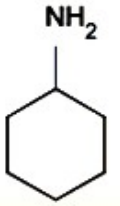
$$R_3C^{\ominus} > R_2-\overset{\ominus}{C}H_2 > \overset{\ominus}{C}H_3 > \overset{\ominus}{P}h > Ph\overset{\ominus}{C}H_2 > \overset{\ominus}{N}H_2 > Ph_2\overset{\ominus}{C}H > Ph_3\overset{\ominus}{C} > \overset{\ominus}{H} > R-C\equiv\overset{\ominus}{C} > RC(=O)\overset{\ominus}{C}H_2$$

$$\approx R-\overset{\ominus}{O}\lesssim \overset{\ominus}{O}H \gtrsim CH_3\overset{\ominus}{O} > CO_3^{2-} > Ph\overset{\ominus}{O} > R\overset{\ominus}{S} > RCO\overset{\ominus}{O} > OC\overset{\ominus}{N} > H_2O > \overset{\ominus}{Cl}$$

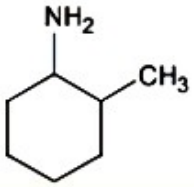
Concept Building Exercise

Compare which is more basic in nature

Q191.



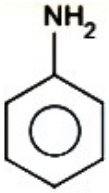
(a)



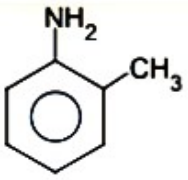
(b)

A191. b

Q192.



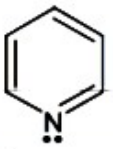
(a)



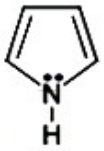
(b)

A192. a

Q193.



(a)



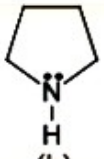
(b)

A193. a

Q194.



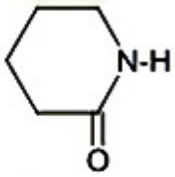
(a)



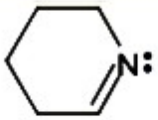
(b)

A194. b

Q195.



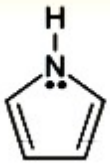
(a)



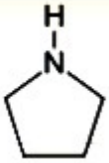
(b)

A195. b

Q196.



(a)



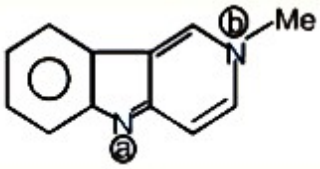
(b)

A196. b

Concept Building Exercise

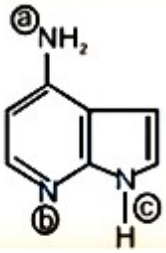
Which Nitrogen is most basic :

Q197.



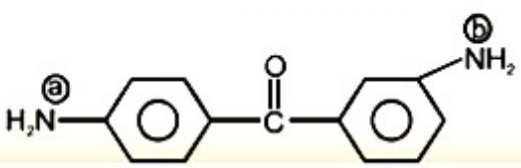
A197. a > b

Q198.



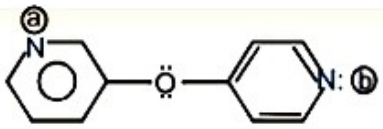
A198. b > a > c

Q199.



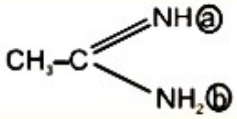
A199. b > a

Q200.



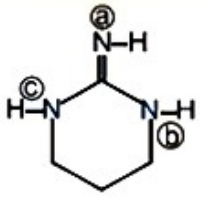
A200. b > a

Q201.



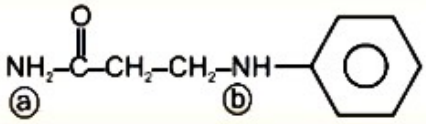
A201. a > b

Q202.



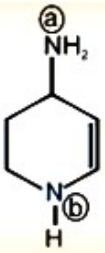
A202. a > b = c

Q203.



A203. b > a

Q204.

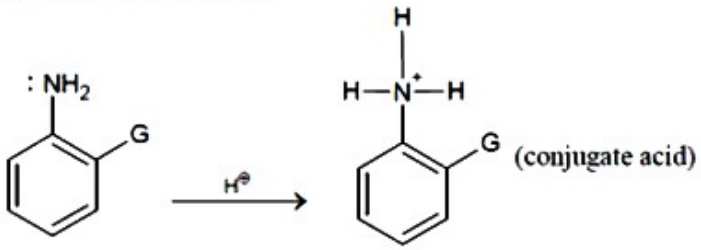


A204. a > b

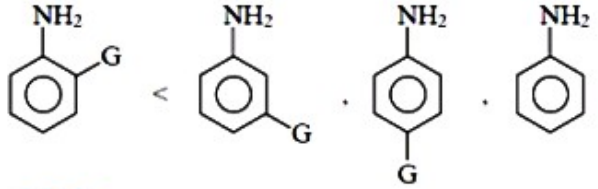
Steric Inhibition in protonation

SIR effect & metal reaction

The presence of bulky group at the ortho position of an aromatic amine provides a steric hindrance and inability during solvation. ortho substituted aromatic amine behaves as a poor base when compare to meta, para unsubstituted aromatic amines.



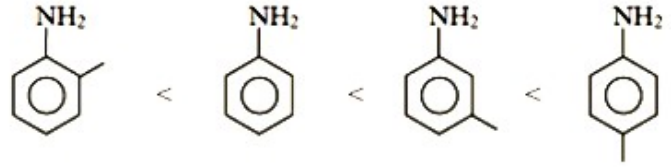
1. Ortho-substituted anilines are mostly weaker bases than aniline itself.
2. Ortho-substituent causes steric hinderance to solvation in the product (conjugate acid i.e. cation).
3. The small groups like $-NH_2$ or $-OH$ do not experience (SIR) due to small size.



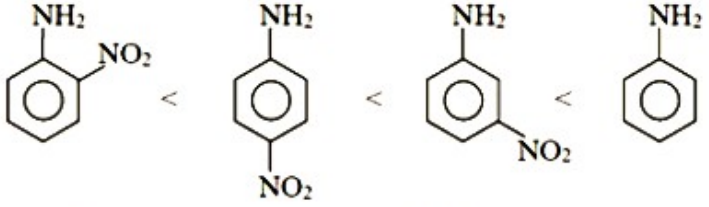
Basic Concpet & Theory :

Example: Basic strength order

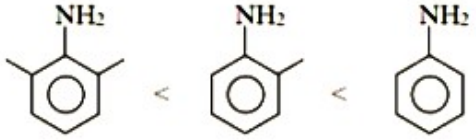
(i)



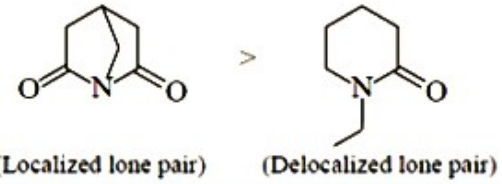
(ii)



(iii)



(iv)

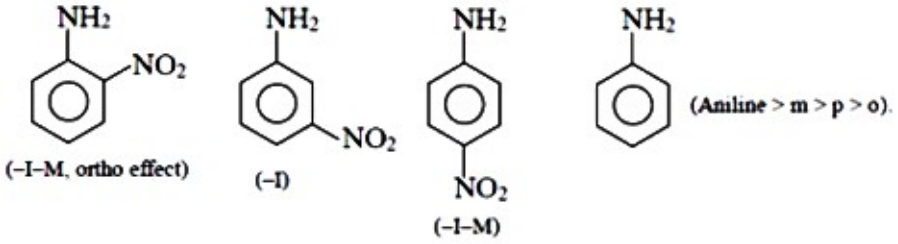


(Localized lone pair) (Delocalized lone pair)

Examples :

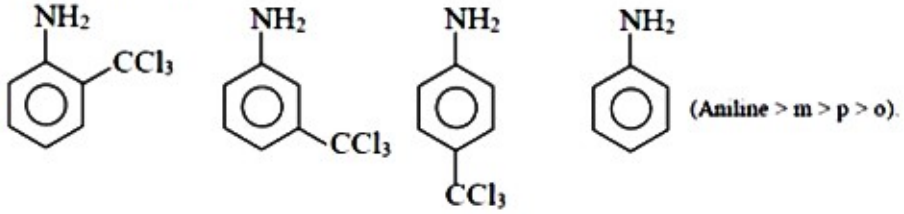
G = (–M, –I); NO_2

(a)



G = (–I); CCl_3

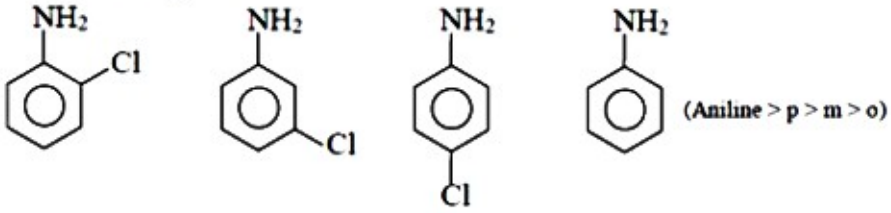
(b)



Only (–I) decides the order.

G=(-I > + M); Cl

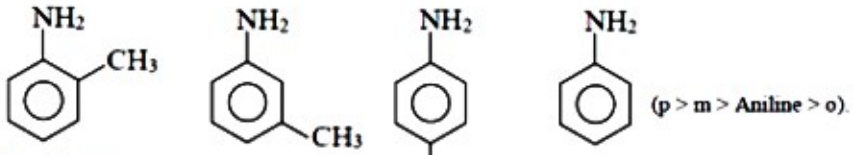
(c)



Only (–I) decides the order

G = (+I, HC); If R = $-CH_3$ (Toluidines)

(d)



Ortho effect.
+I
HC

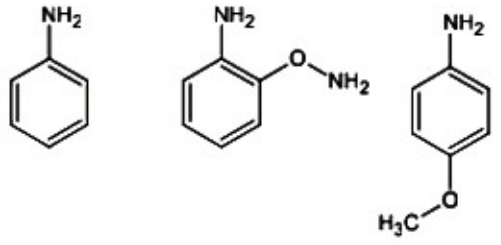
+I

+I

HC → more dominating

G= (+M > –I)

(e)

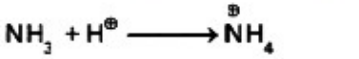


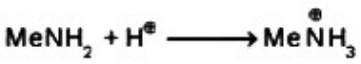
K_b order : p > Aniline > 0 > m

Solvent Effect in bases:

Basic Strength of aqueous solutions

Consider the following amines

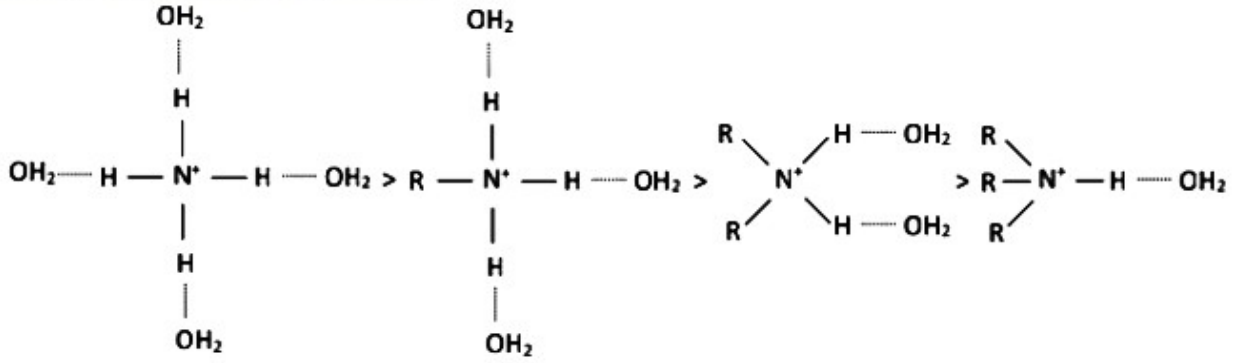






Consider the electron density only, we can clearly observed that $(Me)_3N$ has the highest electron density due to + I-effect of methyl group.

In the aqueous phase, the substituted ammonium cations get stabilised not only by electron releasing effect of the alkyl group (+I) but also by solvation with water molecules. The greater the size of the ion (Alkyl groups are hydrophobic and inhibits H bonding and solvation.), lesser will be the solvation and the less stabilised is the ion. The order of solvation of ions are as follows:



Ammonium ion 1° 2° 3°

Greater is the stability of the substituted ammonium cation, stronger should be the corresponding amine as a base.

Now in aqueous solution, weak also need to consider solvation energy for the amines.

Thus, be consider the solvation energy and find the following analysis.

AMINE	ELECTRON DENSITY	SOLVATION ENERGY
NH_3	×	↑ ↑ ↑
$NH_2(Me)$		↑
$NH(Me)_2$	↑ ↑	
$N(Me)_3$	↑ ↑ ↑	×

Thus, we find the following order in aqueous solution for the basic strength.

$NH(Me)_2$	>	$NH_2(Me)$	>	$N(Me)_3$	>	NH_3
2° amine		1° amine		3° amine		

On the basis of above two sequences, we can say that the basic strength of amines is the combined effect of inductive effect, steric hindrance and solvation.

$R_2NH > RNH_2 > R_3N > NH_3$ (if R = CH_3) .....(1)

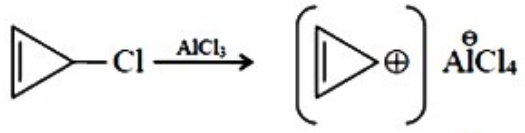
$R_2NH > R_3NH > RNH_2 > NH_3$ (if R = C_2H_5) .....(2)

For Gaseous phase

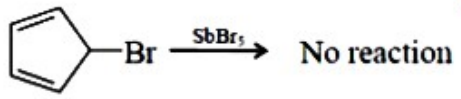
We obtained the order of basic strength as per the order of electron density.

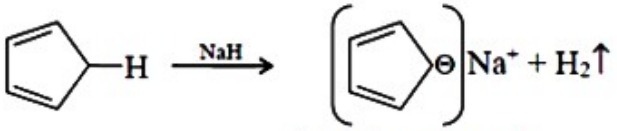
$N(Me)_3$	>	$NH(Me)_2$	>	$NH_2(Me)$	>	NH_3
3° amine		2° amine		1° amine		

SOME REACTIONS :

1. 

id

2. 

3. 

Aromatic compound

Reactions of Bases :

i) **Nature of aqueous solution:** Amines combine with water to form alkyl ammonium hydroxides. This gives hydroxide ions in solution, thus the aqueous solution of amines is basic in

$$RNH_2 + HOH \rightleftharpoons R\overset{+}{N}H_3OH^- \rightleftharpoons [RNH_3]^+ + OH^-$$

1°Amine

$$R_2NH + HOH \rightleftharpoons R_2\overset{+}{N}H_2OH^- \rightleftharpoons [R_2NH_2]^+ + OH^-$$

The aqueous solution of amines behave like NH_4OH and gives the precipitate of ferric hydroxide with ferric chloride

$$3RNH_3OH + FeCl_3 \rightleftharpoons Fe(OH)_3 + 3RNH_3Cl$$

Brown ppt.

ii) Aliphatic and aromatic amines form salt because of their basic nature:

$$RNH_2 + H_2O \rightarrow R\overset{+}{N}H_3\ \overset{-}{O}H$$

Alkylammoniumhydroxide

$$RNH_2 + HCl \rightarrow R\overset{+}{N}H_3\ \overset{-}{C}l$$

Alkylammoniumchloride

$$RNH_2 + H_2SO_4 \rightarrow \left(R\overset{+}{N}H_3\right)_2\ SO_4^{-2}$$

Similarly we get

$$C_6H_5NH_2 + H_2O \rightarrow C_6H_5\overset{+}{N}H_3\ \overset{-}{O}H$$

Aniliniumhydroxide

$$C_6H_5NH_2 + HCl \rightarrow C_6H_5\overset{+}{N}H_3\ \overset{-}{C}l \text{ or } C_6H_5NH_2 : HCl$$

Aniliniumchloride Anilinehydrochloride

Salts of amines are ionic compounds and hence water soluble.

8. Reactions of Acids with salts :

(1) $NaX + HY \longrightarrow NaY + HX$

Salt of Weak acid Strong Acid

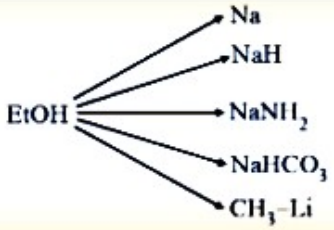
Remark : A stronger acid displaces the weaker acid from weak acid metal salt. The weaker acid is released out as a gas or liquid or precipitates out as a solid. The weaker acid cannot displace the stronger acid from the salt.

1. $2\ NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$
2. $Na_2SO_4 + 2HCl \longrightarrow$ No reaction
3. $CH_3COONa + CH_3SO_3H \longrightarrow CH_3COOH + CH_3SO_3Na$ (feasible)
4. $CH_3COONa + PhOH \longrightarrow PhONa + CH_3COOH$ (not feasible)

Concept Building Exercise

State different type of gases evolved in the following given blocks :

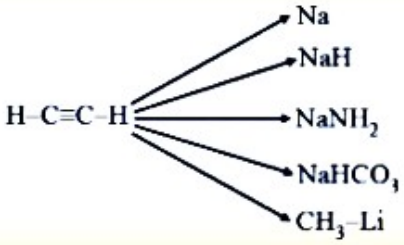
Q205.



A205. $Et\overset{\ominus}{O}\overset{\oplus}{Na} + \frac{1}{2}H_2$ $Et\overset{\ominus}{O}\overset{\oplus}{Na} + H_2$ $Et\overset{\ominus}{O}\overset{\oplus}{Na} + NH_3$ $Et\overset{\ominus}{O}\overset{\oplus}{Na} + H_2CO_3$

$Et\overset{\ominus}{O}\overset{\oplus}{Li} + CH_4$

Q206.

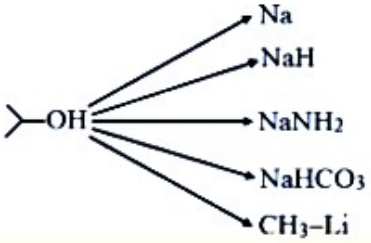


A206. $H-C\equiv\overset{\ominus}{C}\overset{\oplus}{Na} + \frac{1}{2}H_2$ $H-C\equiv\overset{\ominus}{C}\overset{\oplus}{Na} + H_2$

$H-C\equiv\overset{\ominus}{C}\overset{\oplus}{Na} + NH_3$ $H-C\equiv\overset{\ominus}{C}\overset{\oplus}{Na} + H_2CO_3$ Backward

$H-C\equiv\overset{\ominus}{C}\overset{\oplus}{Na} + CH_4\uparrow$

Q207.

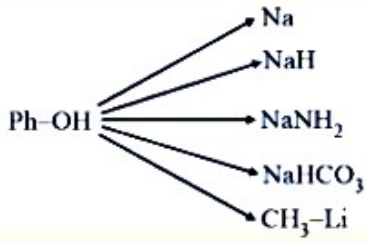


A207. $\overset{\ominus}{O}\overset{\oplus}{Na} + \frac{1}{2}H_2$ $\overset{\ominus}{O}\overset{\oplus}{Na} + H_2\uparrow$

$\overset{\ominus}{O}\overset{\oplus}{Na} + NH_3\uparrow$ $\overset{\ominus}{O}\overset{\oplus}{Na} + H_2CO_3$ (Backward)

$\overset{\ominus}{O}\overset{\oplus}{Li} + CH_4$

Q208.



A208.

$Ph-\overset{\ominus}{O}\overset{\oplus}{Na}+\frac{1}{2}H_2\uparrow$

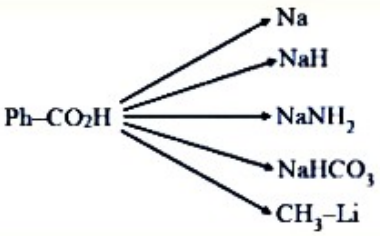
$Ph-\overset{\ominus}{O}\overset{\oplus}{Na}+H_2\uparrow$

$Ph-\overset{\ominus}{O}\overset{\oplus}{Na}+NH_3(g)$

$Ph-\overset{\ominus}{O}\overset{\oplus}{Na}+H_2CO_3$ (Backward)

$Ph-\overset{\ominus}{O}\overset{\oplus}{Li}+CH_4\uparrow(g)$

Q209.



A209.

(i) $Ph-\overset{O}{\overset{\|}{C}}-O^{\ominus}Na^{\oplus}+\frac{1}{2}H_2\uparrow$

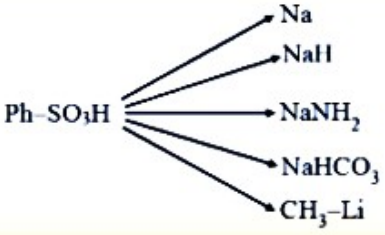
(ii) $Ph-\overset{O}{\overset{\|}{C}}-O^{\ominus}Na^{\oplus}+H_2\uparrow$

(iii) $Ph-\overset{O}{\overset{\|}{C}}-O^{\ominus}Na^{\oplus}+NH_3\uparrow$

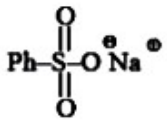
(iv) $Ph-\overset{O}{\overset{\|}{C}}-O^{\ominus}Na^{\oplus}+H_2CO_3(CO_2+H_2O)$

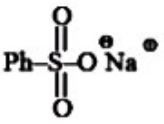
(v) $Ph-\overset{O}{\overset{\|}{C}}-O^{\ominus}Na^{\oplus}+CH_4\uparrow$

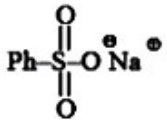
Q210.

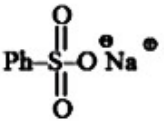


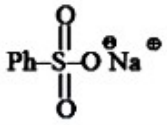
A210.

(i)  $+ \frac{1}{2}H_2$

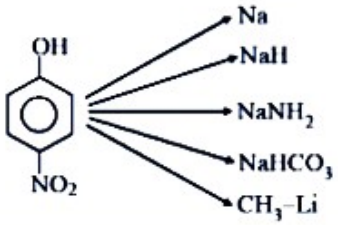
(ii)  $+ H_2$

(iii)  $+ NH_3$

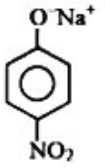
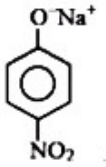
(iv)  $+ H_2O + CO_2$

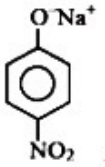
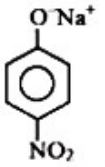
(v)  $+ CH_4$

Q211.

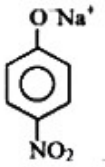


A211.

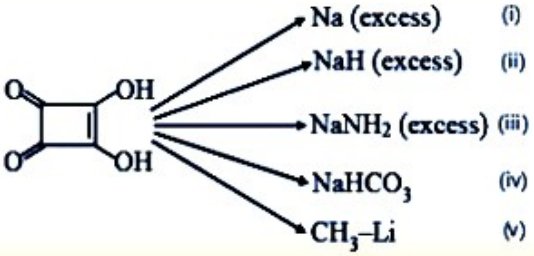
 $+ \frac{1}{2}H_2\uparrow$  $+ H_2\uparrow$

 $+ NH_3\uparrow$  $+ H_2O +$

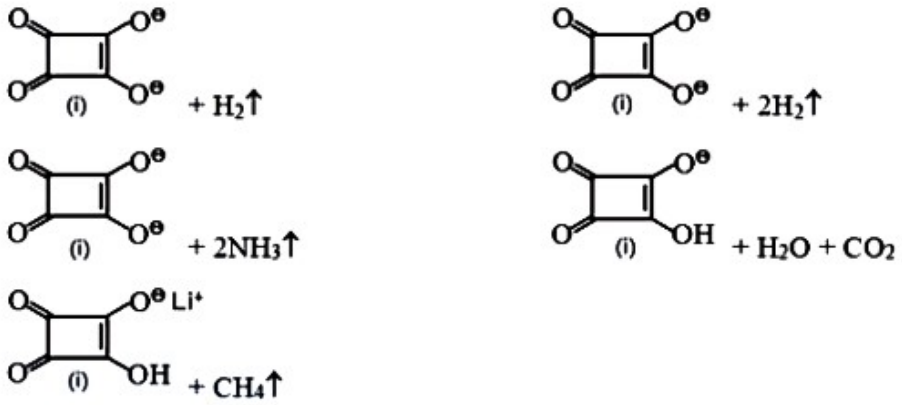
CO_2

 $+ CH_4$

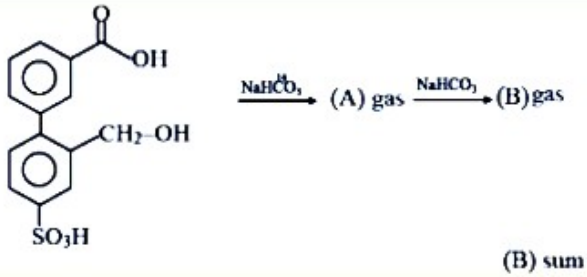
Q212.



A212.

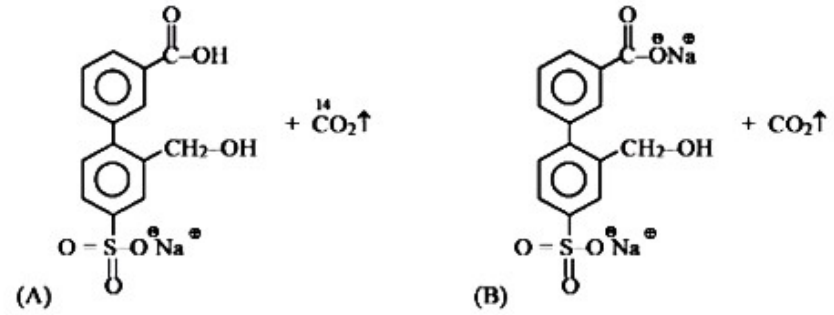


Q213.



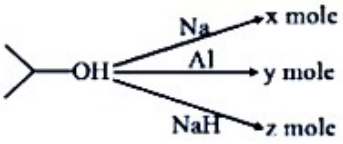
(B) sum of molecular weight of (A + B) ?

A213.



A + B = 44 + 46 = 90

Q214.



Find sum of x + y + z ?

A214.

